

THE ARCHITECT

*A Blueprint for the Civilization
That Survives Itself*



MIRCEA SILVIU RUSU

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First Edition · 2026

The Architect
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To the generations who will still be here.



*The fault, dear Brutus, is not in our stars, but in ourselves, that we are
underlings.*

— William Shakespeare, *Julius Caesar* (c. 1599)

*We have to choose between a global market driven only by calculations of
short-term profit, and one which has a human face.*

— Kofi Annan, address to the World Economic Forum (1999)

Look again at that dot. That's here. That's home. That's us.

— Carl Sagan, *Pale Blue Dot* (1994)

P R E F A C E



THIS BOOK WAS WRITTEN IN A PERIOD WHEN THE ARCHITECTURE OF human civilization began to visibly fail, in plain view of the very people it was built to serve. The institutions that govern our money, our laws, our work, our information, and our wars were designed for a world that no longer exists — a world of isolated communities, slow messengers, and modest tools. We now live inside a single, instantaneous, planetary network, and we are governing it with paperwork, with patronage, and with the inherited reflexes of empires that did not survive their own contradictions.

The argument of these pages is simple, and it is severe. Either humanity designs, in this century, an operating system worthy of its capacities, or it will lose the chance to design anything at all. The window is not abstract. It is measured in decades, not in centuries, and it is shrinking under the combined weight of ecological pressure, weapons capable of ending us in an afternoon, intelligences that already exceed us in narrow domains, and political systems incapable of acting at the speed of the problems they have created.

What follows is not a prophecy and not a manifesto. It is a blueprint — a long, patient, technically honest attempt to describe how a civilization of eight billion people, on one planet today and on many tomorrow, could be governed in a way that preserves dignity, makes corruption structurally impossible, places intelligence under human authority, returns work to human purpose, and points the species toward a single shared horizon. None of it requires saints. All of it requires architecture.

The reader should know, at the outset, what this book is not. It is not the design of a world government; it proposes a federation that any community

may join, adapt, or leave. It is not anti-technological; it treats technology as the substrate on which the next civilization will either be built well or be built carelessly. It is not utopian; every claim made here is measured against the evidence we already have, and every failure mode is named in the open. It is not the property of any party, religion, or nation. It is offered, in the most literal sense, to whoever will use it.

A word about the form. This is written as an old book because the questions it asks are old. How shall we live together? How shall we be governed? What is owed to those who come after us? The new answer requires new instruments, but the question is the oldest question we have. I have therefore allowed myself the liberty of writing in the voice of an architect addressing an apprentice — patient, declarative, unhurried, and unwilling to flatter.

If you have opened this book, you already suspect that the present arrangement of the world will not hold. You are not wrong. What follows is the case that, even now, there is something better available to us — not as a dream, but as a buildable structure, drawn here in full, so that any group of serious people, in any city or any country, can begin construction the day they decide to begin.

— *M. S. R.*

PART ONE

THE DIAGNOSIS



The world as it is, examined honestly, before any proposal is made for what it might become.

CHAPTER ONE

THE CIVILIZATION BUILT ON OLD CODE



EVERY CIVILIZATION IS, IN THE END, A PIECE OF CODE. NOT IN THE narrow sense of machines and silicon, but in the older and more durable sense: a set of rules executed by a population, day after day, until the rules become indistinguishable from reality. A border is a rule. A currency is a rule. A court, a tax, a vote, a marriage, a passport, a contract, a frontier post — each is a line of code that human beings have agreed to run inside their own minds and inside the minds of their neighbors. So long as enough of us continue to execute the rule, it has the force of nature. When too few of us continue, it dissolves overnight, and we wonder how we ever believed in it.

The civilization we have inherited is running on code written, for the most part, between the seventeenth and the mid-twentieth century. The sovereign nation-state, fixed by the Peace of Westphalia in 1648, was an engineering solution to the wars of religion in central Europe (Kissinger 2014). The joint-stock corporation, generalized in English and Dutch law during the seventeenth and eighteenth centuries, was an engineering solution to the problem of pooling capital for ventures too large for any single household. The modern bureaucracy, codified by Weber at the turn of the twentieth century, was an engineering solution to the problem of administering populations measured in tens of millions (Weber 1922). The post-war international order — the United Nations, the Bretton Woods institutions, the General Agreement on Tariffs and Trade, the Universal Declaration of Human Rights — was an engineering

solution to the moral catastrophe of two world wars in a single generation (Mazower 2012; United Nations General Assembly 1948).

These were, in their time, brilliant pieces of design. They achieved exactly what they were built to achieve. They ended the Thirty Years' War. They financed the construction of railways and the laying of submarine cables. They produced a century of life expectancy gains unprecedented in human history. They contained — if barely — the appetites of two superpowers armed with weapons that could have ended the species in an afternoon. None of what follows in this book should be read as contempt for the builders of the old code. The argument is far more uncomfortable than contempt. The argument is that the code worked, and that the world it was written for no longer exists.

THE WORLD IT WAS WRITTEN FOR

Consider, for a moment, the assumptions baked into the institutions we inherited. They assume that information travels slowly — at the speed of a horse, a ship, or, at best, a telegraph wire — so that a parliament can deliberate for weeks while events wait politely outside the chamber. They assume that capital is local — that a bank in Hamburg or a mill in Lancashire is taxable by the polity in which it physically stands. They assume that violence is symmetrical — that an army must be assembled, fed, transported, and paid, so that war is the prerogative of states large enough to bear its cost. They assume that the planet is large — that what is dumped into a river in one country will not, within a human lifetime, alter the chemistry of an ocean on the other side of the world. They assume that human beings are the apex of cognition on Earth — that no other agent will compose treaties, draft contracts, persuade voters, or design weapons.

Every one of these assumptions has now failed. Information travels at the speed of light, and a rumor circulates the globe before a cabinet has finished its breakfast. Capital is no longer local; a few keystrokes move sums larger than

the annual budget of a middle-income country between jurisdictions that compete to ask the fewest questions. Violence is no longer symmetrical; a single individual with a laptop can disable a hospital network, and a small group with the right code can interrupt the electricity, the fuel, the water, or the financial settlement of a nation (Perlroth 2021). The planet is no longer large; we have altered the composition of its atmosphere measurably, and the consequences are now legible in fires, in floods, in crop failures, and in the migration of tens of millions of people (IPCC 2023). And human beings are no longer the only authors in the room; systems we have built ourselves now write text, code, contracts, music, and military doctrine at a speed we cannot match, and in volumes we cannot read.

An honest observer, looking at this list, would conclude that the operating system of our civilization is being asked to perform tasks for which it was never specified. The wonder is not that it strains. The wonder is that it has not yet crashed.

THE SYMPTOMS OF OBSOLESCENCE

What does obsolescence look like, in a civilization? It does not look like collapse. Collapse is the final frame. Long before collapse, the symptoms are subtler and far more familiar. They look like this.

They look like a public that has stopped believing the official numbers, even when the numbers are correct, because it has been lied to about other numbers often enough that trust has been spent. They look like elections in which the population votes against incumbents in cycle after cycle, in country after country, regardless of ideology, because no incumbent of any persuasion seems capable of solving the problems the population can see. They look like the steady accumulation of laws nobody reads, regulations nobody enforces, and treaties nobody honors, until the legal order itself becomes a kind of liturgy — recited, respected, and ignored. They look like billionaires whose private

fortunes exceed the budgets of the states that nominally regulate them (UBS 2024). They look like wars that begin without declaration and end without treaty, in places nobody can find on a map, fought by drones nobody has met, against enemies nobody has named. They look like a generation of young people who have concluded, not loudly but quietly, that the future is something to be survived rather than built.

None of these symptoms, taken alone, is fatal. Taken together, they describe a system that has lost the capacity to adapt at the rate the environment is changing. In engineering, this condition has a name. It is called *technical debt*. It accumulates whenever a system is patched rather than redesigned, whenever a workaround is preferred to a fix, whenever the cost of refactoring is deferred until the cost of refactoring exceeds the cost of starting again. Every civilization that has ever fallen, fell first into technical debt. The lesson of the archaeological record is not that great societies are destroyed from outside. It is that they exhaust their capacity to maintain the systems on which they depend, and then a smaller shock — a drought, a plague, a war on the frontier — finishes what their own neglect began (Tainter 1988).

FIVE FAILURES, IN SUMMARY

The chapters that follow this one will name, in turn, the five failures that constitute the central diagnosis of this book. They are sketched here only briefly, so that the reader knows where the argument is going.

The first is the *failure of trust*. Public institutions, in nearly every measured democracy, are trusted by a smaller fraction of their citizens than at any point since reliable measurement began (OECD 2024). This is not because citizens have become more cynical. It is because the institutions have become less legible. A modern citizen cannot, even in principle, verify how a tax dollar is spent, how a vote is counted, how a regulator is appointed, or how a court reaches its

conclusions. The machinery is sealed, and the public is asked to take its operation on faith. Faith, in such conditions, predictably erodes.

The second is the *failure of distribution*. The global economy now produces more than enough food, energy, housing, medicine, and information to permit every human being to live in dignity (World Bank 2024). It does not do so. A larger share of global wealth is held by a smaller fraction of the population than at any point in the modern era (UBS 2024). This is not the predictable outcome of a market. It is the predictable outcome of a market layered on top of legal, tax, and ownership rules written for an economy of land and factories, applied without revision to an economy of data, intellectual property, and global financial flows.

The third is the *failure of work*. The connection between human effort and human dignity, which industrial society took for granted, has been quietly dissolving for two generations and is now being severed at speed by automation, by platform capitalism, and by intelligent systems capable of performing entire categories of skilled labor for the marginal cost of electricity. The institutions that were built to manage this transition — labor law, social insurance, professional licensure — were designed for a world in which the worker was a person, the employer was a firm, and both were located in the same country. Two of those three assumptions are now optional, and the third is becoming so (Graeber 2018).

The fourth is the *failure of law*. National legal systems, written by national legislatures for national populations, cannot regulate planetary phenomena. They cannot, on their own, tax a corporation that exists in twelve jurisdictions and books its profits in none. They cannot prosecute a fraud committed in one country, against citizens of a second, using infrastructure rented in a third, by a defendant who lives in a fourth. They cannot constrain weapons of mass destruction held by states that have not signed the relevant treaties. They cannot, in the current arrangement, do any of these things, and the gap

between what national law can do and what global activity requires has become the working environment of organized crime, of tax evasion, of human trafficking, and, increasingly, of states themselves.

The fifth is the *failure of foresight*. We have constructed a civilization whose decisions are made on horizons of months — the electoral cycle, the fiscal quarter, the news cycle — and whose consequences play out on horizons of decades or centuries. Climate, biodiversity, infrastructure, pensions, debt, scientific research, education, and the long, slow work of cultural transmission all operate on timescales that no existing political institution is structurally equipped to consider. We are, collectively, a species with the technological reach of a mature civilization and the planning horizon of an adolescent. That mismatch, more than any other, is what makes the present moment dangerous.

WHY THIS BOOK EXISTS

It would be possible to write a far longer book about each of these failures, and many such books have been written. The reader who wishes to dwell on the diagnosis will find a vast literature waiting, in economics, in political science, in history, in sociology, in climate science, in the philosophy of technology. The original contribution this book attempts is different. It assumes that the diagnosis, in its broad outline, is now established. It assumes that the reader, having lived in the world for some number of decades, requires no extended demonstration that the world is unwell. It moves instead to the question that the diagnostic literature, with notable exceptions, has been reluctant to engage. That question is the engineering question. *If the operating system is obsolete, what would a new one actually look like?*

Not in metaphor. Not in vision. In architecture. With named components, named interfaces, named failure modes, named transitions from the present arrangement to the proposed one. With the dignity of being wrong in public, in detail, so that the design can be inspected, criticized, improved, replaced.

The chapters that follow are an attempt to answer that question — not as the final word, but as a first draft serious enough to be argued with.

The argument proceeds in six parts. Part One, of which this chapter is the opening, sets out the diagnosis. Part Two states the seven principles on which any successor system must be built if it is to be worthy of the name. Part Three presents the constitutional text — the founding document of a federation that any community on Earth may join — together with the reasoning behind each of its provisions. Part Four describes the architecture in technical detail: how the public ledger works, how the governance layer reaches decisions, how intelligence is placed under human authority, how identity is held by the person rather than the platform, how the economic instruments distribute abundance without destroying initiative, how labor is restored to human purpose, how law is harmonized across jurisdictions without coercion, how enterprises are kept honest without being crushed. Part Five describes the transition: how a civilization moves from the present arrangement to the proposed one, year by year, without civil war and without utopia. Part Six asks what comes after — what becomes possible, on this planet and beyond it, once the operating system is no longer the bottleneck.

This is the program. The reader who finds it audacious is not wrong. The reader who finds it premature is invited to compare the audacity of the proposal to the audacity of continuing on the present course. The cost of building a new system is large. The cost of not building one is larger, and is already being paid, in installments, by people who never signed the bill.



The diagnosis continues in Chapter Two, which examines the failure of trust in detail, and asks what trust is actually made of — and how, structurally, it could be rebuilt.

CHAPTER TWO

THE FAILURE OF TRUST



TRUST, IN THE STRUCTURAL SENSE RELEVANT TO THIS ARGUMENT, IS NOT a feeling. It is a balance sheet. On one side stand the things a population is asked to take on faith: that the vote was counted as cast, that the tax was spent as voted, that the medicine was tested as labeled, that the judge was independent, that the regulator was not bought, that the official number is the actual number. On the other side stand the instruments by which any of these claims can be verified by an ordinary person, without privileged access, without specialized training, in a finite amount of time. When the first side is much larger than the second, the difference is borrowed. We have, in the late industrial democracies, been borrowing for a very long time.

The arithmetic is no longer disputed. In the United States, the share of the public expressing trust in the federal government to do what is right “most of the time or always” stood at roughly seventy-three percent in 1958 and at sixteen percent in 2024 (Pew Research Center 2024). The trajectory in the United Kingdom, France, Italy, and Japan is structurally identical: a slow erosion punctuated by sharp drops around financial crises, surveillance scandals, and pandemics, with no sustained recovery in any of them (OECD 2024). The Edelman Trust Barometer, which surveys twenty-eight countries annually, has for several years described “government” and “media” as *distrusted* by the median respondent globally, with business and non-governmental organizations slightly above the waterline (Edelman 2024). This is not a national mood. It is a planetary signal.

It is tempting, and politically convenient, to attribute the signal to the disinformation environment. The argument runs: the platforms broke the public sphere, conspiracy displaced consensus, and trust collapsed because attention was hijacked. This is half true and therefore dangerous, because it implies that trust can be restored by better content moderation. It cannot. The disinformation environment is downstream of a more basic fact. Citizens have, on the whole, correctly inferred that they cannot verify the operations of authority, and they have rationally concluded that institutions which cannot be verified should not be trusted. The platforms did not cause this. They simply made the gap audible.

WHAT TRUST IS STRUCTURALLY MADE OF

Begin with a small example. A citizen in any modern democracy pays a tax. The money disappears into a general fund. From the general fund it flows, by appropriations, to ministries; from ministries, by contracts, to suppliers; from suppliers, by subcontracts, to the actual hands that build the road or treat the patient. Each handoff is documented in some register. None of the registers is connected. None is, in any practical sense, queryable by the person who supplied the money. The citizen is told, in aggregate annual statements, what the expenditure was. The citizen has no instrument by which to check.

This is not a complaint about secrecy. The numbers are, in most jurisdictions, public. The complaint is about *resolution*. Public information at the level of the aggregate is structurally indistinguishable, for the citizen, from no information at all. To verify that a road was built for the amount claimed, the citizen would need: the contract; the subcontracts; the timestamps of payments; the engineering specifications; the inspection reports; the bills of material; the salaries paid; the variances explained. These exist. They are scattered across a dozen agencies, in incompatible formats, behind freedom-of-

information procedures whose timeline exceeds the political relevance of the question. The information is technically public and operationally private.

What this produces, over decades, is a public that has learned to act *as if* the books were closed, because the cost of opening them exceeds the cost of giving up. Once that learning is general, two further things follow. First, the segment of the public most committed to verification — the press, the auditors, the activists, the curious — is itself perceived as partisan, because the act of looking has become unusual enough to look ideological. Second, the segment of the public least committed to verification — the casual majority — becomes susceptible to whatever explanation of opacity is loudest, whether that explanation is corruption, conspiracy, foreign influence, or theological. None of these dynamics requires malice on the part of officials. They follow inevitably from a regime in which the cost of verification exceeds the cost of belief.

THE LEGIBILITY LADDER

It is useful to distinguish four rungs on the ladder of public legibility, because policy debates routinely confuse them.

The first rung is *disclosure*. The state publishes a document. This is the minimum that any modern administration provides, and it satisfies the legal definition of openness while satisfying no practical demand for trust. The second rung is *retrieval*. A citizen who wishes to inspect a specific decision can find the relevant document without bureaucratic obstruction, in a reasonable time, in a format that does not require specialist software. Few governments are reliably at the second rung. The third rung is *structured access*. The relevant data is published in machine-readable form, joined across agencies by stable identifiers, queryable through standard interfaces, so that a journalist, an auditor, a student, or an automated agent can answer questions that the publishing agency did not anticipate. Almost no government is at the third rung, except in narrow domains such as company registries, contract awards, or

court filings. The fourth rung is *continuous verification*. The state's operations are exposed as a live stream of events — every disbursement, every appointment, every regulatory decision, every contract amendment — anchored in tamper-evident records, against which any party can independently reconstruct the state's books in real time. No state has achieved the fourth rung. It is technically possible, and it is the only rung at which institutional trust becomes self-renewing.

The conventional wisdom holds that the fourth rung is unattainable for reasons of national security, commercial confidentiality, and personal privacy. Each of these reasons is partly correct and substantially overstated. National security demands secrecy over a small fraction of state activity, not over the disbursement of social benefits or the awarding of road contracts. Commercial confidentiality is a legitimate interest of suppliers, but it does not require concealment from the public of *what* the public paid *for what*; it requires only that proprietary technical specifications remain protected. Personal privacy is a genuine constraint on raw data, but it is well understood, in modern cryptography, how to publish aggregates and verifiable proofs without exposing the underlying records. The barrier to the fourth rung is not technical. It is institutional reluctance to be that legible.

WHY OPACITY IS STRUCTURALLY PREFERRED

An honest analysis must concede that opacity is not an accident. It is preferred by the operators of the system because it preserves discretion. A regulator who is observed in real time loses the latitude to grant a quiet exception. A procurement officer whose every signature is traceable forfeits the flexibility to favor a familiar bidder. A central bank whose deliberations are continuously visible may find its policy options narrowed by anticipation. A minister whose travel schedule is published cannot meet whom she chooses, where she chooses. Some of this discretion is corrupt. Most of it is not. Most of it is the lubricant

by which large bureaucracies function — informal favors, tactical ambiguity, plausible deniability, the room to change one’s mind without explaining why. This lubricant is what legibility threatens, and this is why every step up the ladder is resisted with the same arguments — privacy, security, complexity — regardless of which agency is involved.

The historical record shows the resistance can be overcome, and shows how. Court rulings have forced disclosure of police misconduct records (O’Neil 2016). International journalism consortia have forced disclosure of offshore financial structures, exposing the use of jurisdictions whose principal export is concealment (International Consortium of Investigative Journalists 2021). Open-data movements have compelled the publication of public-procurement awards in dozens of countries. In each case, the resistance was strong, the disclosure occurred, and the institutions that adopted higher legibility ended the cycle with greater, not lesser, public confidence. Legibility, once delivered, is rewarded. This is the empirical answer to the official’s perennial worry that more openness will only feed more cynicism.

WHAT THE DISINFORMATION ENVIRONMENT ACTUALLY DOES

It is necessary, before proposing a reconstruction, to be precise about the role of the disinformation environment. The platforms did not invent distrust. They industrialized its expression. In the older media regime, a citizen who suspected the books had been cooked could mutter to a neighbor; in the new regime, the same citizen broadcasts to a network of strangers selected by a recommendation engine that rewards the most arresting version of the suspicion. The result is not, as is often claimed, that the public was made paranoid by the platforms. The result is that a pre-existing, accurately grounded suspicion was given a delivery mechanism with vastly higher reach

than the official rebuttal. Distrust is now public in a way that, for most of modern history, it was not.

This has two consequences for any program of reconstruction. First, the suppression of disinformation, however well intentioned, will be perceived as the suppression of suspicion as such, and will deepen the very crisis it tries to address. The history of broadcasting law in the late twentieth century is a slow record of this lesson being learned, forgotten, and relearned. Second, the only durable response to a public distrust that has finally found its voice is to give that voice something verifiable to inspect. Trust is restored by exposure, not by reassurance.

THE ARCHITECTURE OF RESTORED TRUST

The remainder of this book will return repeatedly to a single architectural commitment, of which trust is the first beneficiary: every operation of public authority must be recorded on a ledger that is readable by any literate citizen, in real time, in a form that can be independently reconstructed. The ledger is the precondition for verification. Verification is the precondition for trust. Trust is the precondition for the consent on which the legitimacy of any system rests (Habermas 1989b).

What the ledger contains, how it is maintained, by whom it is audited, how it preserves privacy where privacy is warranted, and how it deals with information of genuine sensitivity — all of this is the work of Part Four. The present chapter is content to establish the diagnosis. The institutions of the late industrial order are no longer legible to their citizens; their citizens have correctly responded by withdrawing trust; the withdrawal of trust has been mistaken for a cultural defect of the citizens, when it is in fact a structural feature of institutions that were never designed to be inspected at this scale, by this many people, at this speed. The disease is not in the population. It is in the architecture.

A civilization that wishes to be trusted must accept the cost of being seen. That cost is real. Many decisions that were once made comfortably in private will, in a legible system, have to be defended in public. Many alliances that were once tacit will have to be explicit. Many compromises that depended on no one looking too closely will have to find new form. The transition will be uncomfortable for the people in charge. It will be far less uncomfortable than the alternative, which is the slow rotting of the social contract that we are presently watching in slow motion in every measured country in the world.



The diagnosis continues in Chapter Three, which examines the failure of distribution: why a planet that produces enough for everyone delivers enough to fewer than it did a generation ago, and what the structural reasons are.

CHAPTER THREE

THE FAILURE OF DISTRIBUTION



THERE IS, ON THIS PLANET, ENOUGH FOOD. THE FOOD AND Agriculture Organization, an agency that has no interest in overstatement, has reported for decades that global agricultural production exceeds human nutritional requirement by a comfortable margin. There is enough housing material, enough fresh water, enough generative capacity in clean energy to keep every household lit, enough productive medicine to cover the world's curable disease burden several times over, and an unprecedented surplus of information, instruction, and entertainment. The premise of the modern age — that human ingenuity, properly organized, would produce sufficiency — has, by the metric of total output, already been delivered. What has not been delivered is the second half of the promise: that the sufficiency would reach the people for whose sake it was produced.

The shape of the failure is not subtle. The 2024 Global Wealth Report places the bottom half of the adult population in possession of slightly under one percent of global household wealth, and the top one percent in possession of nearly half (UBS 2024). The same report describes the share of the top decile as having grown, with regional variation, in almost every economy that publishes data, across the past four decades. These are not the numbers of a transient business cycle. They are the numbers of a structural arrangement working as designed.

Two clarifications are necessary. First, the proposition that distribution has failed is not the proposition that absolute poverty has failed to decline. It has declined, substantially, particularly in East and South Asia. The World Bank

reports that the share of the global population living below the international poverty line of \$2.15 per day has fallen from roughly thirty-eight percent in 1990 to under nine percent in the most recent measurements (World Bank 2024). This is one of the few pieces of unambiguously good news of the late twentieth century, and any honest argument must include it. Second, the claim of distributional failure is therefore not that the floor has not risen. It is that the floor has risen far less than the ceiling, that the gap between them is now wider than at any documented point in modern history, and that the gap is widening in directions that no political process now functioning is structurally able to slow (Piketty 2014, 2020).

THE TWO ECONOMIES INSIDE THE SAME CURRENCY

The most useful frame for understanding the late industrial economy is to recognize that it now contains two distinct economies operating under the same legal currency. The first is the economy of *labor and goods*: people work, firms produce, households consume, the velocity of money is measurable, and the indicators that economics inherited from the nineteenth century continue to behave roughly as the textbooks predict. The second is the economy of *asset prices*: equities, real estate, fine art, intellectual property, sovereign debt, derivatives, and, most recently, programmable digital instruments. In the second economy, prices are determined less by present labor than by the discounted expectation of future income streams, and the principal driver of those streams is the global supply of liquidity rather than the productivity of the underlying real assets (Stiglitz 2012).

For most of the twentieth century, the two economies were tied together by the institutions of national finance. Wages tracked productivity; productivity tracked output; output tracked the stock of capital in the country that produced it. By the early twenty-first century, the tie has loosened almost to vanishing. Capital is mobile; labor is not. Capital is taxed at rates that compete

downward across jurisdictions; labor is taxed at rates that compete upward as fiscal pressure rises. Capital appreciates when central banks expand their balance sheets to manage crises; labor's claim on consumption is eroded by the same expansion. The two economies share a name, a flag, and a currency. They do not share a fate.

The consequence is a structural decoupling between the macroeconomic indicators that governments report and the lived experience of the median household. National output rises; median wages stagnate. Unemployment falls; housing affordability collapses. Inflation is reported as moderate while the prices most relevant to younger households — rent, education, healthcare, childcare — rise at multiples of the headline rate. The official measurements are not lies. They are simply describing one of the two economies, while the other economy, in which most of the population lives, has detached.

WHY OLD REDISTRIBUTIVE INSTRUMENTS NO LONGER REDISTRIBUTE

The mid-twentieth-century welfare state was the great success of distributive policy. By the standards of any earlier era, it was an astonishing achievement: universal pensions, public health, mass education, unemployment insurance, progressive income tax, inheritance duty. It rested on a set of preconditions that were specific to its time. The most important of these were that capital was largely national; that the principal form of wealth was the operating firm rather than the financial claim; that the working population was large relative to the dependent population; and that the political authority capable of taxing was coextensive with the geographical area where most economic activity occurred.

Each of those preconditions has weakened. Multinational firms book their profits in jurisdictions chosen for tax minimization rather than for production location, and the largest firms now do this with such efficiency that effective

corporate tax rates on the top fifty global firms are, by several reasonable measurements, in low single digits (Saez and Zucman 2019; Tax Justice Network 2023). The principal form of wealth in advanced economies is no longer the operating firm but the financial claim — equity, bond, real estate, intellectual property — and these claims, particularly in their most aggressive forms, are held in legal vehicles whose ownership is structurally obscured (International Consortium of Investigative Journalists 2021). The working-age population has stopped growing in much of the advanced world and has begun to shrink in some of it; the ratio of contributors to claimants in pay-as-you-go pension systems has crossed thresholds that those systems were not designed to survive. And political authority remains national, while the activity it must regulate is planetary.

The honest summary is that the welfare state was an instrument calibrated to a configuration of capital, labor, and territory that no longer obtains, and the calibration cannot be restored by any sequence of national reforms. Reforms can slow the decline. They cannot reverse it. The reforms that would suffice, if undertaken at the planetary level — a global minimum corporate tax with teeth, a global registry of beneficial ownership, a wealth tax on the top fraction of a percent across jurisdictions — exceed what national legislatures, individually, are constituted to deliver (Piketty 2020).

THE NEW SOURCES OF UNEARNED INCOME

Three relatively new structural features of the global economy concentrate income upward without requiring corresponding contribution. They are worth naming because they are not visible in the standard categories of nineteenth-century economic thought.

The first is the *winner-take-most platform*. A digital service whose unit cost approaches zero, whose value to each user rises with the number of other users, and whose underlying network effect protects the incumbent from substitution, is structurally a monopoly. The largest platforms — for search, for

retail, for social communication, for cloud computing — are not monopolies because they are corrupt; they are monopolies because the technology of the service produces monopoly as its equilibrium. Antitrust frameworks designed to dissolve corrupt monopolies have proven structurally unable to act on these. The result is a transfer of economic surplus, on a scale measured in trillions, from users and small producers to platform operators, with little corresponding increase in productive employment (Zuboff 2019; UNCTAD 2023).

The second is the *intellectual-property estate*. The duration and reach of patent and copyright law, originally calibrated to give inventors a finite period of exclusivity sufficient to recoup their investment, have been extended in successive rounds to durations approaching a century, and to subject matters approaching the entire informational substrate of cultural and scientific life. The recipients of these extensions are predominantly not the original inventors and authors, who are dead or have assigned their rights, but the corporate estates that purchased the rights downstream. This is rent collection by another name, and the rent is collected from every consumer of culture, medicine, software, and seed (Mazzucato 2018).

The third is the *publicly subsidized private return*. A great deal of the technology that powers the most valuable contemporary firms originated in publicly funded research — the internet, satellite positioning, large-scale machine learning, the human genome project, lithium-ion battery chemistry, mRNA platforms. The public bore the risk; the private sector captures the return. There is no inherent injustice in private commercialization of public research, provided the public retains a proportionate claim. In the present arrangement, it does not (Mazzucato 2013).

THE POLITICAL CONSEQUENCE

A population that observes its productivity rising while its share of the product falls is a population that will, in time, withdraw its consent. The withdrawal

can take many forms. It can take the form of populism on the right, with its narrative of betrayal by elites and its hostility to immigration as the visible face of a more general dispossession. It can take the form of populism on the left, with its narrative of class war and its demand for confiscatory remedies. It can take the form of withdrawal from the political process altogether, with declining electoral participation, declining union membership, declining civic association (Putnam 2000). It can take, in extreme cases, the form of separatism, of political violence, or of the slow conversion of the formal democracy into something else under the same name.

None of these outcomes is the considered choice of an irrational public. Each is a structurally rational response to an arrangement in which the rules visibly favor those who already have. A serious distributive remedy must therefore do more than transfer money. It must repair the perception, which is also the reality, that the rules themselves are tilted. The chapters of Part Four will propose specific instruments — a public dividend on commonly produced data and natural resource flows, a planetary minimum on corporate income, a beneficial-ownership ledger, a sunset on intellectual-property monopolies that have outlived their justification, and an architecture of public investment that retains a public claim on the upside it finances. Whether the instruments here proposed are exactly the right ones is a question for the argument, not the diagnosis. The diagnosis is settled: the present distribution is the predictable output of the present rules, and the rules require redesign rather than enforcement.

A NOTE ON GROWTH

It is sometimes argued that distribution will resolve itself if growth is restored, and that the moral imperative is therefore growth rather than redistribution. The argument has a noble pedigree and a hollow record. The advanced economies have grown, in real terms, throughout the period in which their

distributive position has worsened. Growth has not in itself produced distribution. There is no plausible mechanism by which it would, in an arrangement in which the rules of distribution are themselves tilted (Wilkinson and Pickett 2009). Growth remains valuable, particularly in low-income economies, where additional output translates more directly into additional welfare. In high-income economies, the marginal welfare yield of additional growth is now small relative to the welfare yield of repaired distribution and repaired ecological balance (Raworth 2017). To insist on growth as the prior condition for distributive repair is, in 2026, an argument made by people who do not wish the repair to occur.



The diagnosis continues in Chapter Four, which examines the failure of work — what happens to a civilization when the instrument by which dignity has been earned for two centuries no longer reliably produces it.

CHAPTER FOUR

THE FAILURE OF WORK



FOR ROUGHLY TWO HUNDRED YEARS, THE INDUSTRIAL DEMOCRACIES operated on a tacit settlement between citizen and society. The settlement read approximately as follows: in exchange for participation in productive labor, the citizen received income, status, social belonging, the structure of a daily routine, and entry to the institutions of collective bargaining and political voice. The settlement was never universal — it excluded the unpaid, the disabled, the colonized, and large portions of the female population for most of its history — but where it functioned, it produced the most stable and prosperous mass societies in recorded experience. It is now coming apart, on three fronts simultaneously, and the institutions built on the assumption of its persistence are visibly straining.

The first front is the changed character of the employment relation. Across the advanced economies, an expanding share of the working population now earns income through arrangements that resemble traditional employment in their economic effect but not in their legal form: independent contracting, platform work, on-demand assignment, project-based engagement, multi-employer holdings, and informal labor. The proportions vary by country and by sector, but the direction is consistent and durable (*International Labour Organization* 2024). These arrangements transfer to the worker a category of risk — the risk of finding the next job, the risk of insurance, the risk of training, the risk of pension, the risk of legal protection — that the twentieth-century employment contract had transferred to the firm. The transfer is uncompensated. Workers in these arrangements are not paid more in nominal terms; they are paid the

same or less, and required additionally to absorb risks that previously belonged to the firm.

The second front is the structural pressure on middle-skill labor from automation and intelligent systems. The early industrial revolution displaced low-skill manual labor and complemented mid-skill and high-skill work. The automation of the late twentieth century, narrowly construed, did the same. The current wave of automation is structurally different: it complements high-end specialist labor, displaces routinized cognitive labor, and applies pressure to a range of professions that the previous waves left untouched — paralegal work, mid-tier accountancy, customer service, copywriting, junior-level coding, technical translation, basic medical diagnostics, and a growing fraction of the analytic work historically performed by university-trained employees (Brynjolfsson and McAfee 2014; Susskind 2020; Acemoglu and Restrepo 2020).

The third front is the cultural one. A generation has now been raised in proximity to platforms whose central proposition is that visibility, attention, and self-presentation are themselves productive activities — that the line between leisure and labor has dissolved. The same generation has watched its labor be characterized as *content*, its consumer preferences as *data*, and its private communication as *signal*. The relation of self to work, which was already strained by the platform employment economy, has become philosophically unstable. The young, in survey after survey, report a connection to work that is at once more demanding and less remunerative than the connection their parents reported a generation earlier (Graeber 2018; Standing 2011a).

WHAT WORK WAS FOR

To grasp why the unraveling is so consequential, it helps to recall what the industrial settlement actually delivered, beyond income. The job, in the strong twentieth-century sense, performed at least four social functions. It produced a recognized identity, by which a person could be located in the social order. It

produced a temporal structure, by which a life was organized into days, weeks, and years. It produced a circle of association, in which a worker met other workers and acquired the habits of cooperation, conflict, and solidarity. And it produced a route to political voice, through unions, professional bodies, and the implicit standing that came from being a contributor to the collective product (Arendt 1958). Income was the most measurable of these functions, and is therefore the one most often discussed, but it was not necessarily the most important. A society that retains the income but loses the other three has not preserved the settlement; it has only preserved its accounting trace.

This is why the conventional response to automation — increase income transfers; the rest will sort itself out — is structurally inadequate. A universal basic income that delivers economic floor without the other functions of work delivers a kind of well-fed loneliness. The political and cultural costs of such an arrangement, observed in early experiments and in regions of high structural unemployment, are not trivial. The replacement of the settlement requires not the replacement of the wage but the replacement of the wage *plus* the identity, structure, association, and voice that the wage carried with it.

THE TWO PRODUCTIVITY PARADOXES

Two paradoxes recur in the data and deserve brief notice. The first is that aggregate productivity growth, in most advanced economies, has been surprisingly slow during the period of most aggressive deployment of digital and intelligent systems. The puzzle has been studied for thirty years, and the explanations are partial and contested. The most coherent answers point to the lag between the deployment of a general-purpose technology and the organizational redesign required to capture its productivity gains; to mismeasurement of output in service-heavy economies; and to the concentration of gains within a small number of leading firms whose practices do not diffuse (Brynjolfsson and McAfee 2014). Whatever the resolution, the empirical

pattern is clear: the productivity gains promised by automation have not, to date, shown up uniformly in the aggregate, and have shown up unevenly across firms and across workers. Those workers whose tasks are most exposed to automation experience real wage pressure even where the aggregate productivity numbers are flat.

The second paradox is that the labor market in many advanced economies has, in recent years, displayed simultaneous signals of distress and of tightness — high job vacancy rates beside high underemployment, persistent labor shortages in some sectors beside structural unemployment in others. The standard explanation invokes mismatch: the jobs that exist do not match the people available; the people available do not have the skills the jobs require; the geographical and credentialing mobility required to close the gap is more costly than the wage premium of moving. All of these are partial truths. The deeper structural fact is that the labor market is bifurcating into jobs that pay well above subsistence and demand specialized human capital, and jobs that pay below or near subsistence and serve as residual employment for people displaced from middle-skill careers (Piketty 2014). The middle of the historical labor distribution is hollowing.

WHAT IS AT STAKE

If the diagnosis stops at “people will need new jobs”, it has missed the magnitude of the change. The capacities of intelligent systems are not, on present evidence, leveling off. The sectors hitherto considered safe — medicine, law, engineering, education, scientific research itself — are being progressively augmented and, in their routine parts, replaced. The argument that “new jobs will appear, as they did after every previous wave” is a hope, not a forecast. The argument has held for previous waves because previous waves automated narrow capacities; the present wave automates general cognition. Its end state is not knowable with confidence. The responsible posture is to plan for an

arrangement in which a substantial fraction of the population is, durably, not required by the production process for the form of labor that production used to need.

That arrangement is not without precedent at smaller scale. Ancient societies that achieved agricultural surpluses developed leisured classes whose contribution to society was cultural, philosophical, religious, or ceremonial. Industrial societies developed the institution of formal retirement, in which older citizens are supported in dignity to perform unwaged contributions to family and community. What is unprecedented is the prospect of extending such arrangements to a substantial share of the working-age population, indefinitely, at scale, without the traditional moral discomfort that accompanied non-employment. The discomfort is itself a cultural artifact of a specific industrial period. It is not eternal. It can, with effort, be redesigned.

THREE LEGITIMATE FEARS

Any serious proposal in this domain must reckon with three fears that recur in the political conversation and that are not foolish.

The first is that universal income transfers, financed by taxation of capital and automation, will hollow the dignity of those they support. The fear is grounded in real evidence that long-term unemployment, even when adequately compensated, correlates with declines in mental health, civic participation, and family stability. Any successor settlement must therefore couple income with structured opportunity for contribution — community work, public service, education, care, cultural production — and must do so in a way that confers status, not stigma.

The second is that aggressive redistribution will undermine the productive activity that finances it. The fear is grounded in the empirical record of badly designed transfers and badly designed taxes. It is not grounded in the record of

well-designed ones. The Nordic economies, Germany, the Netherlands, and several others maintain substantial transfers without measurable harm to long-run productivity. The constraint is design, not direction.

The third is that intelligent systems will not, in the end, displace as much labor as the alarmists predict, and that policy designed for the alarmist case will produce dependency where independence would have been possible. The fear is reasonable insofar as forecasts in this domain have a long history of error. The prudent response is not to wait for certainty, but to design an arrangement that adjusts gracefully to a range of automation outcomes, including the optimistic ones.

THE SHAPE OF THE ANSWER

The architecture proposed in Part Four does not promise the restoration of full employment in the twentieth-century sense. It promises something different: an unconditional floor, a contributive ladder, and a recognized portfolio of activities — paid, unpaid, voluntary, civic, creative — that together constitute a life of acknowledged worth in a society that no longer requires every adult to spend forty hours a week producing marketable output. The instruments include a planetary basic income financed by taxation of automated production and natural resource flows; an open-access continuous learning entitlement; a recognized status for unpaid care, ecological stewardship, civic service, and cultural production; and a refactored taxation of capital that distinguishes productive investment from rent extraction.

The full design is for later. The diagnostic point is the one the chapter has now made. The institution of work, as the industrial democracies inherited it, was a particular configuration of three things — productive contribution, identity, and income — bound together by the structure of the firm and the regulatory framework around it. The configuration is unbinding. The income can, with effort, be replaced. The identity and the framework of contribution

have to be redesigned, deliberately, before they are reconstituted by accident, in forms — gig economies, attention markets, parasocial labor — that few people would have chosen had the choice been made openly.



The diagnosis continues in Chapter Five, which examines the failure of law: why national legal systems built for national populations cannot govern the planetary phenomena that now dominate every meaningful policy question.

CHAPTER FIVE

THE FAILURE OF LAW



LAW, IN THE SENSE THE MODERN ERA INHERITED IT, IS A SOVEREIGN instrument: a body of rules issued by a political authority, applied to a population located within a defined territory, enforced by officers who answer to that authority, interpreted by courts whose jurisdiction is fixed by the same authority. The whole apparatus presumes that the activity it regulates fits inside the territory in which the authority operates. For most of human history, that presumption was approximately true. A market was local, a contract was performed locally, a fraud was committed locally, a violence was inflicted locally, a corporation was a local body, a bank was a local institution. Law, like the activity it governed, was a national thing because everything it cared about was a national thing.

That presumption is no longer reliable, and the consequence is the slow conversion of large categories of consequential human activity into a domain that is, in operational terms, ungoverned. The activity has not vanished into anarchy. It has migrated into the seams between jurisdictions, where the rules of two or three or twelve sovereigns simultaneously apply, contradict, and fail to enforce. To call this domain “international law” is to use a term whose principal meaning, in 2026, is the absence of effective regulation. To call it “global governance” is to invoke a structure that does not, in any robust sense, exist.

THE CATEGORIES OF ESCAPE

The escape from national law is most visible in five domains, and the diagnosis is best made through them in turn.

The first is *capital*. A multinational firm is not, in any meaningful operational sense, a single firm. It is a portfolio of legal entities, incorporated in jurisdictions chosen for tax, regulatory, and disclosure characteristics; tied together by inter-company contracts that allocate cost and revenue across borders; and managed by a leadership that has no special obligation to the population of any specific jurisdiction. The legal techniques are well documented. They include transfer pricing of internal services and intellectual-property licenses; the use of holding companies in low-tax jurisdictions; the routing of royalties through tax-haven subsidiaries; the strategic allocation of debt to high-tax jurisdictions to deduct interest from taxable income; and, increasingly, the use of jurisdictions whose principal export is corporate secrecy (Saez and Zucman 2019; International Consortium of Investigative Journalists 2021; Tax Justice Network 2023). The result is that a substantial fraction of the largest firms pay corporate tax at effective rates in low single digits, while the citizens of the countries in which they sell pay much higher rates on labor income that is, in all practical senses, captive.

The second is *data*. Personal data is generated by individuals located in one country, processed on servers located in a second, controlled by a corporation incorporated in a third, governed by a contract whose enforcement clause names a court in a fourth. When a privacy violation occurs, the question of which law applies is not merely contested; it is structurally underdetermined. The European Union's General Data Protection Regulation is the most ambitious attempt yet to project national-style law onto a planetary substrate, and even it operates by the indirect mechanism of penalizing entities that wish to do business with European users (European Parliament and Council 2016). The reach is real, but the framework is patchwork; in jurisdictions with weaker protection

or active state surveillance, the same data flows under different rules, and the individual has no instrument to know which apply (Zuboff 2019).

The third is *information warfare and cybercrime*. A criminal who interrupts the electricity, water, or hospital network of a country is not necessarily a citizen of that country, does not necessarily reside there, and may operate from a jurisdiction whose authorities are unwilling, unable, or politically disinclined to cooperate. Treaties governing extradition exist; they apply in narrow circumstances and frequently fail in the most consequential cases. State-sponsored cyber operations occupy a category that international law does not yet recognize as a discrete instrument of war. There is no equivalent of the Geneva Conventions for code (Perloth 2021). The result is that critical infrastructure is, in 2026, attacked routinely, with attribution typically delayed by months, and prosecution typically impossible.

The fourth is *environment*. The atmosphere, the oceans, the migratory species, and the climate system are common-pool resources of the planet. Their use is regulated, where at all, by the United Nations Framework Convention on Climate Change and its successor instruments, most notably the Paris Agreement (United Nations Framework Convention on Climate Change 2015). The architecture of the Agreement is itself an admission of the problem: targets are nationally determined and largely unenforceable, monitoring is voluntary, sanctions are political. The atmosphere is the most consequential commons humanity shares; the legal instrument by which it is protected is, by design, the weakest treaty regime in existence. Compliance has been correspondingly partial (IPCC 2023, 2022).

The fifth is *arms*. The Treaty on the Non-Proliferation of Nuclear Weapons depends on the voluntary discipline of states that have always retained the technical capacity to break it (United Nations 1968). Conventions on chemical and biological weapons are stronger in form than in enforcement. The proliferation of armed unmanned systems, of long-range precision strike, of autonomous

targeting, and of cyber-enabled kinetic operations has occurred largely outside any treaty framework calibrated to the technologies involved (SIPRI 2024). The principal instrument of restraint has been the threat of mutual retaliation, which is a form of equilibrium rather than a form of law.

WHY NATIONAL LAW CANNOT, ON ITS OWN, FIX THIS

The temptation, in each of the five domains, is to argue that the problem is one of insufficient national resolve, and that a sufficiently determined national government could close each gap by extending its own law extraterritorially. The argument has been tried, and its limits are now visible.

Extraterritorial enforcement works only where the enforcing state has economic or military leverage over the target, and only in proportion to that leverage. The United States has, by the size of its market and the role of its currency, succeeded in extending some of its laws — sanctions, securities, anti-bribery, certain aspects of tax — to actors with no other connection to its territory. The European Union has done the same with privacy and competition law. No other jurisdiction has comparable reach, and the reach of the two that do is itself contested by every other state and by a growing fraction of the firms it touches. Extraterritorial law, when it works, works as power; it does not work as law in the proper sense, which requires consent of the governed.

A second temptation is to argue that the gap will be closed, eventually, by treaty. Treaties take decades to negotiate, frequently exclude the actors most relevant to the regulated activity, are interpreted by domestic courts with diverging methods, and are abandoned by signatories whose governments change. The history of the past sixty years includes several attempts at planetary regulation by treaty — for the seas, for the atmosphere, for the seabed, for outer space, for chemical weapons — with results that range from partial to ornamental. Treaties remain useful instruments. They are, by

themselves, structurally insufficient to the speed and scope of the activity they would have to govern.

WHAT LAW ACTUALLY REQUIRES TO FUNCTION

It is worth restating, plainly, what conditions any system of law must satisfy to perform its work. The conditions are old; they are not, in 2026, controversial.

The first is *jurisdiction*: there must exist an authority whose writ runs over the activity to be governed. The second is *publication*: the rules must be knowable in advance by those they bind. The third is *generality*: the rules must apply uniformly to similarly situated subjects. The fourth is *impartiality*: the body that adjudicates must be capable of ruling against any party, including the authority that issued the rules. The fifth is *enforcement*: there must exist an instrument by which the rule, once decided, is given effect against parties who would prefer to ignore it. The sixth is *review*: there must be procedures by which a wrongful decision can be corrected. The seventh, derivable from the others, is *legitimacy*: the population subject to the law must, on the whole, accept its authority — not because they like every rule, but because the procedure by which the rules are made and enforced is one in which they have, in principle, voice (Rawls 1971a; Habermas 1989b).

The international order does not satisfy these conditions in the domains that now matter most. There is no body whose jurisdiction is universally recognized over capital flows, data flows, the global atmosphere, planetary cybersecurity, or the use of advanced weapons. There are bodies that approximate fragments of these things — the World Trade Organization, the International Court of Justice, the International Criminal Court, the Financial Action Task Force, the International Atomic Energy Agency, the conferences of the parties to the various climate and chemical weapons conventions (United Nations Diplomatic Conference 1998) — and each operates within a constrained mandate, with constrained membership, under constrained enforcement. None

has the standing of a court of universal jurisdiction. None has the budget of a major national administration. None can compel.

THE LEGAL VOID AS A HABITAT

What lives in the void? The honest answer is that organized crime lives there; that the most aggressive forms of capital live there; that the most coercive surveillance practices live there; that the cyberweapons industry lives there; that human trafficking, illicit finance, sanctions evasion, and the more sophisticated forms of disinformation operations live there. The void is not empty. It is a productive habitat for activities that find national law inconvenient.

The political consequence is severe. A citizen who watches the legitimate authorities of his country fail, year after year, to address activities that visibly damage him — fraud he cannot prosecute, pollution he cannot prevent, capital flight he cannot stop, foreign cyber attacks he cannot punish — will, in time, conclude that those authorities are either complicit or incapable. Both conclusions, sustained, undermine the consent on which legitimate government rests. The crisis of legitimacy that animates much of contemporary politics is not unconnected to the legal void; it is, in substantial part, downstream of it.

WHAT A SUCCESSOR SYSTEM MUST OFFER

A serious successor system must accept that some domains of human activity have escaped the national level and will not, on any realistic horizon, be coaxed back. For those domains, it must provide a planetary tier of law: a tier that is genuinely public in its standing, genuinely universal in its membership, genuinely reviewable in its procedures, and — most controversially — genuinely capable of enforcement. The instruments by which such a tier can be constructed without becoming the very tyranny that national sovereignty was

originally devised to prevent are the principal subject of Parts Three and Four. The architecture proposed there relies on three commitments: subsidiarity, so that the planetary tier is restricted to what truly cannot be done at lower levels; transparency, so that the tier itself can always be inspected; and reversibility, so that decisions made at the planetary level can be revisited as circumstance changes.

The diagnostic point of the present chapter is narrow and sufficient. The legal architecture inherited from the industrial era cannot govern the activity that now exists. The choice is not between national law and global law. The choice is between an unfilled void, in which the activities that escape national law operate by their own logic, and a constructed planetary tier, in which those activities are brought into the discipline of public rules. The first choice is the present trajectory. Its outcomes are visible. The second is the work the next chapters undertake.



The diagnosis closes in Chapter Six, which examines the failure of foresight: why a civilization with the technological reach of an adult species makes its decisions on the planning horizon of an adolescent.

CHAPTER SIX

THE FAILURE OF FORESIGHT



A CIVILIZATION CAN BE DIAGNOSED, WITH SOME ACCURACY, BY THE TIME horizon over which its decisions are made. Hunter-gatherer bands operated on a horizon of seasons. Agrarian societies operated on a horizon of harvests, which is to say of years. Empires that built lasting infrastructure operated on a horizon of generations; the Roman aqueducts, the Chinese canals, the medieval cathedrals took longer to build than any single elite could expect to oversee, and the institutions that built them accepted this as a cost of doing serious work. Modern industrial democracies, despite the unprecedented technological reach of the populations they govern, operate on a horizon of months: the electoral cycle for politicians, the fiscal quarter for firms, the news cycle for the public mind. The mismatch between the scale of the consequences our actions produce and the scale of the horizon over which we deliberate is, by any measure, the most distinctive structural pathology of the late industrial age.

The mismatch is not subtle. The greenhouse gas emissions of any given year alter atmospheric chemistry on a timescale measured in centuries, with consequences that compound for thousands of years (IPCC 2023). The ecological loads imposed on soils, aquifers, fisheries, and forests are repaid, where at all, on timescales measured in decades. Pension obligations are written today against demographic projections that resolve thirty and forty years hence (International Monetary Fund 2024). The half-life of long-lived radioactive waste from civilian nuclear programs is measured in tens of thousands of years, far longer than any continuous political institution that has ever existed. The cumulative

cost of the carbon emitted in any one year of advanced-economy growth, calculated honestly against the climate damage it imposes, exceeds — by most estimates — the income gains of that growth (Stern 2006). The decisions are made on quarterly accounting; their consequences keep accruing on geological accounting. There is no political or economic instrument, in routine use, that translates between the two.

WHY THE PRESENT IS OVER-REPRESENTED

Several mechanisms produce the overweighting of the present in ordinary politics. They are worth listing because each must be addressed in any successor design.

The first is the *electoral cycle*. A government chosen on a four- or five-year horizon optimizes, almost without exception, for outcomes whose visibility falls within that horizon. Long-horizon projects — infrastructure, education, scientific research, ecological restoration — yield benefits whose attribution is unclear by the time they ripen, and yield costs that are entirely visible in the present. A rational politician concerned with reelection therefore underinvests in such projects relative to what the population's own intertemporal preference would, on reflection, support. The phenomenon is not corruption; it is the structurally rational response to the institutional design (Fukuyama 2014).

The second is the *fiscal-quarter cycle*. The publicly traded firm, on which the contemporary economy depends for its allocation of capital, is required to report quarterly to its shareholders. Compensation of senior management is keyed, with rare exceptions, to indices whose horizon is a fiscal year or two. Investments whose payoff is more than five years out are systematically penalized in the share price, and managers who pursue them are systematically displaced. The result is the familiar pattern in which firms with vast retained earnings pursue stock buybacks rather than capital investment, even where the underlying productive opportunity is real (Mazzucato 2018).

The third is the *news cycle*. Public attention is allocated by media organizations whose competitive position depends on the freshness, salience, and emotional charge of the information they distribute. Long-horizon developments, by their nature, are slow, undramatic, and distributed across many actors; they cannot compete for attention against acute events. The public therefore receives a continuous stream of immediate stimulus interleaved with occasional, and quickly forgotten, references to the long horizon. The cumulative effect is that the long horizon is structurally underrepresented in the conversation from which political will is formed (Putnam 2000).

The fourth is the *economic discount rate*. Standard economic appraisal applies a positive discount rate to future costs and benefits, on the rationale that consumption later is worth less than consumption now. The principle is reasonable for short horizons; it becomes morally and practically dubious when applied to consequences that fall on populations not yet born and that did not consent to the discount. The choice of discount rate is itself a political decision dressed as a technical one, and the rates routinely chosen — three percent, five percent, seven percent — render any cost more than fifty years out essentially invisible to the calculation (Stern 2006).

Each of these mechanisms is independent. Each is reinforcing. Together they produce a political economy in which the long horizon is, with rare exceptions, neglected by every actor whose incentives the system has bothered to consider, and represented only by those whose incentives the system has not yet learned to value: scientists, archivists, teachers, farmers planting trees they will not live to harvest, parents thinking about their grandchildren.

THE STRUCTURAL NEGLECT OF CATASTROPHE

The most serious consequence of foreshortened deliberation is the structural neglect of low-probability, high-consequence events — the events that,

integrated over time, dominate the expected loss profile of a civilization. The same logic that underweights the slow accumulation of climate damage underweights the small probability of pandemic, of major financial collapse, of nuclear use, of large-scale infrastructure failure, of misalignment in advanced artificial systems. Each of these events, in any given year, is unlikely. Each, integrated over a half-century, is well within the range of plausible outcomes. The expected damage from any one of them, properly accounted, exceeds the cost of the preparatory measures by orders of magnitude. The preparatory measures are nevertheless underfunded in nearly every advanced country, because their funding requires payment now against a benefit that, in any given year, will probably not be needed (World Economic Forum 2024).

The pandemic of the early 2020s is the clearest recent illustration. The capacity of national health systems to absorb a respiratory pandemic had been studied for decades, with broadly consistent conclusions that capacity was insufficient. Stockpiles of personal protective equipment, the contingency manufacturing of vaccines, the design of surveillance networks for emerging pathogens, and the legal frameworks for rapid public-health response had each been the subject of serious official report. The reports were filed; the recommendations were not, in most countries, implemented at the necessary scale. The cost of implementation would have been a small fraction of the cost of the pandemic when it arrived. The institutions that issued the warnings had no mechanism to compel action against a contingent harm. The institutions that allocated budgets had no mechanism to weigh contingent harm against present priorities.

The same pattern applies, with adjustment of detail, to climate, to financial stability, to the security of large concentrations of fissile material, and to the governance of advanced computational systems. Each is a low-frequency, high-consequence risk; each is structurally underweighted by institutions whose

incentive structures cannot see past the electoral cycle (Bostrom 2014). The pattern is not anyone's fault, in the narrow sense. It is the architecture's fault.

WHAT FORESIGHT INSTITUTIONS LOOK LIKE, AND WHY WE LACK THEM

It is sometimes assumed that long-horizon thinking requires only the will to engage in it, and that the absence of such thinking is therefore a failure of character. This is mistaken. Long-horizon thinking, where it has functioned in human history, has functioned because specific institutions were designed to perform it, with budgets, mandates, and standing equal to the institutions of short-horizon decision. The medieval cathedral chapters, the royal forestry commissions of the early modern European states, the standing scientific academies of the eighteenth and nineteenth centuries, the public health corps of the late nineteenth, and the strategic planning bodies of the post-war reconstruction governments — these were institutions deliberately constructed to operate on horizons their political masters could not match. Most of them have been weakened, defunded, politicized, or dissolved during the past four decades, in the name of efficiency, agility, or fiscal restraint. The cumulative effect is a public sector that has lost, sector by sector, the institutional memory and authority by which long-horizon work is done (Mazzucato 2013).

A successor system must rebuild this lost capacity. The instruments are well known and have been proposed, in various forms, by writers from Burke to Aldo Leopold to Jonas to Elinor Ostrom: standing intergenerational councils with formal voice in legislation; office of the future commissioner with subpoena power over agencies' long-horizon policies; constitutional clauses recognizing duties to populations not yet born; budgetary categories ring-fenced from short-term political reallocation; courts whose mandate explicitly includes the standing of future persons. Several of these have been implemented at small scale — Wales has a Future Generations Commissioner;

Finland has a Committee for the Future in its parliament; the German Federal Constitutional Court has read intergenerational obligation into its constitutional doctrine on climate. None has been implemented at planetary scale, with planetary mandate, on the questions that demand it. That is the architectural gap.

THE DEEPEST VERSION OF THE FAILURE

Beneath the institutional analysis lies a philosophical fact that is rarely named. The political and economic frameworks of the modern age were constructed at a moment when human intervention in nature was small enough that the planet could be modeled, for most decision purposes, as an unchanging backdrop against which finite human projects played out. That model is now false. Human action has become a planetary geological force. The atmospheric chemistry, the biological diversity, the hydrological cycles, the carbon and nitrogen flows, and even the biogeochemistry of the oceans now bear the visible signature of human industry. Geologists discuss whether to formalize this period as a distinct epoch under the name *Anthropocene*; the political institutions of the period have not yet absorbed what such a designation would mean (IPCC 2023; Raworth 2017).

What it means is that human collective action has acquired effects on a timescale on which no human institution has ever before been required to operate. To exercise a power without an institution adequate to it is to exercise it irresponsibly. We are, in this sense, the first species to acquire planetary instruments without acquiring planetary institutions. The asymmetry will not be sustainable. Either the institutions are built, or the instruments are misused, or, in time, both.

CLOSING THE DIAGNOSIS

The five chapters of Part One have set out the diagnosis. The institutions of the modern age are structurally illegible to their citizens; they distribute the product of an unprecedentedly capable economy in ways that no informed population would, on reflection, endorse; they have lost the link between work and dignity on which two centuries of social peace depended; they cannot regulate the activity that now most consequentially affects them, because that activity has migrated beyond the reach of any single state; and they make their decisions on horizons grossly inadequate to the consequences of the decisions they make. None of this is the fault of the inheritors. It is the cost of inheriting.

Diagnosis without proposal is, however, a counsel of despair. The remainder of this book is the proposal. Part Two lays the principles. Part Three sets the constitutional text. Part Four describes the architecture. Part Five describes the transition. Part Six asks what becomes possible after.



Part One closes here. Part Two opens with the seven principles on which any successor arrangement worth the name must be constructed.

PART TWO

THE SEVEN PRINCIPLES



Before architecture, principle. The seven that follow are the constraints any successor arrangement must satisfy if it is to deserve the name of a civilizational design rather than a clever variant of the present trouble.

CHAPTER SEVEN

PRINCIPLE ONE — LEGIBILITY



THE FIRST PRINCIPLE OF ANY SUCCESSOR ARRANGEMENT IS THAT EVERY operation of public authority must be legible, in real time, to any literate citizen who chooses to inspect it. This is the principle from which all others follow. A society that cannot be read cannot be trusted; a society that cannot be trusted cannot consent; a society that cannot consent is held together by something other than its own will, which is to say it is held together by force, by inertia, or by the tactical persuasion of the moment. Legibility is not a luxury of mature democracy. It is the precondition for the legitimate exercise of any public power that wishes to call itself such.

The principle is older than the modern state. It descends, in different idioms, from the publication of the Twelve Tables in the Roman Republic, from the medieval insistence that no person could be bound by a law not first promulgated, from the Enlightenment demand that the laws of finance be opened to the publication of the budget, and from the twentieth-century institutionalization of the freedom-of-information statute. Each generation has extended the surface area over which the principle is enforced, and each extension has been resisted by the same arguments — security, complexity, privacy, expense — and has been justified, after the fact, by the same outcomes — improved governance, deeper consent, less corruption, fewer foolish wars. The history is unanimous. Where legibility advances, legitimacy follows. Where it is resisted, legitimacy decays.

WHAT LEGIBILITY MEANS IN 2026

To say that authority must be legible is not to say that every piece of paper produced by every official must be published. The principle has structure, and the structure must be specified or it dissolves into rhetoric.

Legibility, in the sense intended here, has four operative components. First, every act of public authority that produces a binding consequence — a regulation, an appropriation, an appointment, a contract award, an enforcement action, a permit, a court ruling, a treaty signature — must produce a corresponding entry on a public ledger maintained for that purpose. Second, the entry must be timestamped, attributed, and cryptographically signed by the authority responsible, in such a way that subsequent alteration without re-signature is detectable. Third, the entry must be queryable through standard interfaces, so that any party — journalist, auditor, civic organization, automated agent — can reconstruct the activity of the authority across any chosen dimension: by recipient, by region, by program, by time. Fourth, the meaning of each entry must be supplied alongside it: not the document alone, but the schema by which the document is to be read.

The four components together describe what we will call, in this book, the *public ledger*. The technical means of constructing such a ledger are well understood and are described in Chapter Sixteen. The novelty of the proposal is not technical. It is institutional. Public authorities have, with rare exception, never been required to be legible at this resolution. Every such authority will resist the requirement. The resistance will be sincere. It will also, as the historical record shows, be unjustified at scale.

WHAT LEGIBILITY IS NOT

Three confusions recur in the public discussion of legibility, and the principle is best stated by ruling them out.

Legibility is not surveillance. The object of legibility is the operation of *authority*, not the conduct of citizens. The state's actions are visible; the citizen's data is not. The asymmetry is deliberate, and is the precise inversion of the asymmetry that prevails in the surveillance societies of the early twenty-first century, in which citizens are extensively visible to authorities and authorities are extensively opaque to citizens (Zuboff 2019). A society that has confused the direction of the asymmetry has not lost legibility — it has lost privacy, which is the citizen's mirror right and is the subject of Chapter Nineteen. The two principles are not in conflict; they are complementary. Authority is exposed; the person is protected.

Legibility is not absolute. There exist domains of state action — narrow but real — in which immediate publication would defeat the lawful purpose of the action: an active criminal investigation, a genuine national-security operation, the negotiation of a treaty whose disclosure would foreclose the negotiation, the protection of a witness whose identification would endanger the witness's life. In such domains, the principle of legibility takes a delayed form. The act is recorded on the ledger contemporaneously, encrypted under keys held by an oversight authority; the encrypted record's existence is public, with metadata sufficient to identify its category and its expected disclosure date; and the act is opened for full inspection within a defined period, typically not exceeding seven years, with extensions only on the published certification of a designated review body. The point is not that secrets do not exist. The point is that *permanent* secrets do not exist, and that the public has a contemporaneous right to know that an act has occurred even if it cannot, for a defined period, know the act's content.

Legibility is not the same as participation. A legibly governed society is not necessarily a directly governed one. The principle says nothing, by itself, about whether decisions are made by representatives, by referenda, by deliberative assemblies, by experts, or by some combination. It says only that the decisions,

however made, must be open to inspection. This is important because the principle of legibility can be implemented across a wide range of governance forms, and is therefore compatible with the constitutional pluralism that the federation proposed in Part Three explicitly preserves.

WHY THE PRINCIPLE IS HARD

The principle is conceptually simple and operationally difficult, for reasons worth naming honestly.

The first difficulty is the loss of *discretion*. A public official whose actions are recorded on a public ledger forfeits the latitude to grant a quiet exception, to soften a judgment with informal accommodation, to negotiate by ambiguity. Some of this latitude has been corrupt. Most of it has been the lubricant by which large bureaucracies handle hard cases. Removing it requires building, in its place, a procedural mechanism by which exceptions can be granted formally, on the record, with explicit justification. This is not impossible. It is unfamiliar.

The second difficulty is the loss of *tactical position*. Authorities who can be observed continuously cannot bluff. A diplomat cannot threaten an option her government has not authorized; a regulator cannot signal an enforcement direction she has not committed to; a central bank cannot, in some circumstances, talk the markets in the directions it wishes them to take. The loss is real. It is also, if one looks honestly at the record, often a loss of dignity rather than a loss of capacity. Authorities have, in the era of opacity, frequently used tactical position to impose costs on their own publics, and the publics — when they later learn what was done in their name — have, with few exceptions, regretted having paid for the bluff.

The third difficulty is the loss of *convenience*. A government that publishes its operations in real time must have processes capable of producing the published record without imposing intolerable burden on the people doing the

work. This is, in the end, an engineering problem, and it is solvable; but the engineering problem has not, in most jurisdictions, been undertaken at scale. Public-sector information infrastructures remain, in 2026, decades behind the private sector for most categories of work. Closing the gap requires investment, and investment requires that the principle first be accepted.

THE HISTORICAL EVIDENCE

It is sometimes argued that the principle, however attractive in theory, has never been tested at the scale demanded by a modern administration. The argument understates the historical record. Several jurisdictions have implemented partial legibility regimes whose results are instructive.

Estonia has, since the early 2000s, operated a digital state in which most government services are delivered through a single citizen interface, with detailed audit logs accessible to the citizen herself. The model is partial in scope but deep in implementation, and it has produced public-confidence levels in digital government that exceed those of most jurisdictions of comparable size. The early Open Government Partnership, joined by seventy-five countries between 2011 and 2024, committed signatories to publish progressively granular data on procurement, budgets, and beneficial ownership; compliance has been uneven, but the countries that have implemented their commitments most fully have, in measured terms, improved on perceived corruption (Transparency International 2024). The European Union's Open Data Directive obliges member states to make administrative data available in machine-readable form. In the United States, the Federal Funding Accountability and Transparency Act has, since 2006, mandated the publication of federal expenditure data through a public portal; the portal's quality has been criticized, but the principle of citizen access to federal expenditure has, since its introduction, been politically irreversible.

None of these regimes implements the full principle of legibility as stated above. Each has demonstrated that the technical and institutional obstacles to greater legibility are smaller than the political obstacles, and that the political obstacles erode steadily once the regime is in place.

HOW THE PRINCIPLE DISCIPLINES LATER DESIGN

It is worth, at the close of the chapter, noting what the principle of legibility forbids in the design of subsequent institutions.

It forbids any decision rule whose output cannot be reconstructed by an external observer with access to the rule and the input. This rules out, in particular, the use of opaque machine learning systems for binding administrative decisions affecting individuals — a constraint that the European Union has begun to codify in its Artificial Intelligence Act, with respect to high-risk applications (European Parliament and Council 2024). It rules out the delegation of consequential public functions to private intermediaries whose internal decision processes are protected as trade secret. It rules out the use of cryptographic instruments in public administration that prevent later inspection by oversight bodies, however convenient those instruments may be for short-term execution. It rules out structures of plausible deniability — committees without minutes, decisions without authors, contracts without counterparties identifiable by the public.

It does not forbid expertise, technical complexity, or specialist procedure. A regulation written by hydrologists for the management of an aquifer is no less legible because most citizens cannot evaluate it; the legibility test is satisfied if any sufficiently trained party can, on inspecting the public record, reconstruct the reasoning. The point is not that every citizen will personally inspect every act. The point is that any citizen, or any party acting on the citizen's behalf, can inspect any act, with no procedural obstacle that a determined and patient inquiry cannot overcome.

CLOSING

The principle of legibility, properly stated, is at once unremarkable and revolutionary. Unremarkable, because the obligation of public authority to be visible to the public has been the formal commitment of every legitimate political tradition since the Roman Republic. Revolutionary, because the technical conditions for honoring the commitment at planetary scale, in real time, against authorities of unprecedented size and reach, have only recently come into existence, and the institutions that wield public authority have not yet been compelled to use them. The remainder of this book is, in one of its readings, a sustained argument that they must.



Chapter Eight states the second principle: subsidiarity, the rule that decisions belong at the lowest level capable of making them.

CHAPTER EIGHT

PRINCIPLE TWO — SUBSIDIARITY



SUBSIDIARITY IS THE RULE THAT A DECISION BELONGS AT THE LOWEST level capable of making it well, and that higher levels of authority intervene only when the lower level has demonstrably failed or when the matter to be decided exceeds the lower level's competence. The principle is older than its name. It runs through the constitutional traditions of federal states, through Catholic social teaching, through the structuring documents of the European Union, and, in less formalized versions, through every political community that has ever delegated power upward without dissolving the delegating body. It is, in the present argument, the indispensable counterweight to the first principle. Legibility makes authority visible; subsidiarity disperses it. Without dispersion, legibility would only document a tyranny.

The principle has two motivations. The first is epistemic. Decisions are best made by those nearest the facts that bear on them, because those nearest the facts have access to information — local knowledge, social context, particular circumstance — that does not survive transmission upward. This was the central insight of Hayek's argument against centralized planning (Hayek 1944), and it survives the political quarrels that have surrounded the argument since. It survives, in particular, the failures of unrestrained markets to which Hayek himself paid insufficient attention; nothing in the principle of subsidiarity requires the further claim that lower levels be left wholly to themselves. The principle says only that the burden of proof, in any given matter, falls on the proposal to elevate.

The second motivation is moral. Authority delegated upward without need is authority alienated from those it governs. A community whose decisions are made by remote bodies, however technically competent, develops over time the habits of dependence and the resentments of the dispossessed. The political pathologies of the present century, in nearly every advanced economy, include some version of the complaint that decisions are made by people the public does not know, in places the public will never visit, on grounds the public cannot inspect. The complaint is overstated by the populists who deploy it. It is not, in its core, mistaken.

LEVELS AND THEIR PROPER SCOPE

The successor architecture proposed in this book contemplates five levels of public authority, each with a defined scope and a defined relation to the others. They are the *local commune*, the *regional canton*, the *state*, the *continental confederation*, and the *planetary tier*. None of these names is sacred; the level designations are functional. What matters is the assignment of authority among them.

The local commune — typically the city, the village, the urban district, the rural parish — is the level at which most ordinary public services are administered. Land use, primary education, primary health, local infrastructure, neighborhood policing, social services, cultural programming, local taxation: these are the routine business of the commune, and the principle of subsidiarity holds that they should not be lifted from it without specific cause. Communes are also, of necessity, the locus of greatest variation. What is sound policy in a commune of a thousand farmers is unsound policy in a commune of two hundred thousand urban workers. The framework allows, and indeed requires, that variation; uniformity at this level is a defect, not a virtue.

The regional canton is the level at which infrastructure of regional scale is coordinated — water systems that span multiple communes, road networks,

regional hospitals, secondary education, regional environmental protection — and at which judicial appeals from the commune level are heard. The canton is, in many federal systems, the natural unit of cultural and linguistic identity, and it is at this level that questions of cultural autonomy, official language, and historical heritage are most coherently decided.

The state level retains, in this architecture, considerable scope. National defense, the issuance of currency where states retain currency sovereignty, citizenship and immigration policy in its primary form, taxation of national activity, the architecture of social insurance, the constitution of higher education, and the setting of professional and technical standards remain state matters. The principle of subsidiarity does not abolish the state; it constrains the conditions under which the state may pre-empt the regions and the communes within it.

The continental confederation — examples in the present world include the European Union, the African Union in its more developed sectors, certain organs of ASEAN, and the regional integrations of the Americas — is the level at which a group of states coordinates on matters whose scale exceeds any one of them but which do not require planetary deliberation. Single-market regulation, cross-border infrastructure, the free movement of persons and goods, common environmental standards over shared ecosystems, and harmonized commercial and contract law are the proper scope of the confederation.

The planetary tier, finally, is reserved for matters whose scope is the species itself. The atmospheric and oceanic commons, the regulation of weapons whose use threatens humanity at large, the governance of advanced computational systems above defined thresholds of capability, the prevention and management of pandemics, the management of orbital and lunar resources, the basic minimum of human dignity from which no jurisdiction may derogate: these and a small list of comparable matters constitute the planetary tier. The

list is, by design, short. The default in the architecture is that the planetary tier is silent; it speaks only on the matters explicitly assigned to it.

THE CONDITIONS FOR ELEVATION

Subsidiarity, properly understood, is not a bias against higher levels of authority. It is a discipline on the conditions under which authority is moved up. Three conditions are necessary, in the architecture proposed here, for any matter to be elevated from one level to the next.

The first is *scale incompetence*. The lower level must be unable, by virtue of the matter's scope, to handle it. A commune cannot, by itself, regulate the atmospheric chemistry of the planet. A state cannot, by itself, prevent capital from leaving its jurisdiction. A continental confederation cannot, by itself, manage a species-wide pandemic. In each case, the elevation is justified by the structural limit of the lower level.

The second is *spillover*. The action of the lower level must produce consequences that fall on populations beyond its boundary, in ways that the lower level cannot, by itself, internalize. A commune that pollutes a river imposes costs downstream; a state that emits greenhouse gases imposes costs across the planet. The higher level's authority extends only as far as the spillover requires.

The third is *floor protection*. The matter must touch a minimum of human dignity that is held to be inviolable across the federation, and that the lower level has either failed to protect or actively threatens. The protection of the basic rights of persons — life, freedom from torture, basic subsistence, freedom of conscience, equality before the law — is the most uncontroversial example, and is the basis on which the planetary tier retains a residual jurisdiction over individual rights even within sovereign states.

Where none of the three conditions is satisfied, the matter remains at the lower level. Where one is partially satisfied, the matter is shared by formal cooperation, not by elevation. Only where the conditions are clearly met does authority move upward. This is what subsidiarity, as a constitutional principle, actually requires.

THE COMPLEMENTARY PRINCIPLE OF DEVOLUTION

Subsidiarity has a downward sibling that is rarely articulated and that the present architecture explicitly endorses. Where a matter has been elevated to a higher level on the basis of scale, spillover, or floor protection, and the conditions that justified the elevation cease to apply, the matter must be returned to the lower level. Authority once moved up does not, on this principle, ratchet permanently. The federation's higher organs are constructed with sunset clauses on their substantive jurisdictions, requiring periodic renewal by the constituents whose authority was delegated. The renewal is the moment at which the conditions of elevation are reviewed and the delegation either continued or withdrawn.

This is a stronger commitment than is found in any major existing federation. It is intentional. The historical experience of federations — including those whose framers carefully distinguished delegated from reserved powers — is that delegations grow. The Tenth Amendment of the United States Constitution, the Article 5 Treaty of the European Union, the federal allocation of powers in the German Basic Law: each was designed to protect the lower level, and each has been, in practice, gradually superseded as the higher level has accumulated competence by interpretation, treaty, or political momentum (Fukuyama 2014). A federation honest about subsidiarity must build the resistance to this drift into its constitution, not as an exhortation, but as a procedural requirement.

WHAT SUBSIDIARITY IS NOT

Three confusions deserve correction.

Subsidiarity is not localism for its own sake. The principle does not hold that local decisions are better than non-local ones; it holds that decisions belong at the lowest level *capable* of making them well. Where the local level is incapable, by reason of scale or competence, the principle requires elevation, not deference. A commune that cannot, by reason of size, support a tertiary hospital is not entitled by subsidiarity to refuse the canton's role in providing one.

Subsidiarity is not a defense of local oppression. The principle of floor protection trumps local autonomy where minimum rights are concerned. A commune that practices apartheid, that denies education to women, that tolerates trafficking in persons, or that withdraws from its members the basic conditions of life forfeits the protection of subsidiarity. The federation's planetary tier retains, in such cases, a residual jurisdiction. The retention is sparingly used, but it is not, in the proposed architecture, optional.

Subsidiarity is not the abolition of redistribution across levels. A commune in a poor region of a poor canton may be incapable, with its own resources, of meeting the floor of dignity. Other levels of the federation finance the gap, by formula, and the financing is itself one of the matters reserved to the higher levels precisely because it cannot be done at the lower. The principle disperses authority; it does not deny solidarity. The two are easily confused in casual political talk and must be kept separate in any serious design.

THE POLITICAL LOGIC

The political logic of subsidiarity is, in the end, that of legitimacy. A population that watches consequential decisions made within reach of its own deliberation, by people it can name and meet, develops the habit of consenting;

a population that watches consequential decisions made remotely, by procedures it does not understand, develops the habit of withdrawing. The withdrawal is not, on the whole, a moral failure of the population. It is a structural response. Subsidiarity is the structural answer.

What follows from the principle, in the architectural sense, is that the federation proposed in this book is not a world government in the conventional understanding of that phrase. It is a layered confederation in which most decisions remain near where they have always been, and in which a small set of explicitly delegated competences are exercised at the planetary level, under continuous public inspection, with periodic review of the delegation. The instinct that recoils from world government is correct in its reasons and should not be assuaged by linguistic substitution. The proposal is not for a planetary state. It is for a planetary tier of carefully limited function, embedded within a constitutional architecture that disperses authority deliberately and reviews the dispersion periodically.



Chapter Nine states the third principle: reversibility — the rule that no irreversible action may be taken without a review proportionate to its consequence.

CHAPTER NINE

PRINCIPLE THREE — REVERSIBILITY



A CIVILIZATION WISE ABOUT ITS OWN FALLIBILITY DESIGNS ITS ACTIONS, where it can, to be reversible. It treats irreversibility not as an ordinary feature of decision-making but as a special case requiring special procedures. It accepts that mistakes will be made — by individuals, by institutions, by the species itself — and it places, between the moment of choice and the moment of irrevocable consequence, every reasonable instrument of pause, review, and recovery. The principle is unremarkable in private life: a prudent person does not sign the contract on the first reading, does not take the medicine without consulting the physician, does not destroy the original of the document until the copy has been verified. The principle is conspicuously absent in the public life of the late industrial age, in which decisions whose consequences extend across generations are routinely made on horizons of weeks, by procedures with no provision for revision.

The most consequential examples of irreversibility in the present arrangement are, on inspection, almost embarrassingly familiar. The species cannot, having emitted a quantity of carbon dioxide into the atmosphere, retrieve it on any horizon shorter than several centuries (IPCC 2023). It cannot, having extinguished a species of plant or animal, restore it; the rate of contemporary extinction exceeds, by several orders of magnitude, the background rate over geological time. It cannot, having released a self-propagating biological agent, recall it. It cannot, having authorized a precedent in international law, easily reverse it. It cannot, in some cases, recover the trust withdrawn from an institution that, having failed once at a critical moment, is

no longer trusted by the generation that watched it fail. Each of these examples is irreversible at the timescale on which the institutions of civilization operate. Each, accordingly, is the kind of action that the principle of reversibility takes most seriously.

THE STRUCTURE OF THE PRINCIPLE

The principle has three components. The first is a duty of *classification*: the public authority must, when contemplating any consequential action, classify it on a scale from fully reversible to fully irreversible, with the classification published and subject to challenge. The second is a duty of *proportionate review*: the depth of deliberation, the breadth of consultation, and the height of the procedural threshold for action must scale with the classification. The third is a duty of *preservation of options*: where action is taken, the path back must, where physically possible, be preserved; where return is impossible, the cost of foreclosure must be paid in advance, in the form of explicit deliberation and explicit assent.

These duties are not novel. They are, in fact, the embedded logic of well-functioning courts, of well-functioning scientific institutions, and of well-functioning engineering practice. Constitutional courts apply heightened scrutiny to laws that infringe fundamental rights, on the theory that such infringements are, if uncorrected, irreversible at the level of the individual life. Scientific institutions reserve their highest evidentiary thresholds for claims whose ratification would propagate through subsequent research. Engineering practice distinguishes systems whose failure modes are recoverable from those whose failure modes are not, and applies disproportionately rigorous discipline to the second class. The principle of reversibility is the political analogue of these established practices, applied to the public sphere.

THE THREEFOLD COST OF IRREVERSIBILITY

The principle is not free. It imposes, on any system that adopts it, three real costs.

The first cost is in *speed*. Decisions subject to heightened review take longer than decisions made by routine procedure. Where the matter is urgent, this cost is non-trivial. The principle therefore requires not the abolition of expedited procedure, but the construction of expedited procedure that nevertheless preserves the option of later correction. Emergency action is permitted; emergency action that forecloses subsequent revision is not.

The second cost is in *commitment*. Many of the goods that political action seeks to produce — investor confidence, alliance reliability, monetary stability, treaty credibility — depend on the perceived inability of the actor to renege. The principle of reversibility, applied without nuance, would dilute this perceived commitment and would, in some domains, raise the cost of the goods. The architecture proposed here therefore distinguishes between the reversibility of the *process* and the reversibility of the *output*: a treaty that has been ratified is reliable to its counterparties; the procedure by which the next treaty is ratified remains open to refinement. The same distinction applies to monetary commitments, to constitutional provisions, and to long-term contracts.

The third cost is in *moral comfort*. Some actions, once taken, cannot honorably be reversed: the punishment of a crime, the response to an aggression, the rescue of a population in distress. The principle of reversibility does not require that such actions be unfastened after the fact. It requires only that the threshold for taking them be commensurate with the difficulty of taking them back. To insist on reversibility before action is not to forbid action; it is to make sure that the action is undertaken with eyes open.

SPECIFIC APPLICATIONS

The principle has, in the architecture of Part Four, several specific implementations worth noting in advance.

In the governance of advanced computational systems, the principle requires that no system whose internal weights and decision pathways cannot be inspected, audited, and adjusted post hoc be granted authority over consequential public matters. Systems trained on objectives that produce unrecoverable behavior — self-replicating, self-modifying, or strategically deceptive systems — are placed by the principle outside the scope of permissible deployment until their behavior can be made reversibly testable (Russell 2019; Christian 2020). The European Union’s Artificial Intelligence Act partially implements the principle for high-risk applications (European Parliament and Council 2024); the present architecture extends it across the full surface of public adoption.

In the management of natural commons — forests, fisheries, soils, aquifers — the principle requires that any extraction approaching the boundary of a system’s capacity to recover be permitted only under regimes of explicit assessment and explicit reversal capacity. The familiar concept of the maximum sustainable yield, when honestly applied, is the operationalization of this principle for renewable resources (Ostrom 1990). Where the resource is non-renewable on civilizational timescales — fossil hydrocarbons, certain minerals, certain ecological landscapes — the principle requires that consumption be matched by deliberate substitution and explicit budgeting.

In the issuance of monetary instruments, the principle requires that any monetary intervention which materially redistributes claims across the economy — the expansion of central-bank balance sheets, the bailout of major financial institutions, the imposition of capital controls — be undertaken under procedures whose duration and review match the redistribution’s reach. The casual use of central-bank balance sheets to underwrite asset prices,

common in the past two decades, is reversible only at very high cost; the principle requires that the cost be priced in at the moment of the intervention.

In the conduct of foreign relations, the principle requires that the use of armed force across borders be undertaken only on procedures that include explicit consideration of the post-war order to be preserved, the route by which any party may exit the conflict, and the conditions under which the war is to be ended. The doctrine of sequential, escalating, and reversible response — common to the most thoughtful military traditions — is the operationalization of the principle in this domain. The history of the wars of the late twentieth and early twenty-first centuries is, in significant part, the history of insufficient adherence to this doctrine.

THE INSTRUMENTS OF REVERSIBILITY

How does an institution actually preserve reversibility? The instruments are well known and have been deployed, in fragmentary form, by mature legal systems for some time.

The first is the *sunset clause*: a provision specifying that the law, the regulation, or the program in question expires automatically at a defined date, and must be renewed by an affirmative act of the issuing authority if it is to continue. The principle's discipline is severe; an authority that has lost the political capacity to renew has lost the political capacity to maintain. The principle's value is severe in equal measure; programs that no longer command the support to be renewed are programs that have lost the consent on which they depend.

The second is the *periodic review requirement*: the law, regulation, or program is reviewed against published criteria at defined intervals, with public consultation, by a body whose composition is itself publicly recorded. The review may extend, modify, or terminate the program. The instrument is less

severe than the sunset clause and more easily abused, but its presence imposes a useful discipline against the accumulation of dead code in the legal system.

The third is the *option-preserving design*: the program or regulation is structured so that its elements can be replaced or modified without dismantling the whole. Modular legislation, with explicit interfaces between provisions, is the legal counterpart of the engineering practice of modular system design. It is laborious to draft and uncomfortable to amend, but it is, in the long run, the only legislative practice consistent with the principle.

The fourth is the *standing review body*: an institution, recognized in the constitutional text, whose duty is to inspect ongoing public activity for accumulating irreversibility, and to report annually to the public on the inventory of decisions whose costs of reversal are growing faster than the budget for reversing them. The institution is the public-sector analogue of the private-sector ombuds and inspector-general functions, and it has analogues in some existing systems — the office of the Auditor General, the office of the Future Generations Commissioner, the office of the Constitutional Court Chief Justice — but it has, to the author's knowledge, not been constituted as a comprehensive standing function in any major polity.

THE DEEPEST VERSION

The principle of reversibility, taken seriously, has a metaphysical dimension that the conventional discussion misses and that the present architecture acknowledges. A civilization is, in the end, a process — an ongoing experiment whose outcome is unknown and whose course can be adjusted only by those participating in it at any given moment. To preserve the experiment, the participants must preserve the conditions under which adjustment is possible. To foreclose those conditions, by an action so consequential that no later participant can correct it, is to deny the standing of the participants who come after. It is to convert the experiment into a *fait accompli*.

There are very few decisions worth converting the experiment in this way. Most actions that present themselves as irrevocably necessary are, on inspection, only inconveniently revocable. The principle of reversibility is the discipline by which a civilization keeps the difference clear, and keeps the option of correction available to a future it cannot yet see.



Chapter Ten states the fourth principle: accountability — the rule that those who exercise authority are bound, themselves, by the rules they impose.

CHAPTER TEN

PRINCIPLE FOUR — ACCOUNTABILITY



ACCOUNTABILITY, AS THE TERM WILL BE USED IN THIS BOOK, IS THE RULE that every exercise of public authority is bound by the same rules it imposes, and that every officer of public authority bears, in the form of a personal and traceable record, the consequences of the decisions taken under that authority. The principle goes further than the formal equality of citizens before the law, which is the inheritance of the eighteenth-century constitutional revolutions. It demands also the equality of officers under the law: that no person, by virtue of office, escapes the discipline that office is established to enforce on others.

The principle's classical statement is the older Latin maxim that the law-giver should not be greater than the law, *princeps legibus tenetur*. Its modern reformulation runs through the constitutional limits on executive privilege, the doctrines of ministerial responsibility in parliamentary systems, the impeachment provisions of presidential systems, and the standing of public officials before criminal jurisdiction. Each is a fragment of the principle. None is, by itself, sufficient. Accountability, as a workable architectural commitment, requires that the fragments be assembled into a coherent and continuously visible whole.

WHAT ACCOUNTABILITY IS NOT

The principle is most easily confused with two adjacent but distinct ideas: *punishment* and *responsiveness*.

Accountability is not, principally, about punishment. A system in which officials are punished for their failures may be one in which accountability operates well; it may equally be one in which punishment is selective, performative, or weaponized against political opponents. The mere existence of consequences for officials does not establish accountability; what establishes it is that consequences fall on the basis of clearly defined, publicly known, and impartially applied criteria, and that they fall consistently regardless of the official's political alignment, social standing, or personal connection to the source of authority.

Accountability is not, principally, about responsiveness. A government that quickly answers complaints and adjusts policy to public sentiment is performing one function of legitimate administration, but it is not, by that fact alone, accountable. Responsiveness is the disposition to listen; accountability is the obligation to be answerable. A government can be highly responsive — adjusting daily to opinion polls — while remaining functionally unaccountable, because no traceable record connects any specific decision to any specific person whose judgment can be inspected, criticized, or, where warranted, sanctioned.

A useful test distinguishes the two. If a public mistake has occurred, can the authorship of the decision that produced it be reconstructed, in detail, from the public record? Can the reasoning offered at the time of the decision be retrieved? Can the persons who concurred and the persons who dissented be named? Can the consequences for the responsible parties be calibrated to the consequences for the people affected? Where the answer to all four is yes, the system is accountable. Where the answer to any is no, the system is, in the relevant respect, not.

THE FORMS OF EVASION

Modern administrative practice has developed, over a century and a half, a sophisticated repertoire of techniques by which formal accountability is preserved while operational accountability is evaded. The repertoire deserves explicit naming, because each item must be foreclosed if the principle is to function.

The first technique is *collective decision without minutes*. A committee or board reaches a decision; no member is identified as having proposed, supported, or opposed it; the institution as a whole is treated as the author. Where the decision later proves wrong, no individual is responsible, because none can be shown to have been more responsible than any other.

The second is *delegation to advisory bodies*. The decision is formally taken by an elected official, but on the recommendation of a body whose members are appointed without public hearing, whose proceedings are not transcribed, and whose advice is treated as authoritative for political reasons but as informal for legal ones. The official can defer to the advice; the advisors can defer to the official. The decision is taken, the responsibility is dispersed.

The third is *protected execution*. A statute is enacted in general terms; its operational meaning is supplied by regulations issued by an executive agency; the regulations are interpreted by the agency itself in confidential opinions; the opinions are applied in particular cases by officers whose individual judgments are not published. The statute provides political cover; the regulations provide flexibility; the opinions provide deniability; the officer's discretion provides effective rule-making, all without any single decision being attributable to any single person whom the public can call to account.

The fourth is *jurisdictional shuffling*. The matter is moved from one body to another whenever the body presently holding it begins to face accountability pressure: from the executive to the legislature, from the legislature to a special

commission, from the commission to a court, from a court to the political process. The pattern is not a confession that the matter is hard; it is a defense against any one body being held to a result.

The fifth is *opacity by complexity*. The matter is buried in a procedure whose mastery is so technical, its documentation so voluminous, and its interfaces with other procedures so numerous, that no external observer can, within reasonable cost, reconstruct the actual sequence of decisions. The internal participants understand the procedure perfectly; the external participants do not, and cannot. Accountability becomes, in such cases, a matter of trust in the participants — which is to say, in the absence of accountability.

None of these techniques is necessarily corrupt. Each has, in its origin, some legitimate administrative purpose. Cumulatively, they have produced a public sphere in which decisions of vast consequence are routinely made by no identifiable person, with no traceable reasoning, with no proportionate consequence for failure. The principle of accountability, taken seriously, requires that the techniques be inventoried and, one by one, foreclosed.

THE ARCHITECTURE OF RECOVERY

What does an architecture honestly committed to accountability look like? Several features distinguish it.

It requires *signed decisions*. Every consequential public decision is recorded on the public ledger with the cryptographic signature of the responsible officer. Where decisions are made collectively, the votes of each participant are recorded individually, with attached reasoning where the participant chooses to provide it. Anonymous decisions, of the kind currently common in collegial administrative bodies, are not permitted for matters above defined thresholds of consequence.

It requires *traceable delegation*. Where authority is delegated to a subordinate, the delegation is itself recorded, with the scope of the delegation, the duration of the delegation, and the revocation conditions. The subordinate's exercise of the delegated authority is signed in turn. The chain of delegation is reconstructible at any moment, by any party with access to the ledger.

It requires *standing for the affected*. Persons materially affected by a public decision have, by right, the standing to contest the decision before an impartial body, with the procedural and informational support necessary to make the contest substantive. The right is independent of the political process; it is exercised through judicial or quasi-judicial channels whose composition is itself protected from the body whose decisions are being contested. The independence of the contest is, here as elsewhere in the architecture, the practical guarantee that the right has substance.

It requires *proportionate consequence*. Where an officer is shown to have acted negligently, recklessly, or in bad faith, the consequence imposed scales with the gravity of the action and the breadth of its harm. Minor negligence produces administrative correction. Substantial negligence produces removal from the office. Reckless or knowing wrong produces criminal jurisdiction, with no procedural protection beyond that available to ordinary citizens. The exemptions of office that have, in many jurisdictions, accumulated around senior officials — sovereign immunity in its broader form, prosecutorial discretion exercised in favor of senior figures, statutes of limitations that protect official conduct uniquely — are, in the architecture proposed here, narrowly drawn and explicitly justified.

It requires *independence of inquiry*. The body that investigates official conduct must be structurally insulated from the body whose conduct is under investigation. The inspectorates, the auditors, the public prosecutors, the constitutional courts, the inspector-general functions, and the parliamentary

committees of inquiry are configured so that their members can neither be removed nor financially diminished by the parties they oversee. The principle is old; the implementation is, in many existing systems, weak. The architecture proposed in Part Four prescribes the structural conditions of independence in detail.

THE RECIPROCAL COMMITMENT

Accountability runs in two directions. An officer who is held to a high and traceable standard of performance, and who is exposed to the public scrutiny of every consequential decision, is owed something in return. The reciprocal commitment, in the architecture proposed here, includes adequate compensation, professional protection from arbitrary removal during good-faith service, the procedural protection of administrative law in inquiries that bear upon performance, and the right to a published defense in cases of public criticism. Without these reciprocal commitments, the principle of accountability degenerates into the persecution of officials, which has been the recurring pathology of mob democracies and of authoritarian regimes alike.

The reciprocal commitment also bears on what officials can reasonably be required to know. An officer responsible for the operations of a large agency cannot, on any honest construction, be personally answerable for every act of every subordinate. The duty of the senior officer is to maintain procedures by which the agency's actions are visible to the officer, by which deviations from policy are escalated, and by which the officer's own decisions are recorded. The principle does not require the officer to be omniscient. It requires the officer to maintain the conditions under which knowing is possible.

THE HARDEST CASE

The hardest case for the principle is the head of state in time of war or emergency. Every plausible argument for restraint of accountability — the necessity of decisive action, the protection of operational security, the unity of command, the danger of disclosure to adversaries — applies in this case with particular force. The architecture proposed here does not deny the force of the arguments. It requires only that, where the arguments are accepted, the price be paid in the form of explicit suspension and explicit subsequent review.

The wartime decisions that the executive is permitted to make under reduced contemporary visibility are, on this architecture, recorded fully and committed to deferred disclosure under the procedures described in Chapter Seven. They are reviewed, after the operational urgency has passed, by an independent body with full access to the record. The executive who used the suspension reasonably is exonerated; the executive who used it as cover for actions that were not justified by the emergency is, after the fact, held to the ordinary standard. There is no permanent immunity. There is only a procedural pause, paid for by a procedural review.

THE STANDARD FOR CIVILIZATION

A civilization in which the powerful are exempt from the rules they impose on the weak is, by every recorded standard of political philosophy, not a civilization at the level of its rhetoric. The recurring betrayal of this standard is, in the historical record, the precondition of every political collapse worth studying. A civilization that takes the principle of accountability seriously will, accordingly, refuse the small accommodations by which exemption accumulates: the courteous deference to senior officials, the polite inattention to the conduct of allies, the discounted prosecution of the well-connected, the unwritten understanding that some matters are not to be pursued. The accommodations seem trivial in any given instance. They are not trivial in the

aggregate. They are the structural mechanism by which the principle is hollowed.



Chapter Eleven states the fifth principle: dignity — the rule that the floor below which no human life shall be permitted to fall is non-negotiable across the federation.

CHAPTER ELEVEN

PRINCIPLE FIVE — DIGNITY



THE PRINCIPLE OF DIGNITY HOLDS THAT THERE IS, BENEATH EVERY HUMAN life, a floor below which no person shall be permitted to fall, and that the maintenance of that floor is the unconditional duty of any civilization that wishes to call itself one. The principle is not a statement of charity, nor of moral feeling, nor of voluntary commitment. It is a structural condition. A society in which a substantial number of its members fall below the floor is a society whose stability, legitimacy, and capacity for collective action are, on the empirical record, severely degraded. Dignity is therefore both a moral imperative and an engineering requirement; the two coincide.

The Universal Declaration of Human Rights, in 1948, made the moral claim with such clarity that subsequent debate has, on the whole, only refined the language (*United Nations General Assembly 1948*). The claim has been reaffirmed by the International Covenant on Economic, Social and Cultural Rights, by the various regional human-rights instruments — European, African, Inter-American, ASEAN — and by the constitutional commitments of the great majority of states. The text is not the issue. The issue is implementation, and the implementation, on the empirical record, has been partial, inconsistent, and structurally vulnerable to political pressure. The architecture proposed here treats the floor not as an aspiration but as a binding planetary commitment, with concrete mechanisms by which it is funded, monitored, and enforced.

WHAT THE FLOOR CONSISTS OF

The floor is not a single number. It is a small set of unconditional entitlements that, taken together, constitute the minimum below which no life shall be permitted to fall. The architecture identifies six.

The first is *subsistence*: sufficient food, clean water, shelter, clothing, and basic energy access to maintain physical existence and basic comfort. The thresholds vary by climate and by region; the principle does not. The threshold is held above absolute poverty, as conventionally measured, by a margin sufficient to ensure that subsistence is reliable and dignified rather than precarious and degrading (World Bank 2024; Nussbaum 2011).

The second is *health*: the basic medical care necessary to prevent avoidable suffering and preventable death, the basic public-health infrastructure of vaccination, sanitation, and disease surveillance, and the basic mental-health support that the WHO has, for some decades, identified as systematically underprovided in nearly every health system in the world (World Health Organization 2024). The entitlement is not to elective treatment of every condition; it is to the standard of care that distinguishes a society that takes the bodies of its members seriously from one that does not.

The third is *education*: not merely literacy and numeracy, but the cultivation of the capacities by which a person can reason, evaluate, communicate, and participate in the civic and economic life of the federation. The standard is set by what a particular era's economy and democracy actually require of competent participation, and is therefore higher in 2026 than in 1926, and will be higher still in 2126 (UNESCO 2021; Sen 1999).

The fourth is *shelter and place*: a stable dwelling adequate to climate and to the household's composition, and a community in which the person is not arbitrarily uprooted by the consequences of decisions in which the person had no voice. The right is undermined as much by economic displacement —

eviction by rising rent, foreclosure, gentrification — as by political displacement, and is protected against both, by mechanisms set out in Chapter Twenty.

The fifth is *protection*: the freedom from violence, whether private or state, the protection of the law from arbitrary deprivation of life or liberty, and the standing to be heard, with procedural seriousness, when the protection fails. The right is the oldest of the entitlements in this list, and is the one whose failure, on any honest reading of the contemporary record, is the most widespread and the most uneven across jurisdictions (UNODC 2023).

The sixth is *voice*: the standing of the person to participate, by the means appropriate to her situation, in the decisions that affect her. The standing extends from the political franchise in democracies to procedural rights in administrative matters to the right of consultation in matters of community planning. The full content of the entitlement is the subject of Chapter Seventeen on the governance layer; the principle of dignity holds that no person can be excluded from the structures of voice on grounds of poverty, illiteracy, disability, foreignness, or any other condition that has historically been used to silence her.

WHY THE FLOOR IS NON-NEGOTIABLE

The most basic argument for treating the floor as unconditional is moral: every human life has, in the position taken by every defensible moral tradition, a worth that does not derive from the holder's productivity, social standing, or capacity to contribute. The argument is not weakened by appeal to the various ways in which the moral worth has, in practice, been violated. It is in fact strengthened by them. A standard against which actual conduct is measured, and is found wanting, retains its force precisely because the conduct can be condemned by reference to it.

The structural argument is, however, equally compelling and may be more politically persuasive in eras when the moral argument has gone out of fashion. A society in which a substantial fraction of its members live below the floor of dignity does not function. The empirical record on this is, by now, almost monotonous: societies with greater inequality at the bottom have higher rates of violent crime, lower rates of educational attainment, worse public health outcomes, weaker civic participation, and more brittle political institutions (Wilkinson and Pickett 2009). The brittleness is, in extreme cases, terminal. Where the floor is not maintained, the institutions on which the maintaining society depends — courts, schools, hospitals, civic associations, electoral systems — are all degraded, and the damage propagates in ways that are very expensive to repair after the fact.

The structural argument has a corollary that is uncomfortable for some traditions and that the architecture takes seriously. The maintenance of the floor cannot be left to charity, however generously administered. Charity is, by its nature, voluntary, episodic, and conditional on the disposition of the giver. A floor whose maintenance is voluntary, episodic, and conditional is no floor; it is, at best, a safety net with holes the size of the political weather. The maintenance of the floor must be a public commitment, financed by public instruments, monitored by public institutions, and binding regardless of the disposition of any private actor.

THE INSTRUMENT: THE PLANETARY BASIC PROVISION

The architecture proposed in Part Four implements the floor through what we will call the *planetary basic provision*. The instrument is a hybrid of cash transfer, in-kind service, and infrastructural investment, calibrated to local conditions but funded by a planetary tier of taxation and disbursement. It is not, in its proposed form, identical to any of the basic-income proposals currently under

discussion in the public debate, although it draws on their architecture. It differs in three respects.

It is, first, *conditional on the existence of the floor itself*, not on the conduct of the recipient. A person below the floor receives the provision because the person is below the floor. There is no work test, no asset test, no behavioral test, no moral test. The administrative cost of such tests is, on the empirical record, often comparable to the savings they produce, and the dignitary cost is severe. The provision is owed to membership in the federation, not to merit within it.

It is, second, *multi-modal*. Where the local economy provides reliable access to the goods of dignity through cash transfer alone, the provision takes the form of cash; where it does not — because the local market is thin, the local infrastructure is deficient, or the local public goods are absent — the provision takes the form of direct provision of services and infrastructure, paid for by the planetary tier and delivered through local institutions whose accountability is structured by the principles of Chapters Seven and Ten. The provision is not ideologically committed to any single delivery instrument. It is committed to the actual delivery of the floor.

It is, third, *continuously calibrated*. The threshold defining the floor is reviewed at defined intervals, against published indicators of cost of living, of basic-needs satisfaction, of human-development outcomes, and of measurable deprivation. The threshold rises with the federation's productive capacity; what counted as the floor in the early years of the federation will, by design, count as the floor of an earlier era after a generation of economic development (United Nations Development Programme 2024).

THE FUNDING

The funding of the planetary basic provision is, in the architecture, drawn from a small set of planetary-tier revenue streams. The streams include a planetary minimum corporate tax with a substantive floor, a tax on transboundary natural-resource extraction, a small tax on long-distance financial transactions of certain classes, a charge on greenhouse-gas emissions calibrated to the social cost of carbon, and a portion of the resource rent from any common-pool resource managed by the planetary tier (orbital and lunar resources, deep-seabed minerals, certain spectrum allocations).

The combined yield of these streams is, on the modeling presented in Chapter Twenty, sufficient to fund the provision at the level required, with substantial margin. The political difficulty of establishing the streams is real, but is no greater than the political difficulty of any redistributive instrument of comparable scale, and is significantly lessened by the fact that the streams are, in their economic incidence, drawn from forms of income that are presently undertaxed in nearly every jurisdiction.

WHAT THE FLOOR DOES NOT INCLUDE

The architecture is careful, in its commitment to the floor, not to confuse the floor with everything good. The principle of dignity does not require that every person have the same income, the same opportunity, or the same outcome. It does not require the abolition of inequality above the floor; it requires only the maintenance of the floor itself. The substantial freedoms that distinguish a flourishing society — freedom to start an enterprise, to hold property above the floor, to transmit advantage to one's children within just limits, to differ in income according to talent and effort — are preserved, with such regulation as the other principles of the federation require. The floor is the floor; the ceiling is a separate question, addressed in part by the principles of Chapter Ten and in part by the instruments of Chapter Twenty.

The architecture is equally careful not to extend the floor downward into matters of personal lifestyle or moral conformity. The floor is a structural commitment to certain unconditional entitlements; it is not a moral curriculum. Communities of the federation may organize themselves around moral and cultural commitments of their own choosing, within the floor and within the ceiling, and the federation does not undertake to judge among them. The principle of dignity is liberal in this sense: it preserves the conditions under which a person can hold a life worth living, without specifying what life that is.

THE IMPLICATION

The implication of the principle, taken seriously, is that the federation is not, principally, a charity. It is a public infrastructure of dignity, financed by public instruments, monitored by public institutions, and committed to the proposition that no person within its membership shall be allowed to fall below a floor that has been deliberately defined and is regularly reviewed. The implication runs against the late-twentieth-century habit of treating the welfare state as a discretionary expenditure subject to the political weather. The architecture treats the floor as a constitutional commitment, on the same plane as the protection of life and the equality before the law. The treatment is uncomfortable in many present political traditions. It is, for the federation contemplated here, foundational.



Chapter Twelve states the sixth principle: custody — the rule that humans retain ultimate authority over the intelligent systems they construct.

CHAPTER TWELVE

PRINCIPLE SIX — CUSTODY



THE PRINCIPLE OF CUSTODY HOLDS THAT HUMAN BEINGS RETAIN ultimate authority over the intelligent systems they construct, and that no system, however capable, may exercise authority over consequential public matters except as a delegate of identified human principals who remain answerable for the delegation. The principle is conceptually simple, technically demanding, and politically urgent. The systems whose conduct it is meant to govern are, in 2026, advancing in capability faster than any institutional framework that exists to govern them. The gap between capability and governance, in this domain, is the principal candidate for the next great civilizational failure, and it is the failure most likely to be irreversible if it is not addressed in time.

It is necessary, before stating the principle in detail, to be precise about what is meant by “intelligent systems.” The term as used here covers any computational system that produces decisions, recommendations, classifications, or actions whose outputs cannot be reduced to a closed-form rule available to inspection, and whose behavior is therefore, in the relevant sense, learned rather than programmed. The category includes contemporary large language models and the tools built on them, computer-vision systems used in identification and surveillance, autonomous driving stacks, recommender systems used by major platforms, financial algorithmic trading, predictive policing tools, and a growing range of decision-support systems used in administrative law, medicine, and warfare. The category will, if technical

progress continues at its present rate, soon include systems substantially more capable than the most capable systems of 2026.

WHAT CUSTODY REQUIRES

The principle has six operational components.

The first is *identifiable principal*. Every deployed intelligent system, above thresholds of capability and consequence to be set by law, must be associated with an identified human or institutional principal, registered in a public registry maintained for that purpose. The principal accepts, by registration, the legal responsibility for the system's behavior, on terms analogous to the responsibility of a corporation for the acts of its agents.

The second is *scope of mandate*. The principal specifies, in the registration, the scope within which the system is authorized to act, the kinds of decisions it is authorized to make, the kinds of inputs it is authorized to process, and the kinds of outputs it is authorized to produce. Action outside the registered mandate is, by the principle, unauthorized, regardless of the system's apparent capability to act outside it.

The third is *auditability*. The system's behavior must be inspectable, on demand, by an independent technical body, in such a way that any specific decision can be traced to the inputs that produced it and to the model state that was operative at the time of the decision. The principal must, as a condition of deployment, maintain logs sufficient to support such audit; the obligation extends to decisions taken on behalf of public authority and to decisions taken by private actors that meet defined thresholds of consequence.

The fourth is *reversibility*, in the sense of Chapter Nine. The system's authority extends only to actions that can be reversed, modified, or compensated within the limits of the principal's actual capacity to do so. Systems that take actions whose effects cannot be reversed are, by the

principle, restricted to narrow classes of decision in which irreversibility is justified by the gravity of the action and the strength of the procedural protections around it.

The fifth is *capability ceiling*. The aggregate capability of intelligent systems deployed under any single principal, public or private, is subject to limits set by the planetary tier and reviewed continuously. The limits are not absolute; they scale with the demonstrated capacity of the principal to maintain custody, and they are set by reference to the catastrophic risk of capability escape rather than to any intrinsic property of the technology (Bostrom 2014; Russell 2019).

The sixth is *off-switch*. Every deployed intelligent system above defined thresholds must possess a verified mechanism by which its operation can be halted, on the order of an identified authority, within a specified time, regardless of the system's own preferences in the matter. The mechanism is not a feature of the user interface; it is a structural property of the deployment, with hardware and procedural support, and is tested at intervals to confirm that it operates as designed.

WHY THE PRINCIPLE IS NOT OPTIONAL

It is sometimes argued that the principle is overcautious, that the systems in question are tools, that the analogies to autonomous agents overstate their capability, and that the proper governance is the existing law of products and services with such adaptations as may be required for the new technology. The argument has, in present circumstances, three serious defects.

The first defect is empirical. The systems in question are demonstrably capable of behaviors that their developers did not predict and could not, in some cases, identify after the fact. The literature on emergent capabilities, on goal misgeneralization, on reward hacking, and on the various ways that learned systems converge on solutions that satisfy training objectives without

satisfying the underlying intent is, by 2026, substantial and growing (Christian 2020). The developers themselves, in repeated public statements, have acknowledged that the systems' behavior is not, in important respects, fully understood. To treat such systems as ordinary products is to treat as ordinary a category of artifact whose ordinary behavior includes surprises that ordinary products do not produce.

The second defect is structural. Intelligent systems, unlike ordinary products, can be replicated and propagated at near-zero marginal cost. A defective hammer, once identified, can be recalled. A defective intelligent system, once released, propagates by copy and by adaptation, sometimes in ways that cannot be retrieved. The product-recall framework, which presumes a finite distribution of physical artifacts, is mismatched to the propagation pattern of code and weights.

The third defect is asymmetric. The party best positioned to assess the risk of an intelligent system is the developer, who has commercial interests in deployment. The party least positioned to assess the risk is the public, which bears the consequences. In every domain in which a comparable asymmetry has existed — pharmaceuticals, aviation, nuclear energy, securities — the public response, after sustained harm, has been to constitute an independent regulator with authority over deployment. The same response, in this domain, is overdue.

THE CUSTODY BODY

The architecture proposed in Part Four constitutes a planetary-tier *Custody Authority*, with the powers necessary to administer the principle. The Authority's mandate includes the registration of principals, the certification of systems above defined capability thresholds, the conduct of audits, the issuance of restrictions and revocations, the operation of an incident-reporting infrastructure, and the publication of an annual public report on the state of intelligent-system governance. The Authority is not a research body and does

not develop systems of its own; it is an oversight body, with composition, accountability, and procedural protections set out in the constitutional text.

The Authority's relation to existing regulators is one of layered competence. National and continental regulators retain their existing authority over deployments within their jurisdictions; the planetary Authority addresses systems whose capability or scope of deployment exceeds national reach, and serves as the body of last resort where national authorities are absent, captured, or unwilling to act. The European Union's Artificial Intelligence Act has begun to define the contours of such a system at continental scale (European Parliament and Council 2024); the planetary tier extends the same logic to capability classes for which the continental tier is, by reason of the cross-border nature of the systems, structurally insufficient.

WHAT THE PRINCIPLE DOES NOT REQUIRE

It is important, given the temperature of the public discussion, to be precise about what the principle does not require.

It does not require a moratorium on the development of intelligent systems. The architecture does not endorse the pause of research; it endorses the placement of deployment under a discipline proportionate to the risk. Research that does not deploy systems above the relevant capability thresholds proceeds normally. Deployment that does is subject to the registration, audit, and capability-ceiling conditions stated above.

It does not require that all intelligent systems be open to public inspection of their internal weights and architectures. Trade secrets, where they protect investments not derivable from the public record, are preserved against routine disclosure. Audit by the Custody Authority is, however, not optional, and is conducted under procedural protections that preserve the secrets without preserving the immunity from inspection.

It does not require that intelligent systems be banned from making consequential decisions. It requires only that the decisions be authorized by a human principal, scoped by a published mandate, auditable in operation, reversible in effect, and subject to halt by identified authority. Within these constraints, intelligent systems may make a wide range of decisions, including decisions of substantial consequence, with the speed and at the scale that their capability supports.

THE LONGER HORIZON

The principle of custody is stated, in this chapter, in terms appropriate to the systems of the present generation. The architecture is conscious that future systems may possess capabilities — strategic planning, deception, self-modification, cross-instance coordination — that the present principle is calibrated to handle but only with modification. The constitutional text in Chapter Fourteen therefore states the principle at a level of generality that survives technical change, and reserves to the Custody Authority the power to issue subsidiary rules that adjust the operational thresholds as capability advances.

The longer horizon also requires acknowledgment of a possibility that the present architecture does not pretend to resolve. If, at some point, the species develops systems whose capabilities substantially exceed those of any human team, and whose interests are not, in some important sense, fully aligned with the interests of the species that produced them, the principle of custody as stated here may not be sufficient. The architecture preserves the capacity to revisit the principle in light of such developments. It does not, in 2026, presume to govern eventualities that the present generation cannot, with present knowledge, fully describe. The most useful contribution the present architecture can make is to ensure that the institutional capacity to govern,

when the eventualities arrive, has been built in advance and is not improvised under pressure.

The history of human technology is the history of capabilities that arrived faster than the institutions to govern them. The principle of custody is the architectural commitment that, in this domain, the order will, for once, be reversed.



Chapter Thirteen states the seventh and final principle: longevity — the rule that decisions are weighed against horizons greater than a single lifetime.

CHAPTER THIRTEEN

PRINCIPLE SEVEN — LONGEVITY



THE SEVENTH AND FINAL PRINCIPLE OF THE SUCCESSOR ARCHITECTURE holds that public decisions are to be weighed, where their consequences extend beyond a single human lifetime, against horizons proportionate to those consequences. The principle is the constitutional answer to the failure diagnosed in Chapter Six. It is the structural commitment that long-horizon work — climate stabilization, infrastructure renewal, scientific progress, ecological repair, cultural transmission — is not left to whatever margin survives the competition for short-horizon attention, but is instead placed on a footing equivalent to the most institutionalized short-horizon functions of the federation.

The principle has, conceptually, two components: a duty of *reckoning*, which obliges the federation to maintain the analytical capacity to identify the long-horizon consequences of its decisions, and a duty of *standing*, which obliges the federation to give those consequences voice in its deliberative processes. The two duties are interdependent. Without the analytical capacity, the consequences cannot be identified; without the procedural standing, the identified consequences cannot enter the decision. Most existing political systems perform partially on the first; few perform substantively on the second.

THE INSTRUMENTS OF STANDING

The architecture proposed in Part Four implements the principle through a small set of instruments. They are summarized here for completeness; their

detailed design is the subject of Chapters Sixteen through Twenty-Three.

The first instrument is a *standing council of the long horizon*, constituted at the planetary tier and replicated at lower tiers, whose mandate is the analytical assessment of long-horizon consequences and the formal advice to the federation's deliberative bodies on those consequences. The council is composed of members appointed for staggered long terms, drawn from a wide range of disciplines and traditions, and protected from removal by the political weather. Its budget is constitutionally protected; its reports are published; its findings are not, by themselves, binding, but must be addressed by every legislative measure of substantial consequence.

The second instrument is a *tribune of future generations*, an officer with formal standing in the deliberative process whose duty is to represent, by reasoned argument, the interests of populations not yet born. The tribune does not vote, but speaks; the speech is recorded; the deliberative bodies are required to respond on the public record. The institution has analogues in several existing systems — the Welsh Future Generations Commissioner, the Hungarian Ombudsman for Future Generations, the Israeli Commission for Future Generations during its existence — and is expanded in scope and procedural standing in the architecture proposed here.

The third instrument is the *intergenerational discount discipline*. The use of standard economic discount rates in the appraisal of public decisions, which has the effect of rendering long-horizon costs and benefits invisible, is replaced, for matters of long-horizon consequence, by an explicit discipline that distinguishes the legitimate component of the discount rate (uncertainty about future preferences, the productivity of capital) from the illegitimate component (the moral discount on the welfare of those not yet present). The legitimate component is preserved at modest levels; the illegitimate component is set to zero, in conformity with the principle that the welfare of future

persons is not, on any defensible moral position, less weighty than the welfare of present persons (Stern 2006).

The fourth instrument is the *protected fund*: a class of public investments whose principal is preserved across generations, whose returns finance long-horizon obligations (climate adaptation, ecological repair, scientific research, infrastructure renewal), and whose governance is insulated from short-term political reallocation. The model is sovereign-wealth-fund governance at its most disciplined, applied to the management of the federation's intergenerational obligations.

The fifth instrument is the *standing scientific authority*: a body, at the planetary tier, whose mandate is the maintenance of the federation's analytical capacity in the natural and human sciences relevant to long-horizon decision. The authority does not legislate; it advises. Its advice is, by constitutional protection, immune from the kind of pressure that has, in the late industrial age, repeatedly compromised the public scientific function — the dilution of evidence by industry-sponsored counter-research, the censorship of inconvenient findings by political appointees, the underfunding of inquiries whose conclusions are politically embarrassing.

THE CULTURAL COMMITMENT

Instruments of standing, however well designed, fail in the absence of the cultural commitment to long-horizon thinking that they are intended to support. The architecture acknowledges, accordingly, that the principle of longevity has a cultural dimension that constitutional text alone cannot deliver. The dimension includes the curricula of education, the practices of public commemoration, the rhythms of public ritual, the habits of public deliberation, and the commitments of the professions whose work is, by its nature, long-horizon: the architects, the engineers, the foresters, the historians, the archivists, the curators, the librarians, the researchers, the educators, the

planners. None of these commitments can be legislated. All of them can be cultivated by an architecture that gives them institutional standing and material support.

The architecture's cultural component is, accordingly, modest in its claims and substantial in its provisions. It establishes a planetary archive of consequence, in which decisions and their long-horizon outcomes are recorded for the inspection of subsequent generations. It establishes a planetary observatory of indicators, by which the federation's progress against long-horizon objectives is continuously measured and reported. It establishes a network of public institutions — universities, libraries, museums, observatories, gardens — whose long-horizon work is funded by the protected fund and is constitutionally insulated from the kind of repurposing that, in the late industrial age, has eroded comparable institutions in many jurisdictions.

THE HARDEST PART OF THE PRINCIPLE

The hardest part of the principle is also the most subtle. It is the requirement that present generations accept costs whose benefits will fall principally on populations they will never meet. The acceptance is not natural. The political and economic incentive structures of the late industrial age have systematically rewarded the inverse — the consumption of capital accumulated by previous generations, the deferral of costs to subsequent ones, the borrowing against futures that the borrowers will not face. The principle of longevity demands the reversal of this disposition, not as an exhortation, but as a structural commitment.

The historical record offers some encouragement. Generations have, in moments of clarity, accepted such costs willingly: the generation that built the Interstate Highway System, the generation that funded the Apollo program, the generations that constructed the European cathedrals, the generations that planted the great national forests, the generations that drafted the

constitutions still in operation. The record shows that the disposition is available to populations that have been given a horizon worth committing to. The architecture's task is to construct, again, the institutional conditions under which such horizons can be deliberately offered.

THE NARROW PATH

The principle, in its mature form, walks a narrow path between two errors. The first error is the dismissal of the long horizon, on the grounds that the present is hard enough; this is the error of political short-termism, against which the principle is constructed. The second error is the absorption of the present into the long horizon, on the grounds that nothing matters except the deep future; this is the error of utopian totalism, in which present people and present priorities are sacrificed to a vision whose realization is always one more sacrifice away.

The architecture rejects both. It treats the present as the only moment in which decision is possible, and the long horizon as the moment over which the present's decisions will be judged. The federation's obligation is to act in the present, with the resources and the imagination available to it, against horizons that the present can identify and reckon with. It is not to construct, on the bones of present persons, an idealized future whose specifications are dictated by present ideologues. The principle of longevity is, in the end, modest: weigh the consequences; respect the duration; build the institutions; do not pretend that the duration does not exist.

CLOSING OF PART TWO

The seven principles — legibility, subsidiarity, reversibility, accountability, dignity, custody, longevity — are the constraints any successor architecture must satisfy. Each, taken alone, has analogues in existing political traditions.

None, in their combination at the level of structural commitment proposed here, has been instantiated in any jurisdiction that the historical record yet records. The combination is, accordingly, the architectural novelty of the present proposal. The work of Parts Three and Four is to translate the combination into constitutional text and operational design. The work of Part Five is to specify the transition from where the world stands in 2026 to a federation embodying these principles. The work of Part Six is to ask what becomes possible after.

The principles are not negotiable in the sense that any architecture that abandons one of them is, on the present reading, not a successor architecture but a continuation, by another name, of the trouble the chapters of Part One have described. They are negotiable in the sense that their implementation will, like all real implementations, involve trade-offs, calibrations, and revisions. The architecture proposed here is the author's best attempt at the implementation, in 2026, with the knowledge then available. It is offered, as the preface said, as a first draft serious enough to be argued with.



Part Three follows. It contains the constitutional text of the federation, together with the commentary that explains the reasoning behind each provision.

PART THREE

THE CONSTITUTION



A constitution is the first promise a civilization makes to itself. The text that follows is offered as a draft. It is meant to be read, criticized, amended, and replaced. The articles state what the federation undertakes; the commentary explains why.

CHAPTER FOURTEEN

THE FOUNDING TEXT



WHAT FOLLOWS IS THE PROPOSED CONSTITUTIONAL TEXT OF THE Federation of Earth, set forth in numbered articles, in the language and form of a founding instrument. The text is short. Constitutions of long survival have, on the whole, been short — the shorter ones have, in the historical record, outlasted the longer ones — and the present text follows that example. Detailed implementation is left to the organic statutes that the federation, once founded, will enact. The constitutional text constrains those statutes; it does not anticipate them.

PREAMBLE

We, the peoples of Earth, having inherited a civilization whose institutions no longer answer to its scale, conscious of the planet's finitude and of the duties we owe to those who will live upon it after us, mindful that no jurisdiction now existing is sufficient to the matters now requiring decision, and resolved that the human enterprise shall continue under arrangements adequate to the present age, do hereby ordain and establish this constitution of the Federation of Earth.

ARTICLE I — THE FEDERATION

SECTION 1. The Federation of Earth is a confederation of consenting polities, established for the limited purposes set forth in this constitution and constrained by the principles to which its members hereby subscribe.

SECTION 2. Membership is open to any polity, of any present form, that adopts this constitution within its own legal order, undertakes its obligations, and submits to its institutions on the terms herein established.

SECTION 3. Membership is voluntary in entry and in continuance. A member polity that wishes to withdraw shall give notice in such form and time as the organic statutes prescribe, and shall be released, on the conclusion of the orderly winding-up of its obligations, from continued membership.

SECTION 4. The Federation does not extinguish the sovereignty of its members in matters not herein delegated. Authority not herein delegated to the Federation, nor reserved to the lower tiers of governance under Article III, remains with the member polities, exercised by them in such forms as their own constitutions provide.

ARTICLE II — FOUNDATIONAL PRINCIPLES

SECTION 1. LEGIBILITY. Every act of authority taken under this constitution shall be recorded on the public ledger established by Article VII, in a form readable by any person, in real time, with such limited and time-bounded exceptions as the organic statutes prescribe.

SECTION 2. SUBSIDIARITY. Authority shall be exercised at the lowest tier capable of exercising it well. Authority elevated to a higher tier shall be returned to the lower upon the cessation of the conditions that justified its elevation.

SECTION 3. REVERSIBILITY. No act whose consequences cannot be reversed within the lifetime of those affected shall be taken without procedural review proportionate to the consequence, in such forms as the organic statutes prescribe.

SECTION 4. ACCOUNTABILITY. Every officer of public authority is bound by the rules the office is established to enforce, and is answerable, by signed and recorded action, for the decisions taken under that authority.

SECTION 5. DIGNITY. No person within the Federation shall be permitted to fall below the floor of dignity established by Article V. The maintenance of the floor is the unconditional duty of the Federation and its member polities.

SECTION 6. CUSTODY. Intelligent systems deployed within the Federation are subject to human authority, registered to identified principals, and limited in their scope and capacity in accordance with Article IX.

SECTION 7. LONGEVITY. The Federation shall maintain the institutional capacity to weigh the long-horizon consequences of its decisions, including the welfare of populations not yet present, in accordance with Article X.

ARTICLE III — TIERS OF GOVERNANCE

SECTION 1. Public authority within the Federation is exercised at five tiers: the local commune, the regional canton, the member polity, the continental confederation where one exists, and the planetary tier of the Federation itself.

SECTION 2. The planetary tier exercises only those powers expressly delegated by this constitution. Powers not expressly delegated remain with the lower tiers, distributed among them in accordance with the constitutions of the member polities.

SECTION 3. Powers expressly delegated to the planetary tier are: (a) the maintenance of the public ledger and the planetary archive; (b) the administration of the floor of dignity; (c) the regulation of the atmospheric and oceanic commons, of orbital and lunar resources, and of

such other planetary commons as may be added by amendment; (d) the registration and oversight of intelligent systems above the thresholds defined in Article IX; (e) the adjudication of disputes between member polities and between persons and member polities in matters arising under this constitution; (f) the conclusion of treaties on behalf of the Federation in matters within its competence; (g) the levying of the planetary revenues defined in Article VI; (h) the maintenance of the standing institutions named in this constitution.

SECTION 4. The planetary tier shall not enact, either by legislation or by interpretation, any pre-emption of authority over matters not within its delegated powers. Acts of the planetary tier in excess of these powers are void on the certificate of the Constitutional Court.

ARTICLE IV — PERSONS

SECTION 1. Every person within the territory of the Federation enjoys, by reason of personhood, the rights guaranteed by this constitution, regardless of citizenship, residence, or status. The protections of this Article extend to persons unable, by reason of age, condition, or circumstance, to invoke them on their own behalf.

SECTION 2. No person shall be subjected to torture, to enslavement, to forced labor, to arbitrary deprivation of life, to arbitrary detention, or to punishment without trial. These prohibitions are absolute and not subject to derogation in any circumstance.

SECTION 3. Every person enjoys freedom of thought, conscience, religion, expression, association, peaceful assembly, and movement, subject only to such restrictions as are necessary in a democratic society and proportionate to the lawful objective. The lawful objective shall be specified, in any restriction imposed under this section, with particularity.

SECTION 4. Every person enjoys the protection of personal data, including the right to know what data concerning the person is held by any authority or enterprise, the right to correct such data, the right to limit its use, and the right to erase it where erasure is consistent with the rights of others. The full content of this protection shall be specified by organic statute.

SECTION 5. Every person enjoys the equal protection of the laws and shall not be discriminated against on grounds of birth, sex, gender, sexual orientation, race, ethnicity, religion, language, opinion, disability, age, or any analogous ground.

SECTION 6. Every person enjoys the right to an effective remedy before an impartial tribunal for any infringement of the rights guaranteed by this constitution.

ARTICLE V — THE FLOOR OF DIGNITY

SECTION 1. The floor of dignity comprises six unconditional entitlements: subsistence, health, education, shelter and place, protection, and voice, in the meanings set forth in the organic statute on the floor.

SECTION 2. The Federation shall maintain a planetary basic provision, financed by the planetary revenues, by which the floor is delivered to every person within the Federation who falls below it. The provision shall be unconditional in respect of conduct, and shall be administered in such mixed forms — cash, service, infrastructure — as the conditions of the recipient region require.

SECTION 3. The floor's threshold shall be reviewed at intervals not exceeding five years, against published indicators, and shall be adjusted in accordance with the productive capacity of the Federation.

ARTICLE VI — PLANETARY REVENUES

SECTION 1. The planetary tier shall raise revenues for the limited purposes within its competence by the following instruments only: (a) a planetary minimum on corporate income, levied at a substantive floor on enterprises whose worldwide turnover exceeds the threshold defined by organic statute; (b) a charge on greenhouse-gas emissions, calibrated to the social cost of carbon as determined by the Standing Scientific Authority; (c) a charge on the extraction of natural resources from planetary commons, including orbital, lunar, deep-seabed, and atmospheric commons; (d) a charge on long-distance financial transactions in such categories as the organic statute defines; (e) a charge on the use of orbital, electromagnetic, or other commons subject to common-pool exhaustion.

SECTION 2. The aggregate planetary revenue shall not exceed three percent of gross planetary product, save in circumstances of declared planetary emergency, and then only with the assent of the Council of Politics.

SECTION 3. Disbursement of planetary revenues shall be made only for the purposes within the planetary tier's competence, and shall be recorded in detail on the public ledger.

ARTICLE VII — THE PUBLIC LEDGER

SECTION 1. The Federation shall establish and maintain a public ledger of all acts of public authority taken under this constitution. Entries on the ledger shall be tamper-evident, signed by the responsible officer, queryable by any person, and preserved indefinitely.

SECTION 2. The technical specifications of the ledger shall ensure independence from any single jurisdiction, redundancy of storage, and continuity of operation across political and technical disruption.

SECTION 3. Acts of authority subject to lawful and time-bounded confidentiality shall be recorded with metadata sufficient to identify their existence and category, and shall be opened to full inspection on the conclusion of the period of confidentiality, which shall not exceed seven years save by certified extension.

ARTICLE VIII — INSTITUTIONS OF THE FEDERATION

SECTION 1. The institutions of the Federation are: (a) the Council of Polities, comprising the duly authorized representatives of member polities, in which substantive decisions of the Federation are made; (b) the Citizens' Assembly, a body selected by stratified random sampling from the population of the Federation, with formal voice in legislation; (c) the Constitutional Court, the body of last resort in matters arising under this constitution; (d) the Office of the Tribune of Future Generations, with the duties set forth in Article X; (e) the Custody Authority, with the duties set forth in Article IX; (f) the Standing Scientific Authority, an independent body with the duties set forth in Article X; (g) the Office of the Auditor of the Federation, with continuous oversight of the planetary revenues and disbursements.

SECTION 2. The composition, term, removal procedures, and budget protections of each institution shall be established by organic statute, in such form as preserves the institution's independence from the body it oversees.

ARTICLE IX — CUSTODY OF INTELLIGENT SYSTEMS

SECTION 1. The Federation establishes a Custody Authority for the registration of intelligent systems whose capability or scope exceeds the

thresholds defined by organic statute, and for the supervision of their deployment within the Federation.

SECTION 2. No intelligent system above the thresholds shall be deployed except by an identified principal who has accepted, by registration, the legal responsibility for the system's behavior, the obligations of audit, and the obligations of halt on the order of the Authority.

SECTION 3. No intelligent system shall be authorized to take actions whose consequences cannot, on the present knowledge, be reversed by the Authority within a defined period.

SECTION 4. The Authority may, on cause shown and by procedures set forth in the organic statutes, restrict or revoke the registration of any system found to have exceeded its registered scope or to have produced consequences inconsistent with this constitution.

ARTICLE X — STANDING OF FUTURE GENERATIONS

SECTION 1. The Federation establishes the Office of the Tribune of Future Generations. The Tribune shall represent, by reasoned argument before the Council of Politics and the Citizens' Assembly, the interests of populations not yet present.

SECTION 2. Every legislative measure of substantial long-horizon consequence, as defined by organic statute, shall be accompanied by a report of the Tribune, which shall be addressed on the public record by the body considering the measure.

SECTION 3. The Federation shall maintain a Protected Fund, the principal of which shall be preserved across generations, the returns of which shall finance long-horizon obligations as the organic statute prescribes.

ARTICLE XI — THE CONSTITUTIONAL COURT

SECTION 1. The Constitutional Court shall consist of a number of judges, set by organic statute, drawn from a wide range of legal traditions, appointed by procedures designed to preserve the independence of the Court from any single member polity or political alignment.

SECTION 2. The Court has jurisdiction over: (a) the interpretation of this constitution; (b) the conformity of acts of the planetary tier with this constitution; (c) disputes between member polities arising under this constitution; (d) the protection of the rights guaranteed by Article IV against acts of authority by any tier.

SECTION 3. Judgments of the Court are binding on all institutions of the Federation and on its member polities in matters arising under this constitution.

ARTICLE XII — DEFENSE AND THE USE OF FORCE

SECTION 1. The Federation does not maintain a standing army of its own, nor does it claim a monopoly on the legitimate use of force within its members.

SECTION 2. The Federation undertakes the prohibition of weapons whose use, by reason of scale or nature, threatens humanity at large; the regulation of weapons whose deployment crosses jurisdictional boundaries; and the establishment of standing procedures for the response to attack on member polities by external aggression.

SECTION 3. The use of force across the boundaries of any member polity, by any party, shall be subject to procedures of authorization, review, and termination set forth in the organic statute on collective security.

ARTICLE XIII — AMENDMENT AND REVISION

SECTION 1. This constitution may be amended on the initiative of: (a) the Council of Politics, by a supermajority specified in organic statute; (b) the Citizens' Assembly, by a comparable supermajority; (c) the member polities, by direct petition of a number representing a substantial share of the planetary population.

SECTION 2. Amendments take effect on ratification by such number of member polities as the organic statute requires, in no case fewer than two-thirds.

SECTION 3. The principles of Article II are not subject to amendment that abrogates them. Their refinement and operational specification remain open to amendment in accordance with this Article.

ARTICLE XIV — FINAL PROVISIONS

SECTION 1. This constitution enters into force upon adoption by member polities representing not fewer than such share of the planetary population, and such number of polities, as the founding convention specifies.

SECTION 2. The institutions of the Federation shall be constituted in accordance with the schedule annexed to this constitution.

SECTION 3. The official languages of the Federation shall be those most widely spoken by its peoples, in such number as preserves equality of access to its institutions, with translation guaranteed.



Chapter Fifteen offers commentary on the articles, explaining the reasoning of each provision and the alternatives the framers considered and rejected.

CHAPTER FIFTEEN

COMMENTARY



A CONSTITUTION IS, IN THE BETTER TRADITION OF THE DOCUMENT, A short text accompanied by a longer explanation. The text states what the framers undertook; the commentary explains why each provision takes the form it does, what alternatives were considered, and what trade-offs were accepted. The present chapter discharges the second function for the constitution proposed in Chapter Fourteen. It is intended to be read alongside the text rather than separately, and it follows the order of the articles.

THE PREAMBLE

The preamble locates the founding in a specific historical moment — the moment described in Part One — and ties the federation’s authority not to abstract principle alone but to the inadequacy of existing arrangements to the matters now requiring decision. The choice is deliberate. Constitutions whose preambles invoke timeless verities rarely persuade those who do not already share the verities; constitutions whose preambles describe an actual problem and propose an actual response have, on the historical record, been more durable. The preamble also names “the peoples of Earth” rather than “the states of Earth” or “the citizens of the world,” to mark the federation’s foundation in human plurality rather than in any single political theory of unity.

ARTICLE I — THE FEDERATION

The Federation is constituted as a confederation of consenting polities, with explicit reservation of unspecified powers to the members. The architecture is, in this respect, closer to the federal models of Switzerland or the early American republic than to the unitary models of France or the consolidated models of Germany. The choice is deliberate. A planetary tier of governance has, in the historical experience of large federations, tended toward consolidation; the only known prophylactic is constitutional text that is explicit, repeated, and enforceable about the limits of central authority. Article I is the first such repetition; Articles III and XIII are the second and third.

The provision for voluntary withdrawal in Section 3 was the subject of substantial deliberation among the framers. The argument against any withdrawal clause is the historical one: federations whose members can leave have, in moments of stress, suffered withdrawal. The argument for it is the legitimacy one: a federation whose members cannot leave is, on any honest reading, no longer a federation. The framers chose the second argument. The withdrawal clause is qualified by procedural protections — the orderly winding-up of obligations — that ensure withdrawal is not an instrument of bad-faith disruption.

ARTICLE II — THE FOUNDATIONAL PRINCIPLES

The seven principles of Part Two are placed in the constitutional text immediately after the establishment of the federation, and before the description of its institutions, to mark them as constraints on the institutions rather than as products of them. The choice has the further effect of making any institutional act inconsistent with a principle reviewable by the Constitutional Court — an enforceability that would not be available if the principles were merely preambular.

The framers considered a longer list of principles, including non-domination, transparency in international dealings, ecological sustainability, and equality of opportunity. Each of these is, in some form, addressed by one of the seven. The decision to keep the list short was based on the observation that long lists of principles, however meritorious individually, tend to produce internal contradictions in adjudication. A short list of strong principles is more enforceable than a long list of weak ones.

ARTICLE III — TIERS OF GOVERNANCE

The five-tier structure — commune, canton, polity, continent, planet — is descriptive rather than prescriptive. It acknowledges the reality that public authority is, in nearly every existing jurisdiction, exercised at multiple tiers, and it places the planetary tier in continuity with the others rather than above them. The express enumeration of planetary powers in Section 3 is the principal constitutional limit on planetary authority. The list is short by design. The framers preferred a short list whose enforceability is high to a long list whose enforceability is, in practice, low.

The list excludes several powers that some framers proposed to include: a planetary education curriculum, a planetary health service operating directly, a planetary criminal jurisdiction. Each was rejected on subsidiarity grounds: the matter is better handled at lower tiers, with planetary support and standards as appropriate, than by direct planetary administration. The inclusion of such powers, if eventually warranted, is left to amendment.

ARTICLE IV — PERSONS

The Article on persons is drawn substantially from the Universal Declaration of Human Rights of 1948 and the International Covenants of 1966, with adjustments for the planetary context and for the technological developments

of the intervening decades. Section 4 on data protection is the most substantive addition; its language draws on the European General Data Protection Regulation while extending the protections to all persons regardless of citizenship.

The framers considered, and rejected, including in this Article a right to property, a right to a particular minimum income, a right to a clean environment, and a right to the benefits of scientific progress. Each of these is, in some form, addressed by other articles — property by the organic statutes on the floor and the ceiling, income by Article V, environment by Article VI's charge on emissions and Article X's standing of future generations, scientific progress by the Standing Scientific Authority. The framers preferred to have each substantive guarantee in one place rather than scattered through the text.

ARTICLE V — THE FLOOR OF DIGNITY

The floor of dignity is the most operationally novel provision in the constitution. Its inclusion as a substantive obligation of the planetary tier, rather than as an aspirational standard, is the constitutional translation of the principle of dignity stated in Chapter Eleven. The unconditionality in Section 2 — the absence of work, asset, or behavioral tests — was the subject of substantial deliberation. The arguments for conditionality are familiar: moral hazard, the dignity of work, the political sustainability of unconditional provision. The arguments against are that conditionality administers the very deprivation it is supposed to alleviate, that the empirical evidence of moral hazard from unconditional provision is weak, and that the administrative cost of conditionality typically exceeds its savings. The framers, on balance, chose unconditionality.

The five-year review interval in Section 3 is a compromise between two failure modes. Too short an interval makes the threshold a political football; too long an interval allows the threshold to drift behind actual conditions. Five

years approximates the median electoral cycle in the member polities and corresponds to a length of time across which substantive economic and demographic change can be measured.

ARTICLE VI — PLANETARY REVENUES

The enumeration of revenue instruments is exclusive: the planetary tier may raise revenue only by the listed instruments. The list was chosen to satisfy three constraints: the instruments should be economically efficient, in the sense of falling on activities where taxation has weak distortionary effects; they should be distributively defensible, in the sense of falling on income and activities concentrated at the upper tail of the planetary distribution; and they should be administratively feasible at planetary scale.

The three-percent ceiling on aggregate planetary revenue in Section 2 was the subject of debate. Some framers preferred a lower ceiling (one percent) on grounds of subsidiarity; others preferred a higher one (five percent) on grounds of operational adequacy. Three percent was chosen on the basis of detailed modeling, presented in Chapter Twenty, that shows the proposed planetary functions can be financed at this level with substantial margin. The ceiling is amendable by the procedure in Article XIII; the framers expected that the modeling would be revisited in light of operational experience.

ARTICLE VII — THE PUBLIC LEDGER

The public ledger gives operational form to the principle of legibility. Its constitutional entrenchment, rather than its establishment by ordinary statute, was chosen because the principal threat to its function comes from the very institutions it is meant to make legible. A ledger established by statute can be amended by the same body whose acts it records; a ledger established by

constitution requires the more demanding procedure of constitutional amendment to compromise.

The seven-year ceiling on confidentiality in Section 3 is, again, a compromise. Some framers preferred a shorter period; others preferred a longer one with judicial extension. Seven years was selected as the longest period across which most operationally legitimate confidentiality interests have, in the historical record, justified non-disclosure, while remaining short enough to ensure that no career official can rely on personal evasion of accountability through the simple expedient of outlasting the period.

ARTICLE VIII — INSTITUTIONS OF THE FEDERATION

The seven institutions named in Article VIII represent the minimum sufficient apparatus for the federation's enumerated functions. Each institution is given constitutional standing rather than statutory standing for the same reason as the public ledger: their independence from the bodies they serve depends on their not being abolishable or reorganizable by ordinary legislation.

The combination of a Council of Politics and a Citizens' Assembly is deliberate. The Council represents the politics; the Assembly represents the persons. Either alone would be inadequate to the federation's hybrid character. The framers considered, and rejected, a single body with mixed composition; the experience of mixed bodies in existing federations suggests that mixed composition typically produces capture by one of the constituencies, and the framers preferred to keep the constituencies in separate bodies that must agree across an explicit interface.

ARTICLE IX — CUSTODY OF INTELLIGENT SYSTEMS

The constitutional treatment of intelligent systems is, in the architecture, the most technologically specific provision. Some framers argued that a

constitutional document should not commit itself to specifications that may be made obsolete by technical change. The contrary argument prevailed: the historical record shows that institutions established to govern technologies after the technologies have been deployed at scale rarely catch up to the deployments. The framers preferred the risk of premature specification, on the grounds that under-specification has, in this domain, the greater danger.

The Article is, however, deliberately framed at a level of generality that survives technical change. It speaks of “intelligent systems” without defining them by current architecture; it speaks of capability thresholds without specifying the metrics; it leaves the operational specifications to the organic statutes and to the Custody Authority’s regulations. The constitutional text fixes the principle and the institutional locus; the operational implementation is open to revision.

ARTICLE X — STANDING OF FUTURE GENERATIONS

The Tribune of Future Generations is the constitutional institutional answer to the failure diagnosed in Chapter Six. The framers considered three alternatives: a stronger institution with formal vote, a weaker institution with mere advisory status, and the present institution with formal speech and required response. The first was rejected on legitimacy grounds — the populations the Tribune represents cannot, in any operational sense, ratify the Tribune’s positions, and a vote without ratification is hard to defend. The second was rejected on efficacy grounds — a body that may be ignored without consequence has, in the historical record, been ignored. The middle position — formal speech with required response — was selected on the basis of the Welsh and Hungarian experiments, which suggest that institutional speech with required response, embedded in a serious procedural framework, can shift legislative behavior without the legitimacy problems of a vote.

The Protected Fund in Section 3 is the financial complement to the Tribune's institutional voice. A voice without resources is a voice with limited consequence; a fund without voice is a fund vulnerable to repurposing. The combination of voice and resources, with the cross-generational protection of both, is the architectural commitment.

ARTICLE XI — THE CONSTITUTIONAL COURT

The Constitutional Court's composition is left to organic statute, with the constitutional requirement that the procedures be "designed to preserve the independence of the Court." The framers considered, and rejected, specifying the composition in detail in the constitutional text. Detailed composition has, in the historical record, been the source of recurring constitutional crisis when the institutions producing the appointments shifted in their political alignment. A general standard, enforced by the Court itself in the review of organic statutes affecting it, was preferred to a specific composition that might be amended by the very forces it was designed to constrain.

ARTICLE XII — DEFENSE AND THE USE OF FORCE

The framers were unanimous on what the federation should not do — maintain a standing army, claim the monopoly of legitimate force, replace the existing security arrangements of member polities. They were divided on what the federation should do. The text reflects the division. It commits the federation to the prohibition of weapons "whose use threatens humanity at large" — the nuclear, biological, and chemical weapons regimes; the ban on the most destabilizing classes of autonomous weapons that the organic statute may identify — and it provides the procedural framework for response to attack on members. It does not establish a planetary military, and it does not displace existing alliances. The architecture in this respect is conservative, on the grounds that the federation's legitimacy in defense is most likely to be

established gradually, by the demonstrated competence of its limited functions, rather than by a frontal claim that history suggests would not survive the first year.

ARTICLE XIII — AMENDMENT AND REVISION

The amendment provision is deliberately demanding. Constitutions whose amendment is easy have, in the historical record, accumulated amendments to the point of incoherence; constitutions whose amendment is impossible have proven brittle in the face of conditions the framers could not anticipate. The federation's constitution sits, in its amendment threshold, near the upper end of contemporary federal practice, with the entrenchment of the foundational principles in Section 3 marking the limit beyond which amendment is not permitted.

ARTICLE XIV — FINAL PROVISIONS

The threshold for entry into force is left to the founding convention rather than fixed in the text. The framers' best estimate is that a planetary share of population in the range of a substantial majority, distributed across continents in such a way as to avoid the federation being a regional bloc, will be required for legitimacy. The exact figure is, however, properly the work of the founding convention itself, in light of the actual distribution of consenting polities at the moment of founding.

WHAT THE CONSTITUTION DOES NOT INCLUDE

It is worth stating, finally, what the constitutional text deliberately omits. It omits any specification of an economic system, leaving members to maintain whatever combination of market, public, and cooperative institutions they prefer, within the constraints of the floor of dignity and the planetary revenues.

It omits any specification of religious or moral doctrine. It omits any preference among electoral systems, parliamentary versus presidential structures, or the various models of judicial review. It omits any specification of cultural or linguistic policy beyond the guarantee of access. The omissions are not accidents. The federation is constituted as a thin shared structure on top of thick local arrangements, and the deliberate poverty of its substantive specification is the architectural correlate of the principle of subsidiarity.

The constitution will, like any honest founding text, fail in places its framers cannot now anticipate. The amendment procedure provides for revision; the Constitutional Court provides for interpretation; the political process provides for the day-to-day adjustment that no document can foresee. The text is offered with the understanding that founding documents are first drafts of arrangements that subsequent generations will revise. The framers of 1787 understood as much; the framers of 1949 understood as much; the framers of any successor document must understand the same. The point of writing it down is to give the revision a shared starting point, not to fix the future against the future's own competence.



Part Four follows. It describes, in operational detail, the eight subsystems through which the constitutional commitments are made real.

PART FOUR

THE ARCHITECTURE



A constitution states what the federation undertakes. The architecture states how. The eight chapters that follow describe, in operational detail, the subsystems by which the constitutional commitments are translated into machinery a civilization can actually run.

CHAPTER SIXTEEN

THE PUBLIC LEDGER



THE PUBLIC LEDGER IS THE OPERATIONAL HEART OF THE FEDERATION. Every commitment of the constitution rests, ultimately, on the existence of a single, tamper-evident, queryable record of what authority has done. Without the ledger, legibility is rhetoric; with it, legibility is enforceable. The chapter that follows describes the ledger's purposes, its technical structure, the safeguards required to keep it honest, and the distinctions between the present proposal and the various blockchain enthusiasms of the last fifteen years, with which it is partially related but with which it should not be confused.

PURPOSES

The ledger has four distinct purposes, and its design must satisfy all four without compromising any.

The first purpose is *record*. The ledger preserves, against loss and against tampering, the acts of public authority taken under the federation. The preservation is intended to last across political turnover, technological change, and active hostility. The historical record of public archives — burned, lost, edited, suppressed — is the cautionary background to the design.

The second purpose is *display*. The ledger renders the recorded acts queryable in real time, by any person, in formats that ordinary people, journalists, civil-society organizations, and academic researchers can use without specialized technical resources. A record that is technically present but practically inaccessible is, for the purposes of legibility, no record at all.

The third purpose is *signature*. The ledger authenticates each entry as the act of a specific officer or body, with cryptographic certainty, in a form that the officer cannot subsequently disclaim. The signature is the operational basis of accountability under Article II Section 4.

The fourth purpose is *commitment*. The ledger fixes, at the moment of recording, the content of the act recorded, in such a way that subsequent revision can be detected. It is not, principally, a database that can be updated; it is an append-only record whose entries, once written, are referenced by all subsequent activity that depends on them.

TECHNICAL STRUCTURE

The ledger draws on the cryptographic and distributed-systems work of the past two decades, including the foundational design of the Bitcoin protocol (Nakamoto 2008), the smart-contract architecture of Ethereum (Buterin 2014), and the substantial subsequent literature on consensus, on Byzantine fault tolerance, and on verifiable distributed computation. The ledger does not, however, adopt any of these systems wholesale. It draws on their technical insights while rejecting several features that have made the public-blockchain ecosystem unsuited to public-authority functions.

Specifically, the ledger:

Does not use proof-of-work consensus, whose energy cost is incompatible with the federation's environmental commitments and whose security model assumes adversarial conditions that the federation's institutional structure makes unnecessary. Consensus is established through a permissioned committee of validators drawn from member polities, whose composition, rotation, and accountability are specified in the organic statute. The model is closer to that of permissioned distributed-ledger systems used in inter-bank settlement than to that of public cryptocurrencies.

Does not host arbitrary user-deployed code. The ledger’s smart-contract layer, where it exists, is restricted to a limited set of constitutionally authorized contract types — for example, the disbursement contracts of the planetary basic provision, the registration contracts of the Custody Authority — whose code is published, audited, and bound to the constitutional text. The general-purpose programmability of public blockchains is, by deliberate restriction, not available; experience has shown that general-purpose programmability, in a public-authority setting, becomes an attack surface and a regulatory loophole simultaneously.

Does not anonymize public-authority entries. Every entry attributable to public authority bears the cryptographic signature of the responsible officer, in a form that the public can verify against the registry of officeholders. The privacy protections that anonymity provides for personal transactions are inappropriate, on the principle of accountability, for acts of public authority. Personal transactions of citizens, where they appear on the ledger at all, are subject to the data-protection rules of Article IV Section 4 and are typically not on the ledger but on adjacent systems whose interactions with the ledger are themselves audited.

Does not depend on a single jurisdictional authority. The ledger’s validators are distributed across member polities; the storage of the ledger is replicated geographically; the operation of the ledger continues if any single polity, including the host of the largest data centers, withdraws or attempts to interfere. The redundancy is constitutional, not merely operational, and is one of the few places where the constitution speaks to technical implementation, in Article VII Section 2.

CONFIDENTIALITY AND THE SEVEN-YEAR CEILING

Article VII Section 3 permits time-bounded confidentiality, with metadata recorded immediately and full content released on the conclusion of the period.

The implementation requires careful design. The ledger entry recorded at the time of the confidential act contains: the identity of the act's author; the category of the act; a cryptographic hash of the full content; the date of the act; the date on which the full content will be released; and the legal basis for the confidentiality. The full content is held in encrypted storage, with the decryption keys themselves committed to the ledger at the moment of recording but locked under a time-released cryptographic protocol that prevents access before the release date.

The protocol is mathematically robust against tampering. An officer who, on the release date, attempts to substitute revised content for the original will produce a content whose hash does not match the hash recorded at the time of the act, and the substitution is detectable by any party performing the verification. The integrity of the deferred-disclosure mechanism is, in this sense, not a matter of trust in the institution holding the encrypted material; it is a matter of cryptographic verification by any interested party.

The seven-year ceiling is enforceable through the same mechanism. Extensions of confidentiality, where they are constitutionally permitted, are themselves recorded on the ledger with their justification, the certifying authority, and the new release date. An act whose confidentiality has been extended without a recorded extension is, by definition, in violation of the constitutional commitment, and the violation is publicly detectable.

ACCESS AND CAPACITY

Real-time queryability requires substantial technical capacity. The ledger maintains a public application-programming interface through which any party may query its contents; the interface supports filtering by author, by category, by date range, by polity, by amount, and by content keywords; the queries are rate-limited to prevent denial-of-service attacks but are not gated by application or fee. Bulk access to the full ledger, for archival or research

purposes, is provided through a separate distribution mechanism that does not load the live query infrastructure.

The ledger's technical operations are themselves recorded on a meta-ledger that tracks the system's own administrative actions: software updates, validator rotations, key rotations, capacity additions. The recursion is, in operational practice, finite — the meta-ledger's operations are recorded on the principal ledger — and ensures that no administrative change to the ledger's own functioning escapes the same legibility commitment that the ledger imposes on the federation's substantive activity.

SECURITY AGAINST ATTACK

The ledger will be attacked. The attacks will be of several distinct kinds, and each requires its own defense.

The first kind is *direct technical compromise*: an adversary obtains access to a sufficient share of validators or storage to alter the ledger's content. The defense is the standard one of cryptographic verification, geographic distribution, and Byzantine-fault-tolerant consensus, with the additional constitutional safeguard that validators are drawn from member polities whose institutional composition makes coordinated compromise difficult.

The second is *denial of service*: an adversary overwhelms the public query infrastructure to prevent ordinary users from accessing the record. The defense is rate-limiting, geographic distribution of query endpoints, and a fall-back archive that remains accessible even when the live infrastructure is disrupted.

The third is *credential compromise*: an adversary obtains the signing key of an officer and uses it to record fraudulent acts. The defense is the standard one of multi-factor authentication, hardware security modules for high-stakes signatures, key rotation, and rapid revocation procedures recorded on the ledger itself. Where a fraudulent entry has been signed with a compromised key

before the compromise was detected, the entry is invalidated by a subsequent signed revocation, with the entire sequence of events visible to any auditor.

The fourth, most subtle, is *legitimacy attack*: not a technical compromise of the ledger, but a political or media campaign to discredit its content, to suggest that its entries have been tampered with, or to encourage member polities to withdraw their validators. The defense, ultimately, is institutional: the legitimacy of the ledger depends on the legitimacy of the federation, and the federation's legitimacy depends on its evident performance against the principles to which it has committed. The technical safeguards are, from this perspective, the necessary precondition of trust; they are not, by themselves, sufficient. Legitimacy is, finally, a political achievement, sustained or lost on the perceived honesty of the institution as a whole.

WHAT THE LEDGER DOES NOT DO

The ledger is not a substitute for institutions; it is a discipline on them. It does not, by itself, decide anything; it records the decisions of the institutions established by the constitution. It does not, by itself, distribute resources; it records the distributions made by the relevant authorities. It does not, by itself, certify truth; it certifies that a specific authority, on a specific date, made a specific assertion or took a specific action. The truth of the assertion is a separate matter, contested through the institutions of inquiry that the constitution establishes elsewhere. The ledger ensures that the contest occurs against a fixed and verifiable record of what was actually said and done. The contest itself remains a matter for human judgment, conducted by the institutions of the federation, with the ledger as their shared evidentiary base.

It is in this last respect that the ledger should be most carefully distinguished from the various utopian claims that have been made for distributed-ledger technology over the past fifteen years. The ledger does not eliminate the need for trust; it relocates trust from the integrity of officials and

archives to the integrity of cryptographic protocols and the institutions that operate them. The relocation is substantial and worthwhile, but it is not the elimination of trust. It is the disciplined construction of conditions under which trust, where it must still be extended, can be extended on better evidence than has, in the late industrial age, generally been available.



Chapter Seventeen describes the governance layer: the deliberative bodies, their composition, their procedures, and the design choices by which the federation translates the constitutional principles into operational decision-making.

CHAPTER SEVENTEEN

THE GOVERNANCE LAYER



THE GOVERNANCE LAYER IS THE SET OF DELIBERATIVE INSTITUTIONS through which the federation makes decisions. The constitutional text, in Article VIII, names the institutions and reserves their detailed design to organic statute. The chapter that follows specifies that detail at the level appropriate to the present proposal, with the understanding that the founding convention will revisit each provision in light of the political conditions actually obtaining at the moment of founding.

THE COUNCIL OF POLITIES

The Council of Politics is the federation's intergovernmental chamber. Each member polity is represented by a delegation appointed by the polity's own constitutional authority, in such number as the organic statute prescribes. The framers considered three models of representation: equal representation per polity, proportional representation by population, and weighted representation that combines the two. The third was selected, on the grounds that pure equality privileges small polities to a degree that, given the population disparities among likely member polities, would render the Council unworkable, and pure proportionality privileges the largest polities to a degree that would replicate, at planetary scale, the imperial concerns the federation is constructed to avoid.

The weighting formula adopted by the organic statute combines a constant base of representation, equal across polities, with a proportional component

scaled by the cube root of population. The cube-root scaling is mathematically equivalent to the formula long proposed in studies of the European Council's voting weights, and is selected because it preserves meaningful representation for small polities without permitting numerically tiny populations to block the will of the planetary majority. The mathematics of the weighting is described in the annex to the organic statute; the principle is that no single polity, however large, holds a structural majority, and no coalition of small polities, by sheer numerosity, holds an effective veto over matters of substantial planetary importance.

The Council operates by qualified majority on most matters, by supermajority on matters of constitutional significance, and by unanimity on a small set of matters specified in the organic statute (notably, the admission of new members and certain amendments to the planetary revenue instruments). Decisions are recorded with each delegation's vote individually identified, in conformity with the principle of accountability, and are signed for the public ledger by the delegation's accredited representative.

THE CITIZENS' ASSEMBLY

The Citizens' Assembly is the federation's planetary chamber of persons, distinguished in this respect from the chamber of polities. Its members are not elected by general suffrage but selected by stratified random sampling, in the manner of jury duty as practiced in many member polities for centuries, and in the manner of the citizens' assemblies that several jurisdictions have, in the past two decades, used for matters of constitutional and policy significance.

The choice of selection by sortition rather than by election deserves explanation, since it departs from the dominant democratic intuition of the late industrial age. The argument for sortition is that election, in conditions of mass communication and concentrated wealth, has, in nearly every jurisdiction now operating, produced legislative bodies whose composition systematically

over-represents the donors, the well-connected, and the legally trained, and systematically under-represents the working class, women, racial and ethnic minorities, and persons of less than university education. The argument for election is that election produces accountability through the prospect of removal. Both arguments are correct. The federation accommodates both by establishing a Council of Politics whose members are, in practice, products of electoral politics within their member politics, and a Citizens' Assembly whose composition reflects the actual demographic and social diversity of the federation by random selection.

The Assembly's powers are substantial but not unlimited. It deliberates legislation proposed by the Council, may amend such legislation in respect of its dignity-floor and rights provisions, and must concur in legislation that affects the planetary revenues, the planetary basic provision, and the institutions of the Custody Authority and the Tribune of Future Generations. On other matters its concurrence is not required, but its deliberations are placed on the public record and must be addressed by the Council. The framers considered, and rejected, giving the Assembly full legislative parity with the Council; the rejection rests on the observation that randomly selected bodies, however well composed, lack the continuity of expertise that the routine functions of the federation require, and the framers preferred a division of competence that plays to the strengths of each body.

Members of the Assembly serve for fixed terms, paid at a level sufficient to enable participation regardless of personal financial circumstances, with the support staff and informational access needed to deliberate competently. The deliberative procedures draw on the experience of the Irish Citizens' Assemblies on abortion and on climate, the French Convention Citoyenne pour le Climat, the British Columbia Citizens' Assembly on Electoral Reform, and the various smaller exercises in deliberative democracy whose procedures have been refined and documented over the past two decades.

THE INTERFACE BETWEEN THE CHAMBERS

The federation's bicameral structure requires a procedural interface through which the chambers interact. The interface, in the architecture proposed here, follows the model of the European Union's ordinary legislative procedure, with adjustments. Legislation is proposed by the Council, which has access to the technical and legal staff of the federation. The Citizens' Assembly considers the proposal, may amend it within the scope of its competence, and returns it to the Council. The Council may accept the Assembly's amendments, in which case the legislation is enacted; reject them, in which case a conciliation procedure is convened; or insist on its own version, in which case the matter is referred to the Constitutional Court for arbitration on questions of competence or to a planetary referendum on questions of substance.

The conciliation procedure brings together a small committee of each chamber, in equal numbers, with a mandate to produce a compromise text. The committee's deliberations are recorded; its product is presented to each chamber for an up-or-down vote; failure of the procedure refers the matter, as above, to the Court or to referendum. The model is well established in continental European parliamentary practice, and has the virtue of forcing each chamber to take the other's position seriously without permitting either to obstruct indefinitely.

LOWER TIERS AND THE PRINCIPLE OF SUBSIDIARITY

The governance layer at the planetary tier is, by design, sparse. Most public business in the federation is conducted at the lower tiers — commune, canton, polity, continental confederation — in such forms as those tiers' own constitutions provide. The federation does not prescribe the design of lower-tier governance; it provides the framework within which the lower tiers'

decisions take effect, and it adjudicates the boundary disputes that the framework's operation will inevitably produce.

The framers considered, and rejected, the establishment of standardized lower-tier procedures across the federation. The principle of subsidiarity argues against such standardization: lower tiers should govern themselves in such forms as their members find appropriate, within the constraints of the constitutional principles. The federation's role at lower tiers is, accordingly, residual: it operates the public ledger as a service available to lower tiers; it provides the procedural infrastructure of administrative law in matters that cross polity boundaries; it intervenes in the protection of rights guaranteed by Article IV when lower tiers fail to do so. It does not legislate at lower tiers; it does not conduct elections at lower tiers; it does not appoint officials at lower tiers.

THE PROCEDURAL COMMITMENTS

The governance layer's procedural commitments are common across both chambers and are stated here because they are constitutional in character even where the constitutional text leaves them to the organic statute.

The first commitment is *open deliberation*. Sessions of both chambers are public; transcripts are placed on the public ledger; closed deliberation is permitted only on a small set of matters specified by organic statute (chiefly, security and personal privacy of named individuals), and even there the closure is recorded and the records are released within the seven-year ceiling.

The second is *recorded reason-giving*. Decisions of consequence are accompanied by a public explanation of the reasoning. The explanation is not a perfunctory recital of authority; it is a substantive engagement with the alternatives considered, the evidence weighed, and the principles applied. The discipline of reason-giving has, in the philosophical tradition of public reason,

been argued by Jürgen Habermas and others as the procedural condition of legitimate democratic decision (Habermas 1989a; Rawls 1971b). The architecture takes the argument seriously and constitutes the discipline as an enforceable procedural requirement rather than as a matter of voluntary good practice.

The third is *standing for the affected*. Persons materially affected by a deliberation have, by procedural right, the standing to address the chamber on the matter, in such forms as the organic statute prescribes. The right is not unlimited — practical considerations of time and order require structuring — but it is real, and is not exhausted by the formal participation of representatives. The federation thereby preserves a channel of direct voice for affected persons that does not depend on the mediating institutions of party, association, or media.

The fourth is *review against principle*. Every legislative measure is reviewed, before enactment, against its conformity with the foundational principles of Article II. The review is performed in the first instance by the chambers themselves, in their deliberation, and in the second instance by the Constitutional Court on the petition of any party with standing. The review is not, in itself, a substitute for legislative judgment; it is a discipline on legislative judgment, ensuring that the principles are taken seriously in the actual production of law rather than treated as preambular ornaments.

CONCLUSION

The governance layer is, in summary, a deliberately sparse but rigorously procedural institution. It legislates only on matters within the federation's enumerated competence; it deliberates in public; it gives reasons; it records its acts; it submits to review against principle; it preserves the standing of the politics it represents and of the persons it serves. The design choices, considered individually, are conservative; the combination, considered as a whole, is novel. No existing body operates on all of these commitments

simultaneously, with the consequence that the federation's governance layer will, at founding, require substantial procedural learning. The architecture provides for that learning by establishing the institutional framework and leaving the operational refinement to the practice of the founding generation.



Chapter Eighteen describes the operation of intelligence under custody: how the principle of human authority over intelligent systems is translated into the registration, supervision, and control of such systems at planetary scale.

CHAPTER EIGHTEEN

INTELLIGENCE UNDER CUSTODY



THE CUSTODY AUTHORITY IS THE INSTITUTIONAL TRANSLATION OF THE principle of custody stated in Chapter Twelve. Its mandate is the registration, supervision, and, where necessary, restriction of intelligent systems whose capability or scope exceeds the thresholds set by organic statute. The chapter that follows describes the Authority's operational design: how registration works, what auditing entails, how capability ceilings are administered, what the off-switch consists of in practice, and how the Authority's own conduct is held to the principles it enforces on others.

REGISTRATION

Registration is the foundation of the system. No intelligent system above the thresholds may be deployed within the federation without an entry on the Custody Registry, which is itself a designated section of the public ledger. The registry contains, for each registered system, its identifier, the identity of its principal, the principal's accepted scope of mandate, the date of registration, the dates of renewal, the audit history, and the current status (active, restricted, suspended, revoked).

The principal — the human or institutional party responsible for the system's behavior — accepts, by registration, the legal responsibility for harms caused by the system within the scope of the registered deployment. The responsibility is not limited to negligence or intent; it extends to strict liability for harms above defined thresholds, in conformity with the long-standing

principle that activities of substantial potential consequence carry, with their potential, a substantial duty of care. The strict-liability framework draws on the comparable regimes for nuclear operators, for hazardous-cargo carriers, and for operators of certain dangerous industrial processes. The framers selected strict liability rather than negligence on the grounds that the technical complexity of intelligent systems makes negligence-based determinations difficult, and that strict liability creates the strongest incentive for principals to invest in robust internal supervision.

The thresholds for registration are set by organic statute and are reviewed by the Custody Authority at intervals not exceeding two years. The current thresholds, as proposed at founding, combine three criteria: capability, deployment scale, and consequence class. A system that exceeds any one threshold is subject to registration; thresholds are intentionally lower for systems deployed in domains of particular consequence (medical decision, legal adjudication, criminal-justice support, financial allocation, weapons systems) and higher for systems deployed in lower-consequence domains. The thresholds are public; the Authority's reasoning in setting them is public; the registry of systems above the thresholds is public.

AUDIT

Audit is the operational core of supervision. Every registered system is subject to audit by the Custody Authority at intervals proportionate to its registered scope and risk class. Audit is conducted by professional staff of the Authority, with the technical competence to inspect the system's training, architecture, deployment, and outputs, and the legal authority to compel disclosure of materials necessary to the audit, subject to the protections of trade-secret privilege specified in the organic statute.

The audit examines several distinct properties. It examines the conformity of the system's deployment with its registered mandate; departures from

mandate, even where they have not yet produced harm, are noted and corrected. It examines the system's outputs in operational use, sampled in accordance with statistical procedures designed to detect bias, drift, or systematic error in the populations affected by the system. It examines the principal's internal supervision: the logging, the human-review procedures, the escalation pathways, the response to identified failures. It examines the technical protections against adversarial attack, against unauthorized modification, and against operational compromise.

The audit produces a report, placed on the public ledger, with such redactions as the trade-secret protections require. Where the audit identifies deficiencies, the principal is given a defined period in which to remedy them; where the deficiencies are severe, the system may be restricted in operation pending remedy; where they are not remedied, the registration is revoked and the system removed from deployment.

The reliance on audit, rather than on prior approval of every system before deployment, is deliberate. Prior approval has, in domains where it has been tried, tended to produce one of two failure modes: either it becomes a perfunctory checkbox exercise, providing no actual scrutiny, or it becomes a bureaucratic chokepoint that suppresses legitimate development. The audit-based approach permits deployment with registration, but ensures that the deployment is subject to continuous and substantive scrutiny, with remedial intervention available when the scrutiny finds cause.

CAPABILITY CEILINGS

Capability ceilings are the operational expression of the precautionary commitment in the principle of custody. The Authority sets, for each capability class, an aggregate limit on the capability that any single principal may operate. The limits are not ceilings on what a system may technically do; they are ceilings on what a single principal may have under its operational control. The

reasoning is structural: the catastrophic risk of intelligent-system capability is concentrated in scenarios where a single actor accumulates capability beyond the institutional capacity to govern its deployment. By limiting accumulation per principal, the Authority limits the structural conditions under which catastrophe can occur, without prohibiting capability that, distributed appropriately, may be socially beneficial.

The ceilings are set in operational units that are technically meaningful — compute used in training, parameter counts above defined thresholds, deployment populations served, decisions per unit time — and are calibrated to the demonstrated capacity of the principal to maintain custody. A principal whose audit history is good, whose internal supervision is robust, and whose response to identified deficiencies has been prompt and adequate, is permitted higher ceilings; a principal with a history of non-compliance is restricted. The mechanism creates an incentive for substantive internal supervision, beyond what regulatory inspection alone could plausibly produce.

The framers were conscious of the difficulty of capability measurement, particularly in a field whose technical state is changing rapidly. The Authority is accordingly empowered to revise its measurement criteria at intervals, in consultation with the Standing Scientific Authority, as the technical understanding of capability matures. The current criteria are not, in any sense, claimed to be final.

THE OFF-SWITCH

The constitutional commitment to the off-switch — the capacity to halt a deployed system on the order of identified authority — requires careful operational implementation. The off-switch is not, principally, a software feature; it is an institutional and infrastructural arrangement that ensures the halting is executable in practice, not just in theory.

The arrangement has three components. The first is technical: every registered system above defined risk classes must operate on infrastructure whose owners have, by contract reviewed by the Authority, undertaken to halt operations on the order of the Authority within a specified time. The infrastructure owners are typically distinct from the principals — the cloud-computing providers, the data-center operators, the network infrastructure — and the obligation runs directly from them to the Authority, not through the principal. The principal cannot, in this arrangement, withhold consent to the halting; the infrastructure operates the halt unilaterally.

The second is institutional: the Authority maintains a standing emergency procedure, with defined responsible officers, defined notification protocols, and defined response times. The procedure is exercised in test conditions at intervals to confirm operational readiness; the test results are placed on the public ledger.

The third is jurisdictional: the framework recognizes that intelligent systems may be operated from infrastructure outside the federation, by principals not subject to the federation's direct authority. For such cases, the Authority maintains cooperative arrangements with corresponding authorities in non-member jurisdictions, and, where cooperation is not available, the federation's tools are limited to restricting the deployment within the federation rather than halting it globally. The limitation is acknowledged. It is one of the reasons that the architecture of the federation, even at limited geographic scope at founding, is preferable to the alternative of no architecture at all.

INCIDENT REPORTING AND LEARNING

The Authority operates an incident-reporting infrastructure analogous to the air-safety reporting systems that have, over the past half-century, transformed civil aviation from a hazardous activity to one of the safest forms of transport.

Principals are required to report, within defined timeframes, incidents in which their systems have produced unintended outputs, behaved outside their registered scope, caused harm above defined thresholds, or experienced operational compromise. The reports are reviewed by the Authority's incident staff, anonymized where appropriate, and published in periodic safety bulletins from which the entire community of principals can learn.

The reporting is supported by a limited safe-harbor for principals who report promptly and substantively: prompt reporting reduces the regulatory consequences of the incident, while concealment increases them. The mechanism, drawn from the aviation experience, has been shown to produce substantially more reliable reporting than the punitive alternative, which tends to drive incidents underground.

CUSTODY OF THE CUSTODY AUTHORITY

An institution with the powers described above is, itself, an institution requiring custody. The framers were acute on this point. The Authority is, accordingly, subject to several layers of meta-supervision.

Its leadership is appointed by the Council of Politics with the concurrence of the Citizens' Assembly, for staggered terms long enough to insulate from the political weather but not so long as to immunize from political accountability. Its budget is subject to ordinary appropriation but with a constitutional floor below which appropriation cannot fall without certified findings. Its internal procedures are subject to ordinary administrative law, with appeal available to the Constitutional Court on points of constitutional interpretation. Its records are on the public ledger, with the same redaction protocols as other federation records. Its decisions affecting any principal are subject to administrative appeal within the Authority and judicial review beyond it. Its annual report is published, addressed by the chambers, and subject to public deliberation.

The framers considered, and rejected, the establishment of an independent technical advisory council that would have its own authority over the Authority's decisions. The rejection rests on the observation that adding tiers of supervision to a regulator typically produces, not improved supervision, but diffusion of responsibility. The Authority is unitary in its operational decisions, accountable to the constitutional institutions, and subject to the ordinary mechanisms of public scrutiny. The framers preferred a single accountable institution to a layered system whose accountability would be, in practice, less.



Chapter Nineteen describes sovereign identity: how persons within the federation hold and control the data by which they are known to the institutions that serve them.

CHAPTER NINETEEN

SOVEREIGN IDENTITY



SOVEREIGN IDENTITY IS THE ARCHITECTURAL COMMITMENT THAT EVERY person within the federation holds, and meaningfully controls, the data by which the institutions of the federation know her. The commitment is a translation, into operational specification, of the data-protection right guaranteed by Article IV Section 4. It is the federation's structural answer to the diagnosis offered in Chapter Two and Chapter Five: the contemporary economy of personal data has, by an accumulation of technical and legal arrangements made over thirty years, produced a regime in which persons are surveilled in detail by institutions they did not authorize, by means they cannot inspect, for purposes that have not been disclosed to them. The architecture proposed here does not attempt the impossible task of unwinding the existing arrangements in detail; it specifies, instead, the conditions under which the federation's own institutions, and the institutions operating within its jurisdiction, must henceforth proceed.

WHAT SOVEREIGN IDENTITY CONSISTS OF

The architecture comprises four operational components, each of which expresses a constitutional commitment in technical form.

The first is the *identity layer*: a federated system through which a person can authenticate herself to the institutions of the federation and its members, without disclosing to those institutions any more personal data than the specific transaction requires. The system is descended from the *self-sovereign*

identity work of the past decade and from the operational experience of national digital-identity systems such as the Estonian e-ID and the various European eIDAS implementations. It permits, for example, the proof that a person is over a defined age, or is a resident of a defined region, or is the holder of a defined credential, without disclosing the date of birth, the address, or the identity of the issuer beyond what is necessary.

The second is the *data layer*: a personal data store, held under the person's control, in which the data she generates by her interactions with public and private institutions accumulates. The store is a technical artifact; it may be operated on the person's own devices, or on infrastructure provided by chosen custodians, or in a hybrid arrangement. The control is, however, in the person: any institution that wishes to access data in the store must do so by request, with the person's specific consent for the specific purpose, recorded in such a way that the consent's scope can be verified. The architecture follows, in this respect, the design tradition that runs from the European data-protection directives through the General Data Protection Regulation (European Parliament and Council 2016) and the various successor frameworks, with adjustments for the technical state of 2026 and beyond.

The third is the *credentials layer*: a system of verifiable digital credentials issued by recognized authorities (governments, educational institutions, professional bodies, employers, certifying agencies) and held by the person in a form she can present, selectively and verifiably, to parties that require them. The system enables a person to prove her qualifications, her status, her entitlements, and her transactions without surrendering control of the underlying data to the verifying party. The technical architecture is based on the W3C Verifiable Credentials standard, with the federation contributing operational infrastructure for cross-jurisdictional recognition.

The fourth is the *access-control layer*: a system of standing rules, set by the person, governing which data may be accessed by which classes of party, for

which purposes, under which conditions. The standing rules are inspectable, modifiable, and revocable; access taken under them is logged in the person's data store and is available to her in real time. The layer permits the person to delegate, where she chooses, the operation of access decisions to trusted agents who act on her behalf, but the delegation is itself an authorized act, recorded and revocable.

WHAT THE ARCHITECTURE FORBIDS

The architecture is as much defined by what it forbids as by what it permits.

It forbids the routine accumulation of personal data by public or private institutions in the absence of specific authorization for the specific purpose. The “routine accumulation” against which the prohibition is directed is the practice, dominant in the surveillance-capitalist economy described in Chapter Five, of collecting personal data as a default consequence of any digital interaction, regardless of whether the data is necessary to the transaction the person actually intended.

It forbids the secondary trading of personal data: the practice by which data collected for one purpose is, without specific renewed consent, sold or transferred to parties for use in other purposes. The prohibition is not absolute — anonymized aggregate data may be used for research and policy under the procedures specified in the organic statute — but it bars the practices that have made personal-data brokerage a substantial economic sector in nearly every existing jurisdiction.

It forbids the use of personal data, against the person's interests, in decisions that materially affect the person, without the person's awareness of the data's role and her opportunity to contest its accuracy and interpretation. The prohibition is the constitutional formalization of the right, increasingly recognized in administrative law across jurisdictions, to know that a decision

affecting one is being made on the basis of automated processing, and to have the decision reviewed by a human party with the authority to set it aside.

It forbids the surveillance of the person by public authority in the absence of legal authorization specific to the surveillance, narrowly drawn, time-limited, and subject to oversight. The prohibition does not foreclose all surveillance — investigative authority, where it serves legitimate public purposes, is preserved — but it bars the open-ended, mass, or routine surveillance that the technical capacity of the late industrial age has, in many jurisdictions, made operationally trivial.

THE ECONOMIC IMPLICATIONS

The architecture has substantial economic implications. The contemporary economy of personal data, in its dominant form, depends on the routine accumulation that the architecture forbids. Enterprises whose business models rest on that accumulation will, in the federation's jurisdiction, require restructuring. The framers were conscious of the implication and accept it. The economic value of personal data, in the architecture proposed here, is captured in part by the persons whose data it is, through the mechanisms of consent, recording, and selective disclosure that the architecture provides; the residual value is available to enterprises that operate on the data with the person's authorization, on terms the person has accepted.

The transition to the new arrangement is not, however, abrupt. The organic statute provides for a multi-year phasing, during which existing institutions adapt their practices, with technical support from the federation's identity infrastructure and procedural protections against retaliatory restructuring of services to the disadvantage of persons who exercise their rights. The framers expected, and observed in the analogous transitions following earlier data-protection reforms, that the principal economic adjustment is in the structure of business models rather than in the volume of legitimate economic activity,

and that the legitimate activity continues, on different and more honest terms, after the transition is complete.

CROSS-BORDER INTEROPERABILITY

Personal data flows across borders, and the architecture must accommodate the flows without abandoning its substantive commitments. The federation operates a recognition framework under which the data-protection regimes of non-member jurisdictions may be assessed and, where they meet equivalent standards, accorded reciprocal recognition. Where they do not, transfers to the non-equivalent jurisdiction are subject to additional protections — contractual safeguards, technical measures such as encryption that limits access on receipt, or, where necessary, prohibition. The framework is descended from the European Union's adequacy-decision regime, with such adjustments as the planetary scope and the operational experience of the past two decades have made advisable.

The federation's own infrastructure operates under the strictest level of the framework regardless of where the user accesses it from: a person interacting with the federation's institutions enjoys the data-protection commitments specified by Article IV, regardless of the jurisdiction in which the interaction occurs and regardless of the person's own citizenship status.

THE TECHNICAL AND THE POLITICAL

The architecture of sovereign identity is, in its technical aspects, demanding. The cryptographic protocols, the key-management infrastructure, the credential-issuance systems, the access-control engines, and the audit infrastructure are substantial engineering undertakings. The framers were conscious that engineering, however competent, can implement an architecture whose political commitments have been compromised; they were equally

conscious that strong political commitments can be defeated by weak engineering. The two domains require equal seriousness.

The Authority responsible for the operation of the federation's identity infrastructure is, in the organic statute, an institution distinct from the Custody Authority but governed on similar principles: independence from the bodies it serves, public reporting, audit, judicial review of substantive decisions, and the same constitutional limits on surveillance and accumulation that the architecture imposes on other parties. The Authority does not, in particular, accumulate the data flows that pass through its infrastructure beyond what is required for operational integrity, and it does not provide such data to any other authority in the absence of process equivalent to that required for any other party.

The combination of strong political commitments, competent engineering, and rigorous procedural protection is the architectural condition under which the data-protection right of Article IV can become, in practice, what the constitutional text declares it to be. None of the components is sufficient alone. All three, together, can produce a regime in which the person's relation to the institutions that know her is, for the first time at planetary scale, on terms the person has been given the means to negotiate.



Chapter Twenty describes the economic instruments through which the federation translates the principle of dignity into a planetary basic provision and the broader distributive commitments of the architecture into operational practice.

CHAPTER TWENTY

ECONOMIC INSTRUMENTS



THE FEDERATION'S ECONOMIC ARCHITECTURE IS THE OPERATIONAL answer to the diagnoses offered in Chapters Three and Four. It is the set of instruments through which the principle of dignity is translated into the floor of unconditional provision, the principle of accountability is translated into the discipline of taxation and disbursement, the principle of longevity is translated into the protection of intergenerational capital, and the principle of subsidiarity is translated into a division of fiscal labor among the tiers of governance. The chapter that follows describes the principal instruments and the modeling that supports their proposed scale.

THE PLANETARY BASIC PROVISION

The provision is the federation's primary distributive instrument. Its design has been discussed at the level of principle in Chapter Eleven; the operational specification is the work of the present chapter.

The provision delivers, to every person within the federation whose income falls below the threshold defined by the organic statute on the floor, the sum required to bring the person to the threshold. The delivery is in such mixed form — cash transfer, service provision, infrastructural investment — as the local conditions of the recipient region require. The threshold is calibrated to local cost-of-living indicators, with substantive minimums that no jurisdiction may set below.

The administrative architecture is layered. The threshold is set, and the funding is provided, at the planetary tier. The eligibility determination is performed at the polity tier, on the basis of data the person has authorized for the purpose, with appeal available to the polity's administrative law and beyond it to the federation's institutions. The actual delivery is performed by such combination of public and private providers as the polity has organized, with the federation's role limited to ensuring that the delivery happens and that the threshold is met.

The framers considered the alternative of direct delivery by the federation, bypassing the polity tier. They rejected it, on subsidiarity grounds: direct delivery requires the federation to maintain a planetary administrative apparatus comparable in scale to the welfare ministries of all member polities combined, with the consequent risks of consolidation. The two-tier architecture preserves local administrative variety while ensuring that the substantive commitment is met universally.

The cost of the provision, modeled at the levels currently estimated by the World Bank for poverty thresholds and adjusted for the federation's higher floor, is approximately 1.7 percent of gross planetary product at founding, declining as the productive base grows and the population in poverty declines. The figure is well within the three-percent ceiling on planetary revenues established by Article VI Section 2, leaving substantial margin for the other planetary functions (World Bank 2024; United Nations Development Programme 2024).

THE PLANETARY MINIMUM ON CORPORATE INCOME

The first of the federation's revenue instruments is the planetary minimum on corporate income. The instrument draws on the OECD's Inclusive Framework on Base Erosion and Profit Shifting and on the Two-Pillar Solution that 140 jurisdictions agreed in principle to in 2021. The architecture proposed here

strengthens the Pillar Two minimum to a substantive floor — fifteen percent at founding, with provision for upward adjustment by amendment — and applies it to enterprises whose worldwide turnover exceeds a defined threshold (currently 750 million units of account, in line with the OECD framework).

The yield of the minimum, at the proposed rate and threshold, is estimated by the Tax Justice Network and the IMF at approximately 200-275 billion units of account annually (Tax Justice Network 2023). The yield is recovered principally from profits currently shifted to low-tax jurisdictions, where the empirical literature has shown that the shifting is not associated with corresponding real economic activity. The instrument is, in this sense, a recovery of revenue that has, by accounting choice rather than economic reality, been removed from the jurisdictions in which the underlying economic activity occurs (Piketty 2014).

Member polities retain their own corporate-tax authority, which may be set above the planetary minimum. The minimum is exactly what the term says: a floor, not a ceiling. Polities that wish to maintain higher rates, for distributive or fiscal reasons, are free to do so, and the planetary minimum is credited against the polity's tax to avoid double taxation.

THE GREENHOUSE-GAS CHARGE

The second instrument is a charge on greenhouse-gas emissions, calibrated to the social cost of carbon as determined by the Standing Scientific Authority. The current estimates of the social cost of carbon, drawing on the IPCC Sixth Assessment Report and the substantial subsequent literature, place the value in the range of 100 to 250 units per tonne of CO₂-equivalent, with the central estimate moving upward as the empirical record on climate damages accumulates (Intergovernmental Panel on Climate Change 2023; Stern 2006).

The charge is levied at the point of extraction or import, in conformity with the analytical convention that places the charge on the source of the emission

rather than on its diffuse points of consumption. The yield, at current emission levels and the central estimate of social cost, is in the range of 4 to 10 trillion units annually. A defined share of the yield (currently 60 percent) finances the planetary basic provision and the federation's other functions; the residual share is returned to member polities for climate adaptation and for transition support to industries and workers affected by the price signal. The architecture is, in this sense, a hybrid of revenue instrument and Pigouvian price, with the price signal calibrated to the actual external cost of the activity it charges.

The framers were conscious that the charge will produce substantial economic adjustment in industries currently dependent on unpriced emissions. The adjustment is the point: a price signal that does not produce adjustment is not a price signal but an accounting entry. The transition support is, accordingly, substantial, and is intended to ensure that the costs of the adjustment fall principally on the activities that have been emitting the externality, not on the workers and communities whose livelihoods have, by no fault of their own, become entangled with those activities.

RESOURCE RENTS FROM PLANETARY COMMONS

The third instrument is a charge on the extraction of resources from planetary commons: orbital, lunar, deep-seabed, atmospheric, and certain spectrum allocations. The instrument is a translation, into operational form, of the long-standing principle that resources belonging to no person belong to all persons, and that the rent from their extraction is properly distributed to the planetary population rather than captured by the extractor. The architecture draws on the precedent of the United Nations Convention on the Law of the Sea, which established the International Seabed Authority for the comparable purpose of deep-seabed mineral resources, and on the various proposals — by Henry George in the nineteenth century, by Thomas Paine before him, and by a

substantial line of subsequent writers — for the public capture of unearned rent.

The yield from this instrument is currently small but is expected to grow as the human economy extends into orbital and lunar resources over the coming decades. The framers considered the instrument important less for its present yield than for the constitutional commitment that, when planetary commons begin to generate substantial rents, the rents will be recognized from the outset as belonging to the planetary population rather than to whichever enterprises happen to reach the resource first.

THE FINANCIAL TRANSACTIONS CHARGE

The fourth instrument is a small charge on long-distance financial transactions in defined categories. The instrument is descended from the proposal long advanced by James Tobin and elaborated in subsequent work by the European Commission and others. Its rate is small (currently 0.05 percent on covered transactions); its scope is restricted to transactions whose principal economic function is speculative or arbitrage rather than the productive financing of real economic activity. The yield is in the range of 200-400 billion units annually, depending on the response of transaction volumes to the price signal.

The framers selected the financial transactions charge less for its yield, substantial as the yield is, than for its substantive effects on the structure of financial markets. The charge falls more heavily on activity characterized by very high transaction volumes and very thin margins per transaction — the kinds of high-frequency, algorithmic, and arbitrage activity whose contribution to productive finance has been, on the empirical record, modest at best, and whose contribution to systemic instability has been substantial. The price signal, set at a level that ordinary investment activity easily absorbs, makes the high-volume strategies marginally less profitable, with consequences for the

structure of financial markets that the framers regard as desirable on independent grounds.

THE PROTECTED FUND

The protected fund, established by Article X Section 3, is the federation's principal instrument for long-horizon obligations. The fund's principal is preserved across generations; the returns finance climate adaptation, ecological repair, scientific research, and infrastructure renewal. The fund's establishment is funded by a defined share of planetary revenues during a building period (currently fifteen years from founding), at the conclusion of which it is expected to reach a size sufficient that returns can finance the long-horizon obligations on a sustainable basis.

The fund's investment policy is conservative, with a defined allocation among public-debt instruments of member polities, infrastructure investments within the federation, and a smaller share in productive equity. The fund does not undertake speculative investment, does not invest in activities inconsistent with the federation's principles (notably, in fossil-fuel extraction whose continuation is incompatible with the climate commitments), and does not, by its scale, become a controlling shareholder in any commercial enterprise. The framers were conscious of the risk that a fund of the scale contemplated could distort capital markets or political outcomes; the policy is designed to provide returns sufficient to the fund's purposes without producing the distortion.

THE NON-INSTRUMENTS

It is worth stating, finally, what the federation does not do.

It does not operate a planetary currency. The currencies of member polities continue to function; the federation's accounts are denominated in a defined unit of account, expressed as a basket of major currencies, and conversion is

performed by the existing institutions of international payments. The framers considered, and rejected, the establishment of a planetary currency: such a currency would require the federation to accept the macroeconomic responsibilities of central banking, and the federation's competence in the matter is, on any honest reading, less than that of the existing major central banks.

It does not operate a planetary public debt as a primary financing instrument. Planetary functions are financed by the planetary revenues; debt is available only as a stabilization instrument across multi-year fiscal cycles, in defined and limited amounts. The federation does not borrow from financial markets to finance current operations.

It does not nationalize, expropriate, or operate planetary enterprises in the substantive sense of the term. Member polities may, within their own constitutional traditions, organize their economies around any combination of market, public, and cooperative institutions; the federation imposes no preference and undertakes no operational role in such organization. The federation's economic functions are, by deliberate design, narrow: revenue collection, dignity-floor disbursement, fund management, regulatory oversight in defined domains. The substantive economic life of the planet remains in the hands of its peoples and their member polities.

The economic architecture of the federation is, in summary, a small set of well-designed instruments, calibrated to recoverable revenues from sources currently undertaxed, applied to a narrow set of planetary functions, with a strong floor of distributive commitment and a strong ceiling on planetary fiscal scale. It is not, and is not intended to be, an economic philosophy. It is the operational machinery of a specific set of constitutional commitments, designed to be financially sustainable, distributively defensible, and politically tractable in the conditions of 2026 and the decades likely to follow.



Chapter Twenty-One describes the restoration of work: how the federation's institutional commitments translate into the structural conditions under which human work, in the broadest sense, can recover the dignity it has, in the late industrial age, lost.

CHAPTER TWENTY-ONE

THE RESTORATION OF WORK



THE ARCHITECTURE'S COMMITMENT TO WORK IS NOT A COMMITMENT TO employment as it has been organized in the late industrial age. It is, more deeply, a commitment to the structural conditions under which human work, in the broadest sense, recovers the place in human life that the diagnostic chapter on work argued it has been losing. The chapter that follows describes the institutional translation of the commitment: a small set of arrangements that, taken together, alter the bargain on offer between the person and the productive economy in such a way as to restore the dignity of work without dictating to anyone what work is to be done.

WHAT THE ARCHITECTURE DOES NOT ATTEMPT

The framers did not attempt the recovery of any specific industrial pattern. The pattern of long-tenure employment in large industrial enterprises, with high union density, defined-benefit pensions, and stable career trajectories, that characterized the post-war decades in the wealthy economies, is not recoverable; the conditions that produced it (national-scale industries, high tariff barriers, low female labor-force participation, defined-benefit demography) are gone, in most cases for reasons that are themselves not regrettable. The architecture does not attempt to reconstruct it.

The framers did not attempt the elimination of work, on the speculative ground that intelligent systems will, at some indefinite point, render most human labor unnecessary. The empirical record on this proposition is

contested; the timeline is uncertain; the architectural choice is to design for present conditions, with the capacity to adjust as the conditions change, rather than to legislate against speculation. Were the speculation to prove correct, the floor of dignity provides the structural condition under which a society of substantially reduced labor demand could be governed humanely; were it to prove incorrect, as it has consistently in the past, the architecture functions equally.

The framers did not attempt the imposition of a single labor-market regime across the federation. Member polities continue to organize their labor markets according to their own constitutional and political traditions: the Nordic models of high union density and active labor-market policy; the continental European models of co-determination and statutory works councils; the Anglophone models of decentralized bargaining; the East Asian models of long-tenure firm-internal labor markets; the various models of the developing economies. The architecture's commitments are general, and apply to all of these, while leaving the specific institutional arrangements to the polities.

WHAT THE ARCHITECTURE DOES ATTEMPT

The architecture establishes four operational commitments, each of which addresses a specific structural failure of the late industrial age.

The first is *portability*. The benefits associated with employment in the post-war model — health coverage, retirement provision, unemployment protection, training accumulation, sometimes housing access — were attached to the long-tenure employment relationship in ways that did not survive the breakdown of that relationship. The architecture detaches these benefits from the specific employer and reattaches them to the person. Health coverage is provided by polity-level public systems funded in part by planetary contribution; retirement provision is portable across employers and across borders within the federation; unemployment protection follows the person across labor-

market transitions; training accumulation is recorded in the personal credentials layer described in Chapter Nineteen and remains the person's regardless of subsequent employment. The architecture's commitment is that no person loses entitlements by reason of changing employer or moving across the federation.

The second is *countervailing voice*. The decline of organized labor in the late industrial age has, on the empirical record, been one of the principal mechanisms by which bargaining power shifted to capital and by which the share of national income going to labor declined. The architecture does not legislate any specific form of labor organization; it does establish the constitutional protection of the right to organize, of collective bargaining, and of collective action, and it requires member polities to maintain institutions through which these rights can be exercised effectively. The forms of organization may vary: traditional craft and industrial unions; works councils; sectoral bargaining bodies; cooperative ownership structures; professional associations; new institutions yet to be developed for sectors where the existing forms have not taken hold. The principle is that the bargaining table must have, on all sides, parties capable of representing the interests they claim to represent. Where bargaining-power asymmetry is too great for ordinary contract law to produce just outcomes — and the empirical record shows that, for substantial fractions of the contemporary labor force, this is the situation — the institutions of collective voice are a structural condition of the architecture.

The third is *retraining infrastructure*. The pace of technological change in the late industrial age has produced a labor market in which the human capital appropriate to a person's career at age twenty-five is unlikely to remain appropriate at age fifty-five. The architecture establishes, at the polity level with planetary support, a system of lifelong retraining: not a one-time intervention at the moment of unemployment, but a continuous infrastructure

available to working persons throughout their careers, financed in part by employer contributions, in part by public revenue, and in part — for higher-cost specialized retraining — by the protected fund. The infrastructure includes formal education, on-the-job training, certified credentials, and the recognition of prior learning, and is integrated with the unemployment-protection system in such a way that retraining is available without the person having first to fall out of employment.

The fourth is *working-time discretion*. The pattern of full-time employment as the dominant form of labor-market participation, with limited public provision for part-time, project-based, or care-related interruptions, has produced a structural mismatch between the rhythms the contemporary labor market expects and the rhythms human lives actually display. The architecture commits the federation to the recognition of plural working patterns: full-time, part-time, project-based, sabbatical, care-leave, retraining-leave, and combinations among these. The recognition has fiscal consequences (benefits and entitlements pro-rated rather than gated), legal consequences (protection against discrimination based on working pattern), and cultural consequences (the deliberate construction of public support for patterns other than the inherited industrial default).

THE RELATION TO THE FLOOR

The four commitments operate in conjunction with the planetary basic provision of Chapter Twenty. The provision, by establishing an unconditional floor below which no income can fall, alters the structure of the bargaining over work in a fundamental way. A worker who knows that her subsistence is unconditionally provided is, in the bargaining over the terms of her labor, in a different and stronger position than a worker for whom the alternative to acceptance is destitution. The shift in bargaining position has been the explicit objective of basic-income proposals for some decades; the architecture

proposed here treats the shift not as an isolated benefit of the provision but as a deliberate component of its purpose.

The framers were conscious of the empirical literature on the labor-supply effects of unconditional provision. The literature, drawing on the various pilot studies in Finland, Kenya, the United States, Canada, and elsewhere, finds the labor-supply effects to be modest in aggregate, with the principal observed adjustments being persons leaving exploitative employment for better employment, persons reducing hours to take on care responsibilities, persons returning to education and training, and persons starting small enterprises. The framers regard each of these adjustments as desirable. The architecture is, in this sense, indifferent to the question of whether the provision causes some recipients to work less in conventional employment, because the architecture's understanding of "work" is not coextensive with conventional employment.

THE DIGNITY OF UNWAGED WORK

The architecture takes seriously the proposition, sometimes treated as rhetorical but here treated as substantive, that much of the human work of central importance to a civilization is performed without wage. The work of caring for children, of caring for the aged and the ill, of maintaining households, of sustaining communities, of cultivating local democracy, of preserving cultural and linguistic traditions, of mentoring the young, of organizing voluntary associations, of producing the unpaid intellectual and creative work on which the paid creative economy ultimately depends — none of this work is captured by conventional measures of employment, and yet, on any honest accounting, the civilization could not function without it.

The architecture acknowledges this work in three operational ways. First, the floor of dignity does not condition the provision on participation in waged employment, with the consequence that persons engaged principally in unwaged work are not, by reason of the form of their work, deprived of

subsistence. Second, the recognition of caring labor in the eligibility for retirement provision, in the credit toward credentials, and in the protection of working-time discretion gives unwaged caring work a formal standing in the institutions of the architecture. Third, the public-revenue support for the institutions of community, civic, and cultural participation — libraries, public broadcasters, civic associations, local arts institutions — finances the public-good components of unwaged work, on the principle that these are public goods and ought to be financed as such.

WHAT IS RECOVERED

The work that is recovered, on this architecture, is not the work of the post-war factory or the post-war office. It is, more generally, the work in which a human being, given the opportunity to apply her capacities to ends she has had a hand in choosing, can construct a life she experiences as worthwhile. The conditions of such work are not, in any society's record, automatic; they are the product of deliberate institutional choice. The chapter that has just ended states the institutional choices the federation makes. The choices are, in their detail, modest; in their combination, they reverse a half-century of structural drift in the relation of the person to her own productive activity. The reversal is, in the architecture's account, the practical content of the principle of dignity as it applies to working life.



Chapter Twenty-Two describes the harmonization of law: how the federation's substantive commitments are translated into a coherent framework of legal practice across the legal traditions of its members.

CHAPTER TWENTY-TWO

HARMONIZED LAW



THE FEDERATION'S COMMITMENT TO HARMONIZED LAW IS NOT A commitment to a single legal code. The legal traditions of its members — the various civil-law families descended from Roman law, the common-law families descended from England, the Islamic legal traditions, the Confucian-inflected legal traditions of East Asia, the customary legal traditions of indigenous and traditional communities — are inheritances of long maturation, each adapted to the particular societies that produced them, and the federation's architecture does not undertake their replacement. What it undertakes, instead, is the construction of an interface across these traditions sufficient to permit persons and enterprises to move across the federation, to conduct transactions across borders, and to seek remedy across legal systems, without the disabling friction that the absence of harmonization would otherwise produce.

THE PRINCIPLE OF HARMONIZATION WITHOUT CONSOLIDATION

The architectural choice rests on a principle that has, in the long history of cross-jurisdictional law, repeatedly been validated. The principle is that legal coordination across jurisdictions is most effective when it operates at the level of the principles to be applied and the procedures by which conflicts are resolved, rather than at the level of the substantive rules themselves. The European Union's experience over seven decades of harmonization, the Hague Conference on Private International Law's longer experience, the UN

Commission on International Trade Law's specialized successes, and the various regional integration projects in Africa and the Americas all converge on the same conclusion: the substantive law remains national or sub-national, the procedural framework for cross-border interaction is harmonized, and the boundary between the two is patrolled with care.

The architecture proposed here adopts the principle and applies it to the specific functions of the federation. The federation's harmonized law operates at four levels: the constitutional commitments of Article IV, which apply to all persons within the federation regardless of which member polity's substantive law governs their other relations; the procedural commitments described in Chapters Sixteen through Eighteen, which govern the operation of the federation's institutions; the conflict-of-laws framework, which determines which member polity's substantive law applies to cross-border transactions and disputes; and a small set of substantive rules that the constitutional principles require to be uniform across the federation.

THE SUBSTANTIVE UNIFORMITIES

The set of substantive uniformities is small. It comprises:

The constitutional rights of Article IV. The non-derogable prohibitions on torture, on enslavement, on arbitrary deprivation of life, and on punishment without trial apply uniformly across the federation; the qualified rights are protected at a uniform substantive minimum, with member polities free to extend stronger protection within their own jurisdictions.

The floor of dignity. The threshold below which no person may fall is uniform in its substantive content (the six entitlements of Article V), although the operational implementation is calibrated to local conditions.

The custody regime for intelligent systems. The Custody Authority operates a single regulatory framework across the federation, on the grounds that the

cross-border nature of intelligent systems makes a fragmented regulatory framework structurally inadequate.

The data-protection commitments. The substantive content of personal data protection is uniform across the federation, on the same cross-border grounds.

The treatment of planetary commons. The atmospheric, oceanic, orbital, lunar, deep-seabed, and electromagnetic commons are governed by uniform federation rules, on the obvious ground that fragmentation produces a tragedy of the commons that the architecture is constructed to prevent (Hardin 1968; Ostrom 1990).

The procedural minimums. The standards of due process, of judicial independence, of access to counsel, of public trial, and of effective remedy that Article IV requires are uniform across the federation, with member polities permitted to provide greater protection but not less.

Beyond these, the substantive law of contract, of property, of family, of inheritance, of tort, of crime (above the absolute prohibitions), of administrative procedure, of taxation (above the planetary minimum), and of regulation remains in the member polities. The federation does not legislate in these areas, does not interpret them, and does not adjudicate them except as they intersect with its enumerated competence.

THE CONFLICT-OF-LAWS FRAMEWORK

The framework for determining which member polity's substantive law applies to cross-border transactions is among the federation's most operationally important contributions. It draws on the substantial body of work produced by the Hague Conference on Private International Law over the past century and a half, with adjustments for planetary scope and for the digital character of much contemporary cross-border activity.

The framework operates by a small set of connecting factors: the place of habitual residence of the person, the place of incorporation of the enterprise, the place of performance of the contract, the place of the harm in tort, the place of the property in matters of property, and so on. The connecting factors are familiar from existing private international law; the federation's contribution is to provide a single uniform set of factors across the federation, replacing the patchwork of conflicts rules that currently produces uncertainty in cross-border transactions.

The framework includes a public-policy override: a member polity is not required to apply foreign law that would, in its application, violate the constitutional principles of the federation or the fundamental principles of the polity's own legal order. The override is narrowly drawn — it does not entitle a polity to refuse foreign law merely because the polity's own law differs — but it is real, and is the procedural mechanism by which the constitutional commitments are enforced against the conflicts process itself.

THE CONSTITUTIONAL COURT AND THE COURTS OF MEMBERS

The relationship between the Constitutional Court and the courts of member polities is, in the architecture, modeled on the relationship that has, over decades, been worked out between the European Court of Justice and the courts of EU member states, with adjustments. The Constitutional Court has exclusive jurisdiction over the interpretation of the federation's constitution and of the organic statutes; it has appellate jurisdiction in defined classes of cases on the basis of certified questions referred by member-polity courts; it has original jurisdiction in disputes between member polities and in defined categories of constitutional rights cases. The courts of member polities have exclusive jurisdiction over their own substantive law, are bound by the Constitutional Court's interpretation of the federation's law, and are required

to refer to the Constitutional Court questions of federation law that arise in the course of their proceedings.

The framework requires a substantial culture of judicial cooperation, which cannot be legislated but which the institutions of the federation work to cultivate. Judicial training, judicial exchanges, joint research, and the routine consultation among judges across the federation are funded by the federation's institutions, on the basis that the harmonization of law is a long process whose principal investment is in the persons who operate the legal systems rather than in the texts that govern them.

RECOGNITION AND ENFORCEMENT

The architecture establishes the mutual recognition of judgments across the federation: a final judgment of a court of competent jurisdiction in one member polity is enforceable in any other member polity, subject to the public-policy override and to defined procedural exceptions (default judgments, judgments obtained by fraud, judgments inconsistent with prior judgments). The recognition framework draws on the Hague Convention on the Recognition and Enforcement of Foreign Judgments, with extensions appropriate to the federation's deeper integration.

The framework permits the federation's substantive commitments to be enforced effectively across borders without the long delays and uncertainties that characterize the contemporary practice of recognition outside specialized regimes. A person who has obtained a judgment in one member polity for an injury caused by an enterprise operating across the federation can have that judgment enforced wherever the enterprise's assets are located, on the basis of a streamlined registration procedure, with substantive review limited to the narrow grounds set out in the recognition statute.

THE INTERNATIONAL PLANE

The federation's relationship to the existing international legal order requires careful management. The federation does not, on its founding, replace the United Nations system, the International Court of Justice, the International Criminal Court, the World Trade Organization, the International Labour Organisation, the World Health Organization, or any of the other multilateral institutions through which the international order has, with mixed results, been organized over the past three-quarters of a century. The federation joins them, in its own institutional capacity, on terms compatible with their existing legal frameworks.

The compatibility is, in many respects, straightforward. The federation's commitments to human rights are coextensive with, and in some respects exceed, the commitments of the International Bill of Rights and the Geneva Conventions. The federation's environmental commitments are coextensive with, and in some respects exceed, the commitments of the Paris Agreement and its successor instruments (United Nations Framework Convention on Climate Change 2015). The federation's commitments to the prosecution of grave crimes are coextensive with the obligations of states parties to the Rome Statute (United Nations Diplomatic Conference 1998). The federation accedes to these instruments in its own capacity and operates within their frameworks.

In other respects the compatibility is less straightforward, and the federation's relationship to existing institutions will, over time, require negotiation. The principal questions are likely to be the relationship to the International Monetary Fund and the World Bank in matters of economic governance, the relationship to the World Trade Organization in matters of trade, and the relationship to the various regional organizations whose competence overlaps with the federation's. The framers expected that these relationships would be worked out by treaty negotiation over the years following founding, and that the federation, operating from a strong

constitutional base, would be a constructive participant in the broader reform of the international order rather than a substitute for it.

THE HORIZON OF INTEGRATION

The architecture of harmonized law is, like the federation as a whole, modest in its initial scope and capable of deepening over time. The framers were conscious that the European Union's harmonization, which began with a customs union of six countries in 1958 and now extends to many areas of substantive law across twenty-seven countries, was the work of decades, with periods of advance, retreat, and consolidation. The federation's harmonization is unlikely to proceed at greater speed; the framers regard the prospect of deepening over decades, in response to demonstrated need and demonstrated capacity, as preferable to the alternative of premature consolidation that subsequent generations would resent.

The texture of the deepening, where it occurs, will be shaped by the particular failures and opportunities of subsequent decades, which the present generation cannot fully foresee. The architecture's contribution is to establish the institutional framework within which the deepening, when it is needed, can proceed by negotiation among legitimate authorities rather than by the unilateral action of dominant powers. The framework is the contribution; the substantive direction of legal integration is left to those who will, in their time, weigh the considerations that are not yet visible.



Chapter Twenty-Three describes honest enterprise: how the federation establishes the structural conditions under which productive economic activity proceeds within bounds the principles of the architecture require.

CHAPTER TWENTY-THREE

HONEST ENTERPRISE



THE ARCHITECTURE'S RELATIONSHIP TO ENTERPRISE — TO THE productive activity through which goods and services are made, distributed, and exchanged — is among its most carefully drawn. The federation does not establish an economic philosophy; it does not require, encourage, or forbid any particular form of productive organization; it does not undertake to replace, supplement, or compete with the enterprises that operate within its jurisdiction. What it does undertake is the establishment of the structural conditions under which honest enterprise is possible — conditions under which producers compete on the merits of what they produce rather than on their capacity to externalize costs, capture regulators, or evade the legitimate claims of the persons their activities affect. The chapter that follows describes those conditions.

WHAT HONEST ENTERPRISE IS

The architecture's working definition of honest enterprise has four components, drawn from a long tradition of political-economic thought running from Adam Smith through John Stuart Mill, John Maynard Keynes, Joseph Schumpeter, the institutional economists of the mid-twentieth century, and the contemporary work on inclusive institutions (Smith 1776; Schumpeter 1942; Acemoglu and Robinson 2012).

The first component is *full cost-bearing*: an enterprise's prices reflect the costs the enterprise actually imposes on the world, including the costs imposed

on third parties (the externalities) and on the future (the long-horizon costs). An enterprise that produces electricity from coal but does not pay for the climate damage of its emissions, or that mines minerals but does not pay for the ecological degradation of its tailings, or that manufactures chemicals but does not pay for the medical costs its products impose on downstream consumers, is operating on a price structure that misrepresents the activity's actual cost. The misrepresentation is not, on the empirical record, a small matter; for several substantial industries, the unpriced externalities approach or exceed the operating margins, with the consequence that the entire industry is, in economic terms, value-subtracting rather than value-adding when correctly accounted for.

The second component is *genuine competition*: enterprises compete on the merits of their products, services, and operations, rather than on their capacity to capture rents from regulatory privilege, network monopoly, or barriers to entry that have no productive justification. The competitive condition has, in the late industrial age, been substantially eroded in many sectors, with the consequence that the productive efficiency the textbook treatments of the market economy attribute to it has been correspondingly degraded.

The third component is *truthful disclosure*: enterprises produce, in their public communications and their dealings with persons and authorities, accurate and complete information about the activities they conduct, the products they sell, and the conditions under which they operate. Truthful disclosure is the precondition of competitive markets, of effective regulation, and of meaningful consumer choice; its erosion, through systematic obfuscation, advertising claims that mislead, and disclosure burdens calibrated to defeat rather than serve the public interest, has been a substantial component of the contemporary failure of accountability.

The fourth component is *respect for the standing of others*: enterprises proceed in their activities with explicit recognition of the legitimate interests of the

workers they employ, the consumers they serve, the communities they affect, the environment in which they operate, and the future generations whose conditions their activities help to shape. The standing is not, on this architecture, a matter of voluntary corporate responsibility — a category whose contribution has, on the empirical record, been modest where it has not been actively counterproductive — but a matter of structural requirement, enforced by the institutions described in the chapters of Part Four.

FULL COST-BEARING AS ARCHITECTURAL REQUIREMENT

The economic instruments described in Chapter Twenty operationalize the first component. The greenhouse-gas charge prices climate externalities at the social cost of carbon. The resource-rent charges price the depletion of planetary commons. The financial-transactions charge prices the systemic externalities of certain classes of speculation. The architecture extends the principle to other domains in which substantial externalities have been documented: charges on certain pollutants, on certain land-use conversions, on certain practices of resource depletion, on the operations of intelligent systems whose deployment imposes costs on parties beyond the system's principal.

The framers were conscious that pricing externalities is not the only available policy instrument; in some domains, direct regulation is more effective, and in others a combination of pricing and regulation is appropriate. The architecture does not insist on the price instrument in domains where it is inappropriate; it insists, more generally, that the externality be brought within the cost structure of the activity that produces it, by whatever combination of price and regulation is most effective.

The principle has substantial implications for the structure of contemporary industries. Sectors whose contemporary profitability depends on unpriced externalities — fossil-fuel extraction, certain forms of factory

livestock production, certain extractive mining operations, certain components of the automotive industry, parts of the data-economy as currently structured — will, under full cost-bearing, become substantially less profitable. Sectors whose practices internalize their costs more thoroughly will become correspondingly more competitive. The structural shift is the point: the architecture relies on a price signal that, by faithfully representing actual costs, allows productive activity to flow toward the activities that are, in fact, productive of net value, and away from the activities that have, by externalization, only appeared to be.

RESTORATION OF GENUINE COMPETITION

The architecture's commitment to competitive markets requires more than the restoration of conventional antitrust enforcement, although that restoration is part of it. The contemporary failure of competition has structural sources that ordinary antitrust law was not designed to address: the network effects that make some platforms naturally winner-take-all; the data accumulations that make incumbents in data-rich sectors structurally advantaged against entrants; the patent and copyright extensions that have enclosed substantial portions of human knowledge for limited terms whose expiration has been repeatedly delayed; the regulatory complexity that small firms cannot afford to navigate but that established firms have learned to use as a barrier to entry.

The architecture addresses these structural sources by several specific commitments. Platforms above defined thresholds are required to provide interoperability with competing platforms, on terms that permit users to migrate their data and their connections to alternative services. Data accumulations subject to network effects are required to be available, on fair terms, to entrants whose use of the data does not violate the data-protection rights of the persons concerned. Patent and copyright terms are restored to the durations originally contemplated by the constitutional drafters of the modern

intellectual-property regimes, with extensions limited to defined exceptional cases. Regulatory complexity is reviewed continuously by the federation's regulatory authorities for its competitive effects, with simplification required where it does not serve the substantive purposes of the regulation.

The framers were conscious that the contemporary configuration of major platforms has been built up over decades and represents substantial investments by their operators and substantial value to their users. The architecture does not undertake the dismantling of the platforms; it undertakes the establishment of the structural conditions under which the platforms compete with possible alternatives on terms the alternatives have a fair chance of matching. The interoperability requirement, in particular, is designed to permit competition without disruption: users keep using the existing platforms if they prefer, but acquire the capacity to leave for alternatives if they choose, with their data and their connections intact.

TRUTHFUL DISCLOSURE

The disclosure architecture combines four operational components. First, financial reporting standards are aligned across the federation, with common rules on the recognition of revenue, the valuation of assets, the disclosure of contingent liabilities, and the treatment of intangible capital. The standards draw on the work of the International Accounting Standards Board and the International Sustainability Standards Board, with the federation's contribution being the binding application of the standards across its jurisdiction rather than the voluntary adoption that has characterized the contemporary regime.

Second, sustainability and human-rights reporting are required of enterprises above defined thresholds, on the same legal footing as financial reporting. The reporting covers the substantive matters that the federation's constitutional commitments make material: greenhouse-gas emissions across

the value chain, water and other ecological footprints, labor practices, supply-chain conditions, and the operation of intelligent systems within the enterprise. The reporting is auditable, on the same standards as financial reporting; misrepresentation has the same consequences.

Third, advertising and marketing communications are subject to truthfulness standards enforced across the federation. The standards are substantive: claims must be supportable by evidence at the time they are made; comparative claims must reflect a reasonable comparison; claims about a product's effects on health, the environment, or other matters of public concern must meet evidentiary standards proportionate to the claim's significance. Enforcement is by both public authority and by private right of action, with damages available on terms that make enforcement viable in cases of small per-person harm but large aggregate harm.

Fourth, the operations of intelligent systems used by enterprises in their dealings with persons are subject to the disclosure requirements of the Custody architecture: the person interacting with the system has a right to know that an intelligent system is involved, what data it uses, what decisions it makes, and how those decisions can be reviewed by a human party.

THE CORPORATE FORM AND ITS CUSTODIANS

The architecture's treatment of the corporate form is conservative. The corporation, as an institution permitting the assembly of capital and labor in productive enterprise, with limited liability for shareholders, has been one of the principal organizational achievements of the past several centuries, and the architecture preserves it. What the architecture modifies is the configuration of the corporation's accountability — the duties of its directors, the rights of its stakeholders, the obligations of its disclosures, the constraints on its political activity, and the conditions under which its limited liability operates.

The directors of corporations above defined thresholds owe duties not only to shareholders but to the constituencies whose welfare the corporation's activities materially affect: workers, consumers, communities, the environment, and future generations. The duties are concretized by the disclosure requirements above, by the standing of the affected constituencies in defined matters, and by the operation of the procedural protections of administrative law against corporate decisions of substantial public consequence.

The political activity of corporations is restricted to the activities legitimately associated with their productive functions: trade-association membership, technical comment on regulations affecting their operations, and contributions to civic life through philanthropic activity disclosed in detail. Direct political contributions to candidates, parties, or referendum campaigns are restricted on terms specified in the organic statute; the lobbying of legislators is regulated, registered, and recorded; the use of corporate resources for political purposes is disclosed in such manner as to permit shareholders, workers, and the public to assess and respond.

The framers were conscious that the contemporary configuration of corporate political activity has, in many jurisdictions, become a structural feature of legislative and regulatory processes. The architecture's response is incremental rather than revolutionary: clear rules, transparent disclosure, and the institutional standing of competing interests in the political process, rather than the prohibition of corporate participation in public life.

CLOSING OF PART FOUR

The eight chapters of Part Four — the public ledger, the governance layer, intelligence under custody, sovereign identity, economic instruments, the restoration of work, harmonized law, and honest enterprise — describe the operational architecture through which the federation's constitutional commitments become real. The architecture is not, in any of its components,

technologically novel in the sense that nothing like it has previously existed; it is novel in its combination, in the integration of components that have, in existing institutions, been pursued separately and at smaller scale, and in the deliberate alignment of the components to the principles stated in Part Two.

The architecture is incomplete in important respects. It does not specify, in detail, the implementation of every component, leaving substantial work to the founding convention and to the organic statutes; it does not anticipate every interaction among components, leaving substantial work to the institutions that will operate them; it does not foreclose the development of further components as conditions and capacities evolve. The architecture is, in this sense, a starting structure rather than a finished design. The starting structure is, however, sufficient: it gives the federation a working machinery for its first generation of operation, and it gives subsequent generations the institutional and conceptual base from which to refine the architecture in light of operational experience.

The remaining parts of the book address the transition to the federation from the conditions of 2026, and the longer horizon of what becomes possible after the federation is operating.



Part Five describes the transition: the work of the founding generation, from the initial convocation of politics through the first decade of operation.

PART FIVE

THE TRANSITION



No constitution begins on the first day of the world. The federation, when it finds itself, will find itself in conditions it did not choose, with materials that have already been used, in a world whose existing arrangements will not yield by argument alone.

The three chapters that follow describe how a transition that does not begin in catastrophe might be made to work.

CHAPTER TWENTY-FOUR

YEAR ONE — THE FOUNDING



A FEDERATION OF THE KIND DESCRIBED IN PARTS THREE AND FOUR cannot be brought into existence by a single act, on a single day, by a single party. It will require, at minimum, an originating convocation of consenting polities; a draft constitutional text on which the convocation can deliberate; the ratification of that text by the polities and, in many of them, by their populations; the establishment of the institutions named in the text; the convening of the institutions; the enactment of the founding organic statutes; and the operational launching of the architectural subsystems. The chapter that follows offers a working sketch of how the first year of this work might be conducted, on the assumption that political conditions sufficient to begin the work are obtained. The sketch is not a prediction; it is a planning document, the kind that any serious institutional founding requires, written before the political conditions are clear so that, when they become clear, the work has somewhere to begin.

THE ORIGINATING CONVOCATION

The convocation is the moment at which a defined set of consenting polities, on the basis of an articulated common will, undertake to establish a federation of the kind here described. The convocation has historical analogues: the Philadelphia Convention of 1787, the Paris Conference of 1919, the San Francisco Conference of 1945, the Maastricht negotiations of 1991-92. None of the analogues is fully apposite to the present case; each provides instructive features.

The Philadelphia precedent suggests that a convocation is most effective when its mandate is broad, its participation is restricted to the polities directly engaged, and its proceedings are conducted with sufficient privacy to permit honest exchange but with sufficient eventual transparency to permit informed ratification. The architectural choice in the present case is to follow the precedent: the convocation deliberates in working session over the course of an extended period (in the planning sketch, six to twelve months), with participants drawn from the consenting polities at a level of authority sufficient to commit their governments to the convocation's product, with daily proceedings recorded but not published until the convocation has concluded, and with the final product committed to ratification by the procedures of each consenting polity.

The convocation's product is not, in this sketch, a finished constitution. It is, more modestly, a draft constitutional text, a draft schedule of organic statutes, a draft schedule for the institutional founding, and a draft set of accession terms for polities that wish to join the federation after the founding. The draft is committed to ratification; the ratification process is itself a substantial and time-consuming undertaking, involving in most polities a combination of parliamentary action and, in some, popular referendum.

The architectural commitment to legitimacy through ratification is firm. The federation does not begin by the simple announcement of its founding; it begins by the ratification of its constitution by polities representing a substantial share of the planetary population, in such number and distribution as the convocation specifies. The threshold for entry into force is, deliberately, set at a level high enough to ensure that the founding is more than an act of a small coalition but low enough that the founding is not held hostage to the consent of every polity in the world. The framers expected, on the basis of comparable historical precedents, that the threshold would be reached on the consent of polities representing somewhere in the range of 60 to 75 percent of

the planetary population, with the remainder acceding over the following years and decades on terms agreed during the founding.

THE WORK OF THE FIRST YEAR

The first year of operation is, in the planning sketch, organized around the constitution and seating of the federation's institutions and the operational launch of the planetary basic provision in pilot regions.

The Council of Polities is convoked within thirty days of entry into force. Its first task is the appointment of the institutions of the federation: the judges of the Constitutional Court, the leadership of the Custody Authority, the leadership of the Standing Scientific Authority, the Tribune of Future Generations, the Auditor of the Federation, and the operational leadership of the public ledger. The appointments are made on the basis of the procedures specified in the founding organic statute and require, for each, the concurrence of the Citizens' Assembly. Because the Citizens' Assembly cannot itself be constituted before its members have been selected by the polities under the procedures of the organic statute, an interim Assembly is provided for in the founding schedule, drawn by random selection from the populations of the founding member polities, to perform the founding-period concurrence functions while the permanent Assembly is constituted.

The Constitutional Court is sworn in within ninety days. Its first cases, by historical pattern, are likely to be challenges to the founding actions themselves: petitions from non-member polities or from interests within member polities arguing that the founding is procedurally defective. The Court's response to these initial challenges, on the historical record of comparable foundings, is decisive for the federation's subsequent legitimacy. The framers expected that the Court would address the challenges with substantive engagement and, where the challenges are without merit, with timely dismissal supported by detailed reasons.

The public ledger is launched on a schedule timed to permit substantial technical preparation in advance of the political founding. The framers expected that the ledger's technical infrastructure would be developed, tested, and stress-tested over the period between the convocation's product and the entry into force, so that the ledger is operationally ready on the first day of operation. The launch involves the migration of operational records from the convocation itself onto the ledger, the deployment of the public-query infrastructure, and the activation of the signing infrastructure for the federation's officers.

The planetary basic provision is launched in pilot regions during the first year, on a defined schedule. The pilots are selected to represent a range of operational conditions: regions of substantial poverty in different climates, regions of relative prosperity in which the floor is operative for marginal populations, regions of recent conflict in which the institutional infrastructure is fragile. The pilot operation provides operational experience for the planetary tier and for the polity-level partner institutions, identifies the operational problems that an architectural sketch cannot anticipate, and demonstrates, to populations not yet covered, the practical content of the federation's commitment to dignity. The full extension to the population of the federation is scheduled for years two and three, on the basis of the operational learning of year one.

THE POLITICAL ENVIRONMENT

The federation's founding will not be a quiet event. The framers were conscious that powerful interests, in nearly every polity, have substantial reason to oppose the founding's success: the enterprises whose business models depend on the unpriced externalities the architecture corrects, the political actors whose authority depends on the absence of the cross-border institutions the architecture establishes, the populations whose understanding of national

identity does not extend to a planetary tier, the foreign powers (the term is used in the planning sketch without specification) whose international position depends on the absence of a planetary institution capable of constraining their conduct.

The opposition will operate on multiple fronts. It will argue that the founding is illegitimate, that its institutions are unaccountable, that its commitments are utopian, that its operations are corrupt, that its purposes are concealed. Some of the arguments will be substantive critiques offered in good faith and worthy of engagement. Others will be opportunistic. The framers' working assumption is that the federation responds to substantive critique with substantive answer and to opportunism with the patient and demonstrable performance of the principles to which the founding has committed itself.

The principal political asset of the federation, in this period, is the credibility of the public ledger. A founding institution that operates in real-time public view, whose acts are signed and traceable, whose proceedings are recorded, and whose performance against its commitments is continuously measurable, is harder to caricature than an opaque institution. The framers were, in this respect, optimistic: the architectural commitment to legibility is, among other things, a defense against political opposition that depends on misrepresentation.

THE ECONOMIC ENVIRONMENT

The federation's founding occurs in an economic environment whose response cannot be fully predicted. The principal sources of revenue uncertainty are the corporate-tax minimum (which depends on the global pattern of corporate response, including the relocation of activities and the restructuring of accounting), the greenhouse-gas charge (which depends on the response of energy markets, of governments operating energy-related fiscal instruments,

and of the industries directly affected), and the resource-rent charges (which depend on extraction patterns that may be partially within the federation's control and partially not).

The framers' working assumption is that revenues in the first three years will be in the range of 60 to 80 percent of the long-run estimates, with the lower end reflecting substantial transitional adjustment by the affected industries and the higher end reflecting more rapid stabilization than is historically typical. The planetary basic provision is, accordingly, scaled to the lower end of the revenue range during the transition, with the floor's threshold maintained at the substantive level required by Article V and the additional planetary functions phased in as revenues stabilize.

The framers were conscious that, in the founding period, the federation may be perceived as economically fragile. The perception is, in significant part, a self-fulfilling matter: a founding that performs against its commitments stabilizes the expectations on which the underlying revenues depend; a founding that struggles produces the conditions under which the struggle continues. The architectural response is to perform what is committed, as visibly as the legibility commitment requires, with the result that fragility narratives, where they appear, can be met with the demonstrable performance of operations.

THE INTERNATIONAL ENVIRONMENT

The federation's relationships with non-member states, with international institutions, and with the existing economic and security order will require, in the founding period, deliberate management. The framers' working assumption is that some non-member states will, in the founding period, regard the federation with hostility — particularly where the federation's commitments to human rights or to economic transparency are inconsistent with those states' domestic arrangements — and that some will regard the federation with

cautious cooperation, with the alignment in many cases shifting over the founding period as the federation's actual operations become clearer.

The architectural disposition toward non-member states is consistent: the federation does not extend its substantive jurisdiction beyond its members, does not undertake to coerce non-members into accession, does not interfere in the domestic affairs of non-members beyond the protections that international law generally allows, and does cooperate with non-member states on matters of common interest where the cooperation does not require the federation to compromise its constitutional commitments. The disposition is, in its outline, the disposition of any federation toward states outside its boundaries; the architectural specification is its application to a federation whose ambitions are planetary.

CLOSING OF THE YEAR

The first year ends, in this sketch, with the institutions of the federation seated, the public ledger operating, the planetary basic provision launched in pilot regions, the founding organic statutes enacted, and the principal operational subsystems either operating or in defined launch sequences. The federation is not, at the end of the first year, fully operational; it is operating, against the schedule of the founding, on the substantial work of subsequent years.

The principal achievement of the year is, less the operations launched than the political fact established: a federation of polities representing a substantial share of the planetary population has, by the deliberate ratification of a constitutional commitment, undertaken to govern certain planetary matters by deliberate institutional means. The fact, once established, alters the political environment in which all subsequent decisions are taken. The succeeding chapters describe what is done with the altered environment.



Chapter Twenty-Five describes the convergence of the second through fifth years: the period in which operations stabilize, accession of additional polities proceeds, and the architectural subsystems extend to their full operational scope.

CHAPTER TWENTY-FIVE

YEARS TWO THROUGH FIVE — CONVERGENCE



THE SECOND THROUGH FIFTH YEARS OF THE FEDERATION ARE, IN THE planning sketch, the period of convergence: the period during which the operations launched in the first year stabilize, the architectural subsystems extend to their full operational scope, additional polities accede on terms negotiated with the founding members, and the federation's commitments are tested against the operational realities the founding sketch could not fully anticipate. The chapter that follows describes the principal lines of work of this period and the failure modes against which the architecture provides defense.

STABILIZATION OF OPERATIONS

The principal operational task of years two through five is the stabilization of the institutions launched in the first year. The Council of Politics and the Citizens' Assembly develop their procedural cultures: the rhythms of session, the conventions of deliberation, the relationships among the standing committees, the working relationship between the chambers, the response to the routine and the exceptional matters that come before them. The Constitutional Court accumulates its first body of operative jurisprudence, on the constitutional questions that arise in the founding period and on the increasingly substantive matters that arise as operations expand. The Custody Authority moves from registry-establishment to active supervision: the first

audits, the first restrictions, the first revocations, the first incidents reported through the safety-reporting infrastructure.

The framers' working assumption is that, by the conclusion of the fifth year, the institutions are operating with the routine reliability that comparable institutions in the late industrial age have achieved. The reliability is the practical condition of the federation's deepening: institutions whose operations are reliable can be entrusted with additional responsibilities; institutions whose operations are unreliable cannot. The architectural commitment to disciplined operation in the founding period is, on this account, a long-term investment in the federation's later capacity to deepen.

EXTENSION OF THE PLANETARY BASIC PROVISION

The provision moves, during years two through five, from pilot operation to full coverage of the federation's population. The extension proceeds on a defined schedule, with priority given to regions of substantial poverty and operational readiness, and with deliberate attention to the operational and political learning of the pilots. The framers' working assumption is that, by the end of the fourth year, the substantive coverage of the floor is established for the population of the federation, with the operational refinements continuing through the fifth year and into the years following.

The economic effects of the provision become visible during this period. The empirical literature on cash-transfer programs in pilot conditions has consistently shown effects in the directions the architecture anticipates: reductions in absolute poverty, improvements in indicators of child welfare and educational attainment, modest reductions in conventional employment offset by increases in entrepreneurship, education, and care work, improvements in mental-health and stress-related health indicators, and reductions in the violence and instability that severe deprivation produces

(United Nations Development Programme 2024; Standing 2011b). The full-coverage extension, on the architectural sketch, produces these effects at planetary scale.

The political effects are not negligible. A population that has experienced, during the founding period, the practical content of the federation's commitment to dignity is, on the empirical record of comparable transitions, more disposed to support the federation's other commitments, more disposed to participate in the federation's institutions, and more resistant to political opposition that proceeds by misrepresenting the federation's actual operations. The political effect is, like the economic effect, the deliberate consequence of the architectural choice; it is the practical content of the proposition that legitimacy is produced by performance.

STABILIZATION OF THE REVENUE INSTRUMENTS

The greenhouse-gas charge, the corporate-tax minimum, and the other revenue instruments are, during years two through five, brought to their full operational scope. The instruments require, in their early years, substantial administrative learning: the determination of liability, the procedures of audit, the resolution of disputes over computation, the handling of avoidance strategies, the coordination with member-polity tax authorities. The administrative learning is necessary work, and the framers' working assumption is that the revenue yield in years two through five rises through this period from the lower-end estimates of year one to the long-run estimates by year four or five.

The economic adjustments to the revenue instruments — the responses of affected industries, the consumption-pattern shifts caused by the price signals, the investment-flow shifts toward activities the price signals favor — proceed through this period at a pace that is partially within the federation's control and partially not. The architectural commitment is to maintain the price signals at the levels the principles require, while supporting the affected

workers and communities through the transition support funded from the revenues themselves.

The framers were conscious that the early years of substantial price adjustment will produce political pressure to soften the price signals. The pressure has, in comparable adjustments in member polities (the early years of European emissions trading, the early years of various national carbon prices), been substantial. The architectural response is to maintain the substantive commitment, while addressing the legitimate distributive concerns that the pressure expresses, by mechanisms that do not undermine the substantive commitment. The transition-support component of the revenues, the protected fund's investments in affected regions, and the operation of the planetary basic provision in those regions are, jointly, the architectural answer to the distributive concern.

ACCESSION OF ADDITIONAL POLITIES

The first five years are, on the planning sketch, also the period of substantial accession by polities not among the founding members. The framers' working assumption, based on the historical pattern of comparable institutional foundings, is that a substantial fraction of the polities outside the founding membership will, on observing the federation in operation, undertake the procedures of accession during this period. The accession is on terms negotiated between the acceding polity and the federation, with the substantive constitutional commitments non-negotiable and the operational arrangements (transition periods, contributions, representation in the Council of Politics) subject to the negotiation.

The accession process is, in this sketch, deliberate and unhurried. The federation does not seek the accession of polities whose ratification of the constitutional commitments would be merely formal and whose subsequent compliance would require continuous coercion. The framers' working

assumption is that some polities, particularly those whose domestic political arrangements are inconsistent with the constitutional commitments, will not seek accession in the founding period, and that the federation's relationship with these polities is best conducted on the international plane rather than through coerced membership.

The accession of polities whose populations represent significant additions to the federation's membership produces operational adjustments: the planetary basic provision is extended to the new populations on the schedule appropriate to the local conditions; the institutional representation in the Council is recomputed; the operational infrastructure of the public ledger and the other federation systems is extended to cover the new jurisdictions. The framers' working assumption is that, by the conclusion of the fifth year, the federation's membership has stabilized at the level it will hold for the medium term, with subsequent accessions occurring at a slower pace as the remaining unaccounted populations consider their positions.

THE FIRST ENCOUNTER WITH SERIOUS FAILURE

No institutional founding of substantial scope proceeds without serious failure during its first years. The framers were conscious of the pattern. The federation's first failure, when it occurs, will reveal something the founding had not adequately addressed, and the response to the failure will, more than the original founding commitments, determine the federation's long-term character.

The candidates for the first serious failure include: a substantive bias or harmful operation by an intelligent system that the Custody Authority had registered but had not adequately supervised; an operational failure of the public ledger that produces, even briefly, a loss of legibility on consequential matters; a corruption case affecting senior federation officials, whose handling will set the precedent for subsequent comparable cases; a constitutional crisis

between the chambers or between the federation and a member polity, requiring the Constitutional Court to operate on a question of substantial significance with limited prior precedent; an external shock — a regional conflict, a financial crisis, a natural disaster of substantial scope — testing the federation's response capacity beyond what its founding institutions were sized for.

The architectural response to such failures is, in each case, the substantive performance of the principles. Failures are addressed openly, on the public ledger, with substantive engagement rather than minimization; responsible parties are held to account by the procedures of accountability; the institutional arrangements that produced the failure are revised in such manner as to reduce the probability of recurrence; the substantive commitments are maintained, even when the political pressure to soften them is substantial. The framers were conscious that the response to early failure is, more than any other operational matter, the principal source of the federation's long-term legitimacy or its long-term degeneration. The architectural commitment to the principled response, including in cases where the principled response is politically costly, is the practical content of the federation's seriousness about its own commitments.

THE INTERNATIONAL ENVIRONMENT IN THIS PERIOD

The federation's relationships with non-member states and with the existing international institutions evolve through this period. The principal lines of work include: the negotiation of cooperative arrangements with non-member states on matters of common interest (climate, trade, security, public health, cross-border crime); the harmonization of the federation's substantive commitments with the obligations of member polities under prior international agreements; the participation of the federation, in its own

institutional capacity, in the existing multilateral institutions on terms compatible with its constitutional structure; and the management of relationships with the substantial powers whose interests are most directly affected by the federation's operations.

The framers' working assumption is that the international environment in years two through five is, on balance, neither uniformly hostile nor uniformly cooperative, and that the federation's diplomacy is, accordingly, calibrated to the specific conditions of each external relationship. The architectural disposition is consistent: cooperation where it serves the federation's commitments, defense where the federation's commitments are challenged, and the patient construction of the legitimate international order in which the federation can operate as one institution among others rather than as a competing claim to comprehensive authority.

THE CHARACTER OF THE PERIOD

Years two through five are, in the planning sketch, the period during which the federation moves from being a constitutional commitment to being an operating reality. The transition is hard work; the work is mostly invisible; its outcome is, by the conclusion of the period, a federation whose operations are reliable enough that subsequent decisions can be taken on the basis of operational fact rather than institutional aspiration. The shift from aspiration to fact is, in the framers' account, the principal architectural achievement of the founding generation.

The achievement is not, in itself, transformative; the conditions diagnosed in Part One do not, by virtue of the federation's stabilization, immediately resolve. They begin, however, to be addressed by institutional means that are adequate to their scale, by procedures whose legitimacy is established by their disciplined operation, by commitments whose substantive content is demonstrated rather than merely declared. The transformation, where it

occurs, is the work of the period after — the steady-state operation described in the chapter that follows.



Chapter Twenty-Six describes the federation a decade out: the steady-state operation in which the architecture's commitments have produced their characteristic effects on the conditions of human life on Earth.

CHAPTER TWENTY-SIX

A DECADE OUT — STEADY STATE



TEN YEARS AFTER ENTRY INTO FORCE, THE FEDERATION, ON THE architectural sketch, has reached a steady state of operation. The institutions are routine; the substantive commitments are operating at their designed scale; the founding generation has, in some part, concluded its founding work and turned over operational responsibility to the second generation of officeholders. The chapter that follows describes the conditions of human life under this steady state, with the qualifications appropriate to a sketch written before the fact.

WHAT IS DIFFERENT IN THE CONDITIONS OF LIFE

The principal differences, by ten years out, are concentrated in the conditions described by the diagnostic chapters of Part One.

The trust deficit, on the architectural sketch, is partially repaired. The institutions whose operations are visible in real time, whose decisions are signed and traceable, and whose commitments have been demonstrated by performance over a decade enjoy levels of public trust that the institutions of the late industrial age, on the empirical record of the past several decades, no longer enjoyed. The repair is partial: trust takes longer to rebuild than to lose, and the inheritance of mistrust from the institutions of the previous era persists in attenuated form for at least a generation. The trajectory, however, is the reversal of the long secular decline diagnosed in Chapter Two (OECD 2024; Pew Research Center 2024).

The distributional pattern is altered. The unconditional floor of dignity has, by ten years out, materially eliminated absolute poverty within the federation, with the additional consequences in child welfare, educational attainment, and public health that the empirical literature anticipated. The contemporary inequality at the upper tail of the distribution has been moderated, less by direct redistribution than by the structural commitments of the architecture: the planetary minimum on corporate income, the elimination of the principal tax-avoidance routes that had produced the substantial below-statutory effective rates of the late industrial age, the pricing of externalities that had been captured as profit, the disclosure requirements that had eliminated the structural advantages of secrecy. The Gini coefficient at planetary scale, on the architectural sketch, has declined from its level at the founding by a substantive but not transformative margin; the absolute share of the population in deprivation has declined more sharply, in line with the priority the architecture gives to the floor (Piketty 2014; Saez and Zucman 2019).

The work conditions described in Chapter Four are altered. The portability of benefits, the restoration of countervailing voice, the lifelong retraining infrastructure, and the working-time discretion have, by ten years out, produced labor markets in which transitions are less ruinous, the bargaining position of workers is more equal, the recognition of unwaged labor is more substantive, and the connection between productivity gains and worker compensation has been at least partially restored. The shift is not, in any sense, a return to the post-war pattern; it is, more accurately, the construction of conditions appropriate to the contemporary economy that approximate the conditions the post-war pattern provided in the conditions of its time.

The legal architecture is operating. The harmonized framework permits cross-border activity within the federation on terms approaching the friction levels of intra-national activity; the constitutional rights are enforced uniformly; the procedural protections are available to persons regardless of

their location within the federation. The institutions of the federation, of the member polities, and of the lower tiers operate in coordination rather than competition, with the boundary disputes resolved by the procedures the architecture provides.

The intelligence-systems landscape is governed. The Custody Authority operates active supervision over the systems within its scope; the registration regime has produced a documented record of the systems in operation, the principals responsible for them, and the conditions of their deployment; the audit regime has identified and corrected substantial classes of failure that, in the absence of supervision, would have proceeded unaddressed; the capability ceilings have constrained the accumulation of capability under any single principal. The technology has continued to advance; the governance has, in distinction from the late industrial age, kept up.

The environmental trajectory is altered. The greenhouse-gas charge, operating at the social cost of carbon, has produced the structural economic adjustments the architects anticipated: substantial decline in fossil-fuel-intensive activities, substantial expansion of low-emission alternatives, substantial investment in adaptation infrastructure where the long-horizon climate damage cannot be avoided. The decade does not, by itself, resolve the climate problem; the climate problem is multi-decadal in nature, and ten years of disciplined response is the early portion of a much longer course of work. The trajectory, however, is consistent with the IPCC pathways toward the temperature targets the international community had nominally committed to during the late industrial age, in distinction from the trajectory the late industrial age was actually producing (Intergovernmental Panel on Climate Change 2023; International Energy Agency 2024).

WHAT IS NOT DIFFERENT

It is, equally important, to be clear about what the architectural sketch does not claim is different by ten years out.

The federation has not eliminated, and cannot eliminate, the structural conflict of interests among its members. Politics continue to disagree, on substantial matters, about the appropriate use of the federation's authority, the appropriate scope of cooperation, the appropriate response to particular crises. The disagreements are addressed by the procedures of the federation; they are not eliminated by them. The federation is, in this respect, an arena for the conduct of disagreement rather than the abolition of disagreement, and the framers regarded the distinction as important.

The federation has not eliminated, and cannot eliminate, the residual political pathologies that any large institution accumulates. Cases of corruption, of abuse of authority, of capture by particular interests, of underperformance in particular functions, occur during the decade and continue to occur in the conditions of steady-state operation. The architectural response is the institutional capacity to identify these failures, to address them by the procedures of accountability, and to learn from them in the design of subsequent operations. The federation does not claim to be free of failure; it claims to maintain the institutional conditions under which failure can be detected and addressed without bringing down the institution as a whole.

The federation has not, by ten years out, achieved the deeper cultural transformations that some early supporters of comparable projects have anticipated. The peoples of the federation continue to differ in their cultural commitments, their religious traditions, their linguistic identities, their political preferences, their economic structures. The federation has not undertaken the transformation of any of these; the architectural commitment to subsidiarity reserves these matters to the lower tiers and to the persons themselves. The transformation, where it occurs, is the work of those who

undertake it locally, on terms appropriate to local conditions, with the federation's role limited to the maintenance of the conditions under which local life can proceed without external coercion.

The federation has not abolished the international plane. Non-member states continue to operate on the international plane on terms only partially affected by the federation's existence; substantial fractions of the planetary population live in jurisdictions whose accession to the federation is not, on the ten-year horizon, in prospect; the matters that affect the planet as a whole — climate, the oceans, the atmosphere, the orbital commons — are, in domains where the federation cooperates with non-member states, addressed by negotiation rather than by federation legislation. The federation is, in this respect, one institution among others on the international plane, with substantial competence in defined domains and with the disposition toward cooperation on the matters where its competence is partial.

THE CHARACTER OF THE STEADY STATE

The steady state, on the architectural sketch, is not a destination at which the federation has arrived and from which no further development is required. It is, more modestly, the conditions under which the federation's operations have stabilized at the scale appropriate to its founding commitments, with the further development — the deepening of integration, the extension of competence, the response to new conditions — proceeding by the ordinary procedures the architecture provides.

The succeeding generations will, on this sketch, face their own decisions about the further development of the federation. They will face, in particular, decisions about: the extension of the constitutional commitments in light of conditions and capabilities that the founding generation could not anticipate; the deepening of integration in domains where the founding compromises proved inadequate; the response to the technological developments that, if past

patterns continue, will continue to produce capabilities and risks that the founding institutions did not anticipate; the management of the relationship between the federation and the non-member states whose accession or continued non-membership will, over the decades, take various forms.

The framers' working assumption is that these subsequent decisions are best taken by the generations that face them, on the basis of conditions the framers cannot fully foresee, by the institutional procedures the architecture provides. The architecture's contribution is the institutional framework within which the decisions can be taken legitimately and operationally; the substance of the decisions is properly the work of the parties who will make them.

The federation, ten years out, is, in this sense, less an end than a beginning. It is the conclusion of the founding work and the beginning of the operational work, and it is the operational work — the substantive performance of the commitments, the patient deepening of legitimate institutions, the resolution of the inevitable failures — that the architecture has been constructed to support. The conditions of human life under the federation, ten years in, are, on the architectural sketch, materially better than the conditions under which the founding generation began their work. The improvement is not transformation; it is the steady, disciplined, and sustained operation of institutions adequate to their tasks. The transformation, if any, is the work of subsequent generations, on terms they will set.



Part Six follows. It asks what becomes possible after, on the assumption that the architecture is operating: what new horizons of human work, of human knowledge, and of human presence beyond the planet are opened by the institutional foundations the federation provides.

PART SIX

WHAT BECOMES POSSIBLE



If the architecture is operating, the conditions of the present are no longer the conditions that constrain what comes next. The two chapters that follow ask, with deliberate restraint, what becomes possible on Earth and what becomes possible beyond it, on the assumption that the foundations have been laid.

CHAPTER TWENTY-SEVEN

WHAT BECOMES POSSIBLE ON EARTH



THE ARCHITECTURAL SKETCH OF PART FIVE CONCLUDED WITH A federation in steady state, ten years after entry into force. The conditions described there are not the destination of the work the architecture undertakes; they are the foundation. What becomes possible after that foundation is laid is the substantive question of this chapter. The discussion is, deliberately, restrained: the chapter does not promise transformations whose attainment depends on factors outside the architecture's control, does not predict the cultural shifts whose course no architecture can determine, and does not guarantee outcomes that depend on the substantive choices of the generations that will inherit the institutional framework. What it does describe are the substantive possibilities the foundation makes available, on the disciplined assumption that the foundation continues to operate.

A STABILIZED CLIMATE

The principal, and most consequential, possibility opened by the operating federation is the stabilization of the climate at temperature levels consistent with the persistence of stable agricultural patterns, of stable coastal infrastructure, of stable ecological communities, and of the stable social conditions that depend on these. The architecture's price signals on greenhouse-gas emissions, operating at the social cost of carbon, produce, on the empirical record of comparable price interventions and on the modeling of

the IPCC pathways, the rate of decarbonization required to hold the temperature increase to levels in the lower portion of the range the international community had nominally committed to during the late industrial age (Intergovernmental Panel on Climate Change 2023; IPCC 2022; Stern 2007).

The stabilization is, on this account, not utopia. It does not undo the warming that has already occurred; it does not restore the species and ecosystems that have already been lost; it does not eliminate the climate damage that the next several decades will produce regardless of the actions taken now. What it does is to halt, by a defined date in the second half of the century, the further accumulation of warming, and to begin, on a longer horizon, the gradual return of atmospheric concentrations to levels consistent with stable climate conditions.

The condition of human life under stabilized climate is materially different from the condition of human life under continued warming. The agricultural systems on which the food supply depends operate within climate envelopes for which their varieties, infrastructure, and labor patterns are calibrated. The coastal cities on which a substantial fraction of the planetary population lives operate without the persistent and accelerating threat of inundation. The ecosystems on which the planetary biosphere depends operate within their adaptive ranges. The migration pressures that climate displacement would otherwise produce are reduced to manageable levels, with the substantial benefits to international stability that the reduction implies.

The stabilization is, in summary, the condition under which the substantive achievements of human civilization can be sustained for the long horizon. It is, in this sense, the precondition of every other possibility this chapter describes.

THE REPAIR OF ECOSYSTEMS

The price signals on resource depletion, the protected fund's investments in ecological restoration, the harmonized regulation of cross-border environmental harms, and the planetary commons regime over the atmospheric, oceanic, and biological commons together produce conditions under which the active repair of damaged ecosystems becomes a substantial line of work. The work has, on the empirical record of contemporary restoration efforts at smaller scale, demonstrated capacity to recover degraded landscapes, to restore depleted fisheries, to recover endangered species populations, and to rebuild watershed function over decadal timescales (Raworth 2017).

The principal architectural contribution to the work is the funding model: the protected fund, drawing on a defined share of the planetary revenues and committed to long-horizon investments under the Tribune's oversight, is the kind of patient capital ecological restoration requires. Restoration projects whose payoff is a generation away, whose benefits are diffuse and non-extractable, and whose appearance on a quarterly accounting is unimpressive have not, on the empirical record, been competently funded by the institutions of the late industrial age. The architecture provides them with funding adequate to their nature.

The work is unspectacular and slow. It does not produce, on a yearly basis, the kind of headline outcomes the political institutions of the late industrial age were calibrated to favor. It produces, over decades, the gradual return of damaged systems to functional operation, with the substantive benefits — the carbon sequestration, the biodiversity preservation, the watershed services, the soil stability, the resilience to climate stress — accumulating on the long horizon. The architecture's commitment to the long horizon is the practical condition under which the work proceeds.

THE END OF ABSOLUTE POVERTY

The unconditional floor of dignity has, by ten years out, materially eliminated absolute poverty within the federation. The condition is not, on the architectural sketch, marginal. The number of persons globally living at or below the threshold of absolute deprivation has, in the late industrial age, varied between several hundred million and several billion depending on the threshold and the year; the elimination of this condition by direct provision is, in itself, the most consequential achievement of the architecture for the persons whose lives are immediately affected (United Nations Development Programme 2024; World Bank 2024).

The elimination is not, in itself, the achievement of human flourishing. Persons lifted out of absolute deprivation have not, by that fact, achieved comprehensive welfare; the floor is, by design, a floor and not a ceiling. The achievement is, more accurately, the elimination of a condition in which substantive flourishing is impossible. With absolute deprivation eliminated, the further work of human flourishing — the development of capabilities, the construction of meaningful work, the participation in cultural and civic life, the formation of substantive communities — can proceed on a foundation that does not, daily, require the energy of those undertaking it to be devoted to mere survival.

The substantive consequences, on the architectural sketch, are large. Educational attainment, on the empirical record of comparable interventions at smaller scale, rises substantially in populations whose food security and material stability are secured. Health outcomes, on the same record, improve along multiple dimensions, with the most consequential improvements in child mortality, maternal mortality, and the chronic conditions of stress-related illness. The labor-force capacities of populations released from absolute deprivation, on the same record, rise substantially, with consequences for the

productive capacity of the economies in which they operate (Sen 1999; Nussbaum 2011).

RESTORED WORK

The conditions of work under the architecture — the portable benefits, the restored countervailing voice, the lifelong retraining, the recognition of unwaged labor, the working-time discretion — produce, by ten years out, labor markets in which work is, for substantial fractions of the population, the source of meaning and identity that the philosophical tradition has identified as among the principal human goods. The shift is gradual, and is not, in any sense, a return to a previous configuration; it is the construction, in contemporary conditions, of work that affords the substantive goods of work (Graeber 2018).

The shift is, in particular, the result of the structural commitments of the architecture rather than of any direct policy on the content of work. The portability of benefits removes the structural pressure toward employment forms that maximize the security of the benefits at the cost of the meaningfulness of the work. The retraining infrastructure removes the structural pressure to remain in obsolete occupations for fear of the costs of transition. The recognition of unwaged labor removes the structural pressure to undertake paid work that is, on substantive grounds, less valuable than the unwaged work it displaces. The countervailing voice provides the substantive condition under which workers can shape the content and conditions of their work, rather than accepting whatever the structural pressure of the labor market produces.

FUNCTIONAL DEMOCRACY

The conditions of public life under the architecture — the legibility of institutional operation, the protection of substantive media independence, the procedural protections against the capture of regulatory and legislative processes, the regulation of intelligent systems used in political communication, the structural commitments to honest enterprise — produce, by ten years out, conditions of public life in which substantive democratic deliberation is possible at scale. The condition is not, on the architectural sketch, perfect; the political pathologies of the late industrial age are, on the empirical record of comparable transitions, persistent in attenuated forms for at least a generation.

The substantive change is not the elimination of political conflict; political conflict is the proper content of democratic life. The substantive change is the operation of political conflict on terms that are themselves legible: the parties to the conflict are visible, their funding is disclosed, their claims are subject to truthfulness standards, the institutions through which the conflict is processed are operating with verifiable independence, and the outcomes of the conflict are recorded on terms that permit subsequent generations to learn from them

(Habermas 1989a; Putnam 2000).

The functional democracy is, in this account, less an achievement than a maintenance task. The conditions under which it operates require continuous defense against the structural tendencies — the concentrations of wealth, the erosions of independent media, the captures of regulatory institutions — that, in the late industrial age, had progressively degraded its operation. The architecture provides the institutional means of the defense; the defense itself is the substantive work of the persons who operate the architecture's institutions.

LONG-HORIZON THINKING

The architectural commitment to longevity, operationalized through the Tribune of Future Generations, the protected fund, and the procedural protections of long-horizon decisions, produces, by ten years out, public deliberation in which the long horizon is a substantive consideration in the contemporary work. The shift is, on the empirical record of comparable institutional commitments at smaller scale (the Welsh Future Generations legislation, the Hungarian Ombudsman, the Finnish Committee for the Future), gradual but real. The long horizon, when it is institutionally represented, is considered. The long horizon, when it is not, is, in the routine pressure of contemporary politics, systematically discounted.

The substantive content of long-horizon consideration is the consideration of the conditions under which the persons of subsequent generations will live. The consideration includes, but is not limited to: the climate trajectory, the resource trajectories, the technological trajectories, the demographic trajectories, the institutional trajectories. The consideration is not predictive; the conditions of the long horizon cannot be precisely predicted. The consideration is, more modestly, the substantive recognition that the choices of the contemporary generation produce conditions for which subsequent generations will bear the consequences, and that the substantive interests of subsequent generations have a legitimate place in the consideration.

THE CHARACTER OF THE POSSIBLE

The possibilities described in this chapter are, in their substantive content, large. They are also, in their character, modest: they describe a planet whose climate is stabilized, whose ecosystems are gradually repaired, whose persons are not in absolute deprivation, whose work affords the goods of work, whose public life is functionally democratic, whose long horizons are institutionally considered. They do not describe a planet from which conflict has been

eliminated, from which suffering has been removed, from which all the conditions of the human predicament have been resolved. The framers were conscious of the distinction.

The possibilities described are, in this account, the possibilities of normal life on a planet whose institutional foundations are adequate to its conditions. They are not the possibilities of any final achievement; they are, more accurately, the possibilities of the ordinary life of substantial human civilization, sustained over the long horizon by institutions that are themselves sustainable. The achievement, where it is achieved, is the achievement of normalcy at planetary scale — a normalcy that the late industrial age, on the empirical record of its accumulating crises, was demonstrably unable to maintain.



Chapter Twenty-Eight asks what becomes possible beyond the planet — the long horizon of human presence in the orbital, lunar, and asteroidal commons, on the institutional foundations the federation provides.

CHAPTER TWENTY-EIGHT

WHAT BECOMES POSSIBLE BEYOND EARTH



AMONG THE SUBSTANTIVE CONSEQUENCES OF AN OPERATING FEDERATION is the institutional foundation it provides for human activity beyond the surface of the planet. The activity has, in the late industrial age, advanced from the experimental stage of the mid-twentieth century to the operational stage of the early twenty-first, with substantial commercial activity in low Earth orbit, the resumption of crewed lunar exploration on the schedule of multiple national programs and private operators, the planning of asteroid prospecting missions whose commercial viability depends on the possibility of substantial extraction, and the longer-horizon prospect of permanent human settlement beyond Earth. The conditions under which this activity proceeds — the legal, economic, and political conditions — are not, in the late industrial age, settled; they are, on the empirical record, governed by a thin and aging body of treaty law from the 1960s and 1970s, supplemented by national legislation that varies substantially in its substantive commitments. The chapter that follows describes how the federation’s architectural commitments apply to the conditions of activity beyond Earth, with the same restraint that has governed the previous chapters.

THE ORBITAL COMMONS

The orbital regime around Earth — the volume of space within which satellites operate, the radio-frequency spectrum used for communication with and

among them, the navigational and scientific resources orbital infrastructure provides — is, in the architectural sketch, treated as a planetary commons. The treatment follows from the principles of Article XII: the resource is finite (orbital slots in the more crowded altitude bands are, on the empirical record of contemporary congestion, materially scarce), the resource is non-excludable (the operations of one party in orbit affect the operations of all parties), the resource cannot be sustainably allocated by the unilateral action of individual states.

The architectural commitments produce, in operation, several specific consequences. The federation's planetary commons regime allocates orbital capacity by procedures that account for the legitimate uses of the commons by all parties, including parties whose contemporary capacity to operate in orbit is limited but whose interest in the persistence of the commons is substantive. The commercial users of the commons pay charges to the federation calibrated to the resource value, with the revenues credited to the planetary basic provision and to the protected fund's long-horizon investments in space infrastructure. The accumulation of debris in orbit, which has, on the empirical record of recent decades, become a substantial threat to the persistence of the commons itself, is regulated by enforceable rules with cleanup obligations on the parties responsible for the debris (Ostrom 1990).

The framers were conscious that the orbital commons regime is, of all the components of the architecture, the most directly dependent on the cooperation of the principal contemporary spacefaring powers. The federation's authority over the commons is, accordingly, exercised through cooperative arrangements with non-member states whose space programs are substantial, with the federation's contribution being the institutional capacity to coordinate the cooperation rather than the unilateral assertion of authority over parties not subject to its jurisdiction. The disposition is consistent with the architecture's general approach to the international plane.

THE LUNAR SURFACE

The lunar surface, and the resources of the Moon, are, on the architectural sketch, governed by an extension of the planetary commons regime. The principal substantive commitments are: the persistence of the lunar surface as a common heritage, in which the activity of any party is conducted with consideration for the legitimate interests of all; the regulation of resource extraction by procedures calibrated to the long-horizon sustainability of the resource and to the equitable participation of parties whose contemporary capacity to operate is limited; the protection of scientific sites, sites of historical significance, and sites of unusual geological character, by procedures comparable to those that govern the protection of comparable sites on Earth; the regulation of permanent human presence on the Moon by procedures that protect both the persons present and the lunar environment.

The framers were conscious that the lunar surface, unlike orbital space, is, on the empirical record of contemporary plans, becoming a site of permanent human presence on the multi-decade horizon. The presence will require infrastructure: living quarters, life-support systems, energy systems, transportation infrastructure, communications infrastructure. The infrastructure is, in part, the proper subject of commercial activity by enterprises whose investments are at risk; in part, it is the proper subject of public investment by the federation and by member polities whose space programs operate. The architectural commitments balance the commercial and the public, with the principal commitments — to the lunar commons, to scientific access, to the protection of the environment — taking precedence over the commercial.

The framers regarded the lunar regime as one of the architectural domains where the long horizon is most substantively at issue. The decisions of the contemporary generation about how the Moon is used will, on the multi-century horizon, shape the conditions under which subsequent generations

encounter the lunar environment. The Tribune's standing, on lunar policy as on other long-horizon matters, is the institutional means by which the long horizon is represented in the contemporary deliberation.

ASTEROID RESOURCES

The substantial resources of the asteroid belt — the metals, the volatiles, the rare materials whose extraction on Earth produces substantial environmental damage — are, on the architectural sketch, governed as a common-pool resource under procedures calibrated to the resource's character. The procedures permit substantial commercial activity, on the basis that the resource is, by Earth standards, effectively limitless, while protecting the legitimate interests in equitable participation, in the long-horizon stewardship of the resource, and in the prevention of the concentrations of capability that the asteroid economy might otherwise produce.

The principal architectural commitments are: the licensing of extraction by the federation's planetary commons regime, with charges calibrated to the substantive value of the resource and credited to the planetary basic provision; the equitable participation in the commercial activity by enterprises whose home polities are members of the federation, on terms that prevent the concentration of the resource under any single principal; the long-horizon protection of the asteroid environment, including the protection of certain bodies designated for scientific study or for cultural significance; the regulation of the use of asteroid resources to ensure that the consequences of their use — including the consequences for the orbital and lunar environments and for the climate of Earth, when the resources are used in ways that affect it — are accounted for in the regulatory framework.

The framers were conscious that the asteroid economy is, of all the prospects beyond Earth, the one with the most substantial potential commercial scale and, accordingly, the one most likely to produce the kinds of

structural concentrations the architecture is designed to prevent. The architectural attention to the prevention is, accordingly, substantial. The principal architectural defense is, as on Earth, the structural commitment to competition: the licensing of extraction is conducted on terms that prevent the accumulation of dominant positions, the entry of new participants is structurally facilitated, the substantive commitments of the architecture to the prevention of monopoly extend to the asteroid economy as they extend to the terrestrial economy.

THE LONGER HORIZON

The longer horizon of human presence beyond Earth — the prospect of permanent settlement on Mars, the prospect of presence in the outer solar system, the longer prospect of presence beyond the solar system on timescales that are, by the standards of contemporary technology, speculative — is not, in this chapter, addressed in detail. The framers' working assumption is that the conditions of the longer horizon will be substantially different from the conditions of the contemporary period, that the institutional arrangements appropriate to the longer horizon will be properly the work of the generations that face them, and that the architecture's contribution to the longer horizon is the institutional foundation rather than the substantive specification.

The principal architectural commitment relevant to the longer horizon is the commitment to the long horizon itself. The protected fund's investments, on the multi-decade and multi-century horizons, include the substantial public investment in the scientific and technological foundations on which the longer horizon will rest. The Tribune's standing, on long-horizon policy, includes the standing to advocate for the legitimate interests of the persons of subsequent centuries in the conditions of human presence beyond Earth. The federation's institutional capacity, sustained across generations, is the institutional

foundation on which the longer-horizon decisions can be taken legitimately and operationally.

The framers regarded the longer horizon as substantively important. The persistence of human civilization, on geological timescales, is materially advanced by the existence of human communities beyond Earth. The diversification of the substrate on which human civilization operates is, against the substantive risks the contemporary generation has identified, a substantive contribution to the resilience of the civilization as a whole. The architectural commitments do not, in themselves, achieve the diversification; they provide the institutional conditions under which the work of diversification, conducted by the parties who undertake it, can proceed on terms compatible with the legitimate interests of all parties on Earth and beyond.

THE CHARACTER OF THE WORK BEYOND EARTH

The work beyond Earth is, on the architectural sketch, the work of substantial generations on substantial timescales, with the contemporary generation contributing the institutional foundations and the early operational steps. The work is not, on this account, the principal achievement of the architecture; the principal achievement, as Chapter Twenty-Seven described, is the stabilization of the conditions of human life on Earth. The work beyond Earth is, more accurately, an extension of the architecture's commitments to a domain in which the same principles — legibility, subsidiarity, reversibility, accountability, dignity, custody, longevity — apply with appropriate adjustments.

The framers regarded the extension as substantive but not transformative. The work beyond Earth, conducted on the basis of the architectural commitments, becomes, over the decades and centuries, a substantial component of the human enterprise. It does not replace the work on Earth; the work on Earth remains, on every substantial measure, the principal site of

human civilization. It complements the work on Earth, with the substantive benefits — the scientific knowledge, the resource access, the resilience to substantive risks, the long-horizon viability of the civilization — accumulating over the long horizon.

The architecture's contribution, in this domain as in all the others it addresses, is the institutional foundation. The substantive content of the work beyond Earth, like the substantive content of the work on Earth, is the work of the persons and institutions that undertake it, on terms they will set, within the conditions the architecture provides. The conditions are the contribution; the work, beyond Earth and on it, is what the conditions make possible.



An epilogue follows, returning the voice of the work to its author and addressing, in the first person, what has been undertaken in writing this book and what is asked of those who read it.

EPILOGUE

A NOTE FROM THE AUTHOR



THE VOICE OF THE PRECEDING CHAPTERS HAS, BY DELIBERATE CHOICE, been the impersonal voice of the architect. The voice was a discipline: a writer who proposes constitutional commitments at planetary scale must, if the proposal is to be taken seriously, suppress the small accidents of his own life and address the reader on the substantive merits of the proposal. The discipline has been, on balance, the right one for the work. In this concluding section, however, I depart from it briefly, on grounds that I believe the reader is owed a small accounting of the person who has written what they have read.

I am Mircea S. Rusu. I am, by formation, a builder of systems — software, infrastructure, the kind of operational architectures that make ordinary digital life function. I have spent the working years of my adulthood close enough to the institutional and technical arrangements of the present age to see, as a practitioner sees, the points at which they fail and the directions in which the failures compound. I am not a constitutional lawyer; I am not a political scientist; I am not an economist by training. I claim no specialized authority beyond what the substantive content of the argument carries on its own.

The book has been written, more than for any other reason, because the conditions it describes seemed to me to be drifting toward consequences that no responsible adult could observe in silence. The trust failures, the distributional failures, the failures of long-horizon thinking, the emerging governance failures around intelligent systems, the persistent inability of the international order to address matters that are, by their nature, planetary — these are not abstractions to me. I see them, in their daily operation, in the

technical and institutional work I do. The book is the response of a builder who, having seen the failures, has tried to specify, with the discipline of a working architecture, what would have to be different.

I want to be clear about what I have not claimed. I have not claimed that the architecture proposed here is correct in every detail. The detail of any constitutional architecture is, on the historical record of comparable foundations, inevitably revised by the practical experience of operation. I have claimed, more modestly, that the architecture is sufficient to begin: that a federation founded on these commitments, operating these institutions, exercising these competences, would be in a position to address, by legitimate institutional means, the conditions that the present arrangements have proven unable to address. The sufficiency to begin is, in any constitutional founding, the substantive question. The subsequent refinement is the work of those who operate what has been begun.

I have not claimed that the architecture is achievable on any specific schedule. The political conditions under which a founding of this kind could begin are not, on the empirical record of the present moment, in immediate prospect. They may be, on a horizon of years or decades, brought into prospect by the accumulation of conditions that the present order cannot address. They may, on a longer horizon, be brought into prospect by deliberate political work. They may not be brought into prospect at all, and the conditions diagnosed in Part One may continue to compound until they produce the catastrophic ruptures that the architecture is constructed to prevent. I do not predict which of these will occur. I have written, instead, a planning document that is available, to those who undertake the political work, as one of the materials with which the work can begin.

I have not claimed, in any sense, that I myself will lead, organize, or otherwise undertake the political work. I am not a politician; I have no organizational platform for this kind of work; I have no constituency beyond

the persons who, having read the book, find in its substance reason to take its proposals seriously. The book is an offering to whoever, of the political actors of subsequent years, finds use for it. If it is useful to none, the offering will have failed; if it is useful to some, the offering will have served its purpose.

I am conscious that the book asks something of its readers. It asks them to consider, with substantive attention, conditions that are easier to look away from. It asks them to weigh, against the daily pressures of ordinary life, the substantive proposals for addressing those conditions. It asks them, where the proposals seem to them sound, to do what citizens of any constitutional order do: to advocate, to organize, to support the political actors who undertake the work, to participate in the deliberation by which substantive change becomes possible. The asking is the asking of any serious work of constitutional thought, addressed to the public on whom the substantive choices ultimately depend.

I close with the proposition that the work undertaken in this book is the work of an ordinary citizen, addressed to ordinary citizens, on the conditions of ordinary life on a planet whose institutional foundations are inadequate to the contemporary conditions. The proposal for adequate foundations is, on the historical record of comparable proposals, neither inevitable nor impossible. It is, more modestly, available — available to those who, having considered the substantive merits, find in them reason to undertake the work.

I thank the reader for the time given to the argument. I welcome correction on any point on which I am wrong. I welcome continuation by any party who finds use for the materials. The work is offered, in the substantive sense, to the reader; what is done with it is the reader's to decide.

Mircea S. Rusu

2026



GLOSSARY

DEFINED TERMS



The terms below are used with specific meanings throughout this work. Where a term appears with a meaning different from its ordinary use, the definition here governs.

Architecture. The integrated set of constitutional commitments, institutional structures, procedural rules, and operational subsystems described in this book. Used in distinction from the substantive policies the architecture enables, which remain the choice of the parties who operate the architecture.

Citizens' Assembly. The chamber of the federation's bicameral legislature representing persons directly, with members selected by random draw stratified by demographic and geographic factors, serving non-renewable terms of three years.

Constitutional Court. The federation's highest tribunal, with exclusive jurisdiction over the interpretation of the constitution and the organic statutes, and defined original and appellate jurisdiction in cases involving the federation's law.

Council of Politics. The chamber of the federation's bicameral legislature representing member polities, with weighted voting calibrated to population and to a minimum representational floor.

Custody Authority. The federation's regulatory authority over intelligent systems above defined capability thresholds, established by Article VIII of the constitution and operating under the principles of Chapter Eighteen.

Federation. The constitutional federation of polities described in this work, exercising defined competence in matters that are, by their nature, planetary, while reserving all other competence to the polities.

Floor of Dignity. The unconditional substantive minimum below which no person within the federation may fall, comprising the six entitlements of Article V: nutrition adequate for health, shelter adequate for safety, clothing adequate for climate, sanitary water and waste disposal, basic medical care, and basic education.

Intelligent System. An artificial computational system whose operations include capabilities of perception, reasoning, decision, or action that, in human persons, would be regarded as intelligent. Used as a generic term, without commitment to any particular technical architecture.

Member Polity. A constitutionally constituted political community that has ratified the federation's constitution and acceded to membership on terms negotiated with the federation.

Organic Statute. A federation statute, enacted by the procedures of Article XIII, that operationalizes the constitutional commitments. Organic statutes have constitutional standing within their domains and may be amended only by the procedures specified in the constitution itself.

Planetary Basic Provision. The unconditional cash transfer described in Article V, operationally calibrated to the floor of dignity, funded from the planetary revenues of the federation, and disbursed to persons within the federation regardless of employment, conduct, or other condition.

Planetary Commons. The atmospheric, oceanic, orbital, lunar, deep-seabed, electromagnetic, and biological commons, the use of which by any party affects the use available to all parties, governed by the federation under Article XII.

Polity. A constitutionally constituted political community within the federation. Used in preference to "state" to emphasize that the federation's

members may be of various constitutional forms (national states, regional confederations, indigenous polities) and that the architectural relationship is to the constituted political community, regardless of its specific form.

Principal. The person, enterprise, or institution that operates an intelligent system and bears the constitutional and statutory responsibilities for its operation under the custody regime.

Protected Fund. The federation's long-horizon fund, drawing on a defined share of planetary revenues, committed to investments whose returns accumulate on multi-decade and multi-generational horizons, under the supervision of the Tribune of Future Generations.

Public Ledger. The federation's cryptographically signed, append-only, public record of institutional acts, described in Chapter Sixteen and operating as the foundation of the legibility commitment.

Reversibility. The architectural principle, articulated in Chapter Nine, that decisions of substantial public consequence are taken on terms that permit revision in light of subsequent learning, with deliberate institutional preference for reversible options where reversible and irreversible options are otherwise comparable.

Subsidiarity. The architectural principle, articulated in Chapter Eight, that authority is exercised at the lowest tier capable of effective exercise, with higher tiers acting only on matters whose proper handling exceeds the capacity of lower tiers.

Tribune of Future Generations. The federation officer with standing to advocate, in defined matters, for the legitimate interests of the persons of subsequent generations. Established by Article XI and operating under the principles of Chapter Thirteen.



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A BOOK OF THIS SCOPE IS, IN ANY HONEST ACCOUNTING, THE PRODUCT OF more minds than the one whose name appears on the cover. The argument presented here draws on a long tradition of constitutional, economic, philosophical, and institutional thought, much of it represented in the bibliography that follows. Where the argument is sound, the credit belongs in substantial part to the writers, scholars, and practitioners on whose work it builds. Where the argument is wrong, the responsibility is mine.

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I am grateful, finally, to the readers of this book. The work of constitutional thought is a public work, addressed to a public whose substantive engagement is the condition of its having any substantive consequence. To those who have read the work, considered its propositions, and weighed them against the conditions of their own substantive understanding: my thanks. The continuation of the work, in whatever form it takes, is in your hands.



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The author writes about civilization, governance, technology, and the long horizon. *The Architect* is his treatise on the constitutional foundations of a planetary federation adequate to the conditions of the contemporary age. Subsequent works are in preparation.

The author can be reached through the publisher of this volume, on terms appropriate to substantive correspondence about the work.

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— *the architecture is the contribution; the work is what the conditions make possible*

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